ἄρα ἀπὸ τῆς ΔΒ πενταπλάσιόν ἐστι τοῦ ἀπὸ τῆς ΔΓ· ὅπερ ἔδει δεῖξαι.

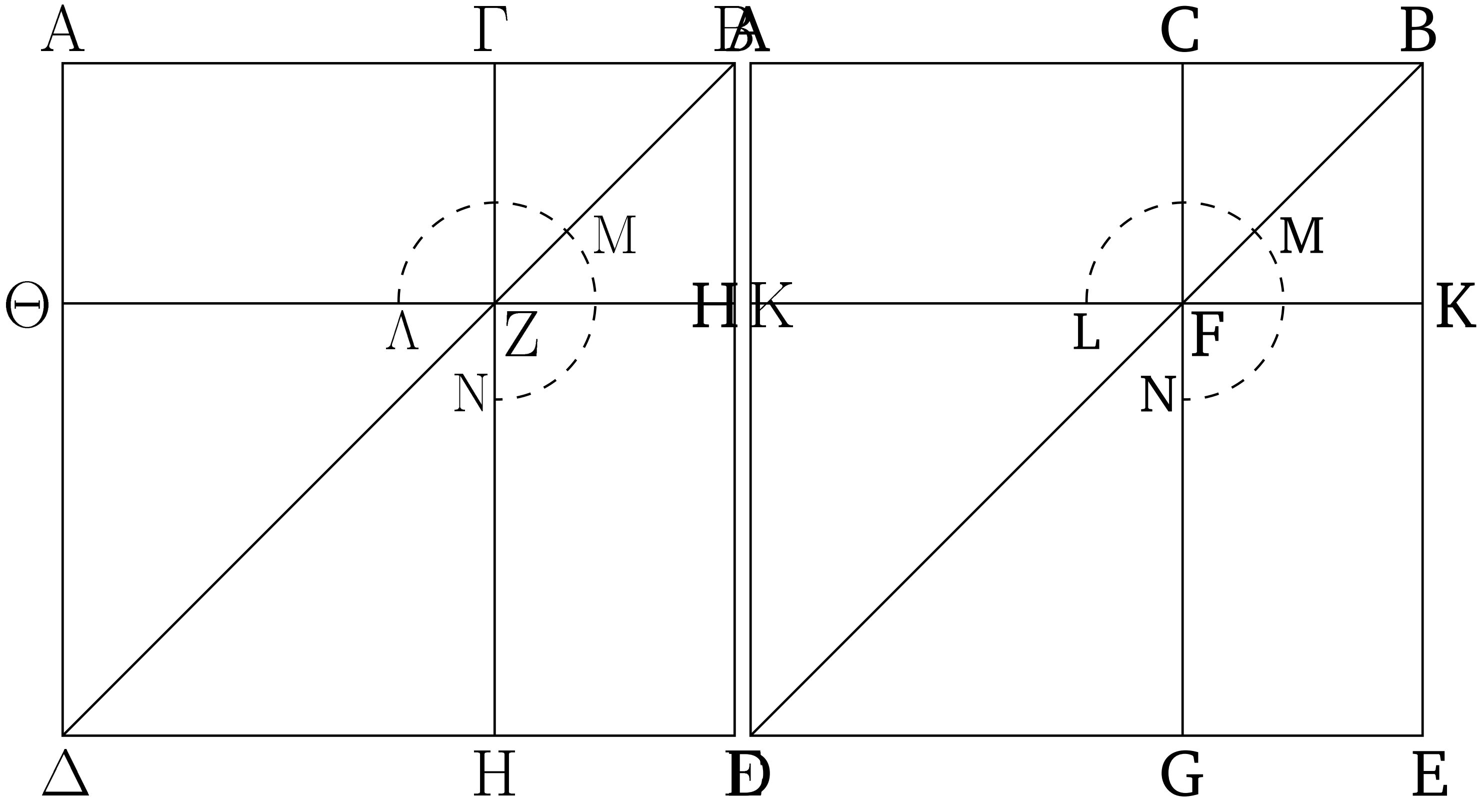
4.

Ἐὰν εὐθεῖα γραμμὴ ὁάκρον καὶ μέσον λόγον τμηθῇ, τὸ ἀπὸ τῆς ὅλης καὶ τοῦ ἐλάσσονος τμήματος, τὰ συναμφότερα τετράγωνα, τριπλάσιά ἐστι τοῦ ἀπὸ τοῦ μείζονος τμήματος τετραγώνου.

is five times the (square) on DC. (Which is) the very thing it was required to show.

Proposition 4

If a straight-line is cut in extreme and mean ratio then the sum of the squares on the whole and the lesser piece is three times the square on the greater piece.



Ἔστω εὐθεῖα ἡ ΑΒ, καὶ τετμήσθωὁάκρον καὶ μέσον λόγον κατὰ τὸ Γ, καὶ ὁέστω μεῖζον τμῆμα τὸ ΑΓ· λέγω, ὅτι τὰ ἀπὸ τῶν ΑΒ, ΒΓ τριπλάσιά ἐστι τοῦ ἀπὸ τῆς ΓΑ.

Ἀναγεγράφθω γὰρ ἀπὸ τῆς ΑΒ τετράγωνον τὸ ΑΔΕΒ, καὶ καταγεγράφθω τὸ σχῆμα. ἐπεὶ οὖν ἡ ΑΒὁάκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ Γ, καὶ τὸ μεῖζον τμῆμά ἐστιν ἡ ΑΓ, τὸ ἄρα ὑπὸ τῶν ΑΒΓ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΑΓ. καί ἐστι τὸ μὲν ὑπὸ τῶν ΑΒΓ τὸ ΑΚ, τὸ δὲ ἀπὸ τῆς ΑΓ τὸ ΘΗ· ἴσον ἄρα ἐστὶ τὸ ΑΚ τῷ ΘΗ. καὶ ἐπεὶ ἴσον ἐστὶ τὸ ΑΖ τῷ ΖΕ, κοινὸν προσκείσθω τὸ ΓΚ· ὅλον ἄρα τὸ ΑΚ ὅλῳ τῷ ΓΕ ἐστιν ἴσον· τὰ ἄρα ΑΚ, ΓΕ τοῦ ΑΚ ἐστι διπλάσια. ἀλλὰ τὰ ΑΚ, ΓΕ ὁ ΛΜΝ γνώμων ἐστὶ καὶ τὸ ΓΚ τετράγωνον· ὁ ἄρα ΛΜΝ γνώμων καὶ τὸ ΓΚ τετράγωνον διπλάσιά ἐστι τοῦ ΑΚ. ἀλλὰ μὴν καὶ τὸ ΑΚ τῷ ΘΗ ἐδείχθη ἴσον· ὁ ἄρα ΛΜΝ γνώμων καὶ [τὸ ΓΚ τετράγωνον διπλάσιά ἐστι τοῦ ΘΗ· ὁώστε ὁ ΛΜΝ γνώμων καὶ] τὰ ΓΚ, ΘΗ τετράγωνα τριπλάσιά ἐστι τοῦ ΘΗ τετραγώνου. καί ἐστιν ὁ [μὲν] ΛΜΝ γνώμων καὶ

Let AB be a straight-line, and let it be cut in extreme and mean ratio at C, and let AC be the greater piece. I say that the (sum of the squares) on AB and BC is three times the (square) on CA.

For let the square $ADEB$ be described on AB, and let the (remainder of the) figure be drawn. Therefore, since AB has been cut in extreme and mean ratio at C, and AC is the greater piece, the (rectangle contained) by ABC is thus equal to the (square) on AC [Def. 6.3, Prop. 6.17]. And AK is the (rectangle contained) by ABC, and HG the (square) on AC. Thus, AK is equal to HG. And since AF is equal to FE [Prop. 1.43], let CK be added to both. Thus, the whole of AK is equal to the whole of CE. Thus, AK plus CE is double AK. But, AK plus CE is the gnomon LMN plus the square CK. Thus, gnomon LMN plus square CK is double AK. But, indeed, AK was also shown (to be) equal to HG. Thus, gnomon LMN plus [square CK is double HG. Hence, gnomon LMN plus] the squares CK

τὰ ΓΚ, ΘΗ τετράγωνα ὅλον τὸ ΑΕ καὶ τὸ ΓΚ, ὁάπερ ἐστὶ τὰ ἀπὸ τῶν ΑΒ, ΒΓ τετράγωνα, τὸ δὲ ΗΘ τὸ ἀπὸ τῆς ΑΓ τετράγωνον. τὰ ἄρα ἀπὸ τῶν ΑΒ, ΒΓ τετράγωνα τριπλάσιά ἐστι τοῦ ἀπὸ τῆς ΑΓ τετραγώνου· ὅπερ ἔδει δεῖξαι.

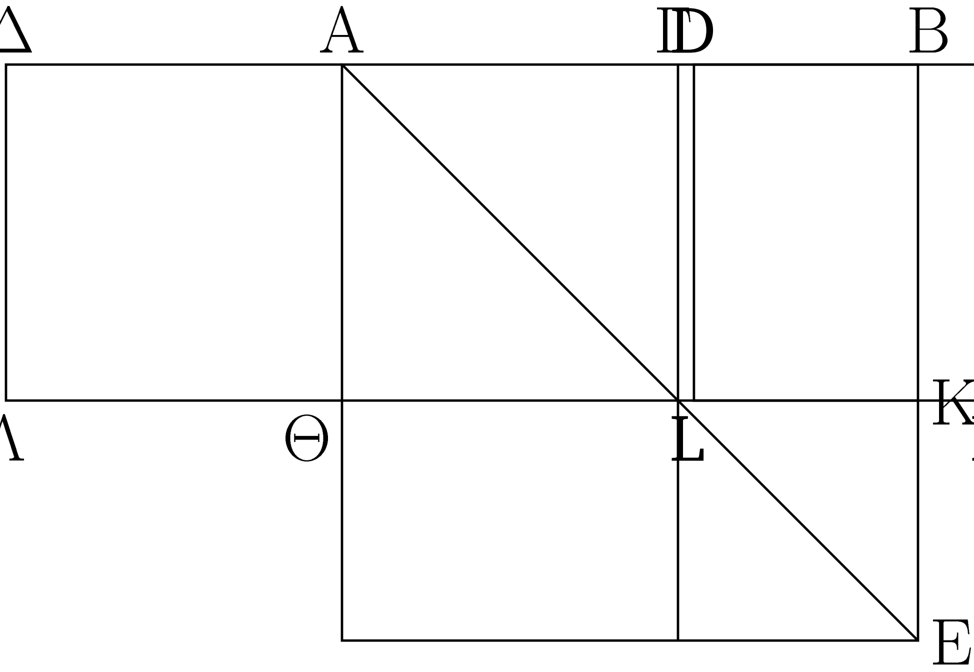
and *HG* is three times the square *HG*. And gnomon *LMN* plus the squares *CK* and *HG* is the whole of *AE* plus *CK*—which are the squares on *AB* and *BC* (respectively)—and *GH* (is) the square on *AC*. Thus, the (sum of the) squares on *AB* and *BC* is three times the square on *AC*. (Which is) the very thing it was required to show.

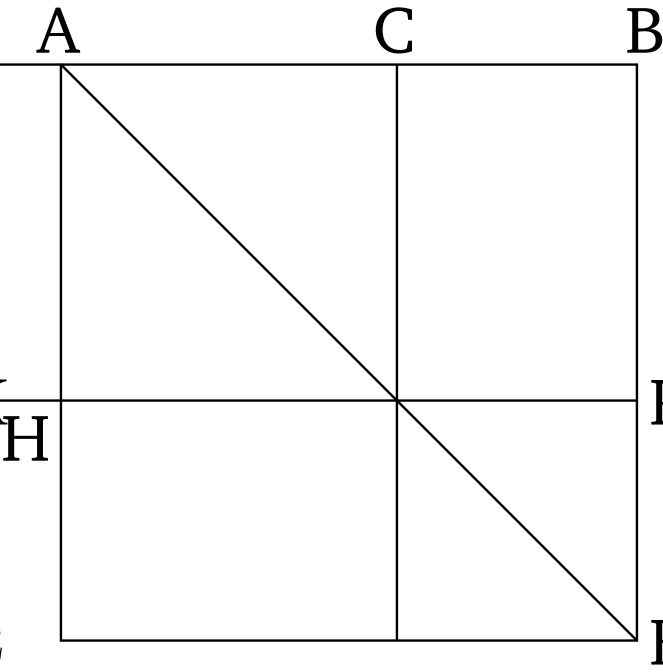
5.

Proposition 5

Ἐὰν εὐθεῖα γραμμὴ ὁάκρον καὶ μέσον λόγον τμηθῇ, καὶ προστεθῇ αὐτῇ ὁίση τῷ μείζονι τμήματι, ἡ ὅλη εὐθεῖαὁάκρον καὶ μέσον λόγον τέτμηται, καὶ τὸ μεῖζον τμῆμά ἐστιν ἡ ἐξ ἀρθῆς εὐθεῖα.

If a straight-line is cut in extreme and mean ratio, and a (straight-line) equal to the greater piece is added to it, then the whole straight-line has been cut in extreme and mean ratio, and the original straight-line is the greater piece.





Εὐθεῖα γὰρ γραμμὴ ἡ ΑΒὁάκρον καὶ μέσον λόγον τετμήσθω κατὰ τὸ Γ σημεῖον, καὶ ὁέστω μεῖζον τμῆμα ἡ ΑΓ, καὶ τῇ ΑΓὁίση [κείσθω] ἡ ΑΔ. λέγω, ὅτι ἡ ΔΒ εὐθεῖαὁάκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ Α, καὶ τὸ μεῖζον τμῆμά ἐστιν ἡ ἐξ ἀρθῆς εὐθεῖα ἡ ΑΒ.

Ἀναγεγράφθω γὰρ ἀπὸ τῆς ΑΒ τετράγωνον τὸ ΑΕ, καὶ καταγεγράφθω τὸ σχῆμα. ἐπεὶ ἡ ΑΒὁάκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ Γ, τὸ ἄρα ὑπὸ ΑΒΓ ἴσον ἐστὶ τῷ ἀπὸ ΑΓ. καί ἐστι τὸ μὲν ὑπὸ ΑΒΓ τὸ ΓΕ, τὸ δὲ ἀπὸ τῆς ΑΓ τὸ ΓΘ· ἴσον ἄρα τὸ ΓΕ τῷ ΘΓ. ἀλλὰ τῷ μὲν ΓΕ ἴσον ἐστὶ τὸ ΘΕ, τῷ δὲ ΘΓ ἴσον τὸ ΔΘ· καὶ τὸ ΔΘ ἄρα ἴσον ἐστὶ τῷ ΘΕ [κοινὸν προσκείσθω τὸ ΘΒ]. ὅλον ἄρα τὸ ΔΚ ὅλῳ τῷ ΑΕ ἐστιν ἴσον. καί ἐστι τὸ μὲν ΔΚ τὸ ὑπὸ τῶν ΒΔ, ΔΑ· ὁίση γὰρ ἡ ΑΔ τῇ ΔΛ· τὸ δὲ ΑΕ τὸ ἀπὸ τῆς ΑΒ· τὸ ἄρα ὑπὸ τῶν ΒΔΑ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΑΒ. ὁέστιν ἄρα ὡξ ἡ ΔΒ πρὸς τὴν ΒΑ, οὑύτως ἡ ΒΑ πρὸς τὴν ΑΔ. μείζων δὲ ἡ ΔΒ τῆς ΒΑ· μείζων ἄρα καὶ ἡ ΒΑ τῆς ΑΔ.

For let the straight-line *AB* be cut in extreme and mean ratio at point *C*. And let *AC* be the greater piece. And let *AD* be [made] equal to *AC*. I say that the straight-line *DB* has been cut in extreme and mean ratio at *A*, and that the original straight-line *AB* is the greater piece.

For let the square *AE* be described on *AB*, and let the (remainder of the) figure be drawn. And since *AB* has been cut in extreme and mean ratio at *C*, the (rectangle contained) by *ABC* is thus equal to the (square) on *AC* [Def. 6.3, Prop. 6.17]. And *CE* is the (rectangle contained) by *ABC*, and *CH* the (square) on *AC*. But, *HE* is equal to *CE* [Prop. 1.43], and *DH* equal to *HC*. Thus, *DH* is also equal to *HE*. [Let *HB* be added to both.] Thus, the whole of *DK* is equal to the whole of *AE*. And *DK* is the (rectangle contained) by *BD* and *DA*. For *AD* (is) equal to *DL*. And *AE* (is) the (square) on *AB*. Thus, the (rectangle contained) by *BDA* is equal to the (square) on *AB*. Thus, as *DB* (is) to *BA*, so *BA* (is) to *AD* [Prop. 6.17]. And

Ἡ ἄρα ΔΒὁάκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ Α, καὶ τὸ μεῖζον τμῆμά ἐστιν ἡ ΑΒ· ὅπερ ἔδει δεῖξαι.

DB (is) greater than *BA*. Thus, *BA* (is) also greater than *AD* [Prop. 5.14].

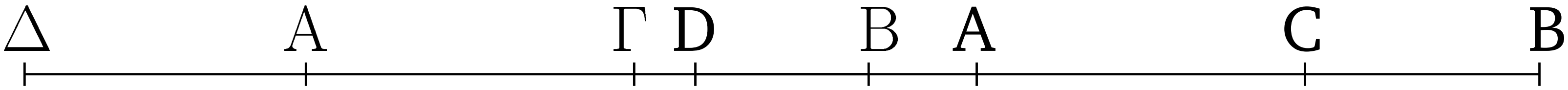
Thus, *DB* has been cut in extreme and mean ratio at *A*, and the greater piece is *AB*. (Which is) the very thing it was required to show.

6.

Proposition 6

Ἐὰν εὐθεῖα ῥητὴὁάκρον καὶ μέσον λόγον τμηθῇ, ἑκάτερον τῶν τμημάτωνὁάλογόξ ἐστιν ἡ καλουμένη ἀποτομή.

If a rational straight-line is cut in extreme and mean ratio then each of the pieces is that irrational (straight-line) called an apotome.



Ἔστω εὐθεῖα ῥητὴ ἡ ΑΒ καὶ τετμήσθωὁάκρον καὶ μέσον λόγον κατὰ τὸ Γ, καὶ ὁέστω μεῖζον τμῆμα ἡ ΑΓ· λέγω, ὅτι ἑκατέρα τῶν ΑΓ, ΓΒὁάλογόξ ἐστιν ἡ καλουμένη ἀποτομή.

Ἐκβεβλήσθω γὰρ ἡ ΒΑ, καὶ κείσθω τῆς ΒΑ ἡμίσεια ἡ ΑΔ. ἐπεὶ οὐῦν εὐθεῖα ἡ ΑΒ τέτμηταιὁάκρον καὶ μέσον λόγον κατὰ τὸ Γ, καὶ τῷ μείζονι τμήματι τῷ ΑΓ πρόσκειται ἡ ΑΔ ἡμίσεια οὐῦσα τῆς ΑΒ, τὸ ἄρα ἀπὸ ΓΔ τοῦ ἀπὸ ΔΑ πενταπλάσιόν ἐστιν. τὸ ἄρα ἀπὸ ΓΔ πρὸς τὸ ἀπὸ ΔΑ λόγον ἔχει, ὃν ἀριθμὸξ πρὸς ἀριθμόν· σύμμετρον ἄρα τὸ ἀπὸ ΓΔ τῷ ἀπὸ ΔΑ. ῥητὸν δὲ τὸ ἀπὸ ΔΑ· ῥητὴ γάρ [ἐστιν] ἡ ΔΑ ἡμίσεια οὐῦσα τῆς ΑΒ ῥητῆς οὐύσης· ῥητὸν ἄρα καὶ τὸ ἀπὸ ΓΔ· ῥητὴ ἄρα ἐστὶ καὶ ἡ ΓΔ. καὶ ἐπεὶ τὸ ἀπὸ ΓΔ πρὸς τὸ ἀπὸ ΔΑ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν, ἀσύμμετρος ἄρα μήκει ἡ ΓΔ τῇ ΔΑ· αἱ ΓΔ, ΔΑ ἄρα ῥηταί εἰσι δυνάμει μόνον σύμμετροι· ἀποτομὴ ἄρα ἐστὶν ἡ ΑΓ. πάλιν, ἐπεὶ ἡ ΑΒὁάκρον καὶ μέσον λόγον τέτμηται, καὶ τὸ μεῖζον τμῆμά ἐστιν ἡ ΑΓ, τὸ ἄρα ὑπὸ ΑΒ, ΒΓ τῷ ἀπὸ ΑΓ ἴσον ἐστίν. τὸ ἄρα ἀπὸ τῆς ΑΓ ἀποτομῆς παρὰ τὴν ΑΒ ῥητὴν παραβληθὲν πλάτος ποιεῖ τὴν ΒΓ. τὸ δὲ ἀπὸ ἀποτομῆς παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν πρώτην· ἀποτομὴ ἄρα πρώτη ἐστὶν ἡ ΓΒ. ἐδείχθη δὲ καὶ ἡ ΓΑ ἀποτομή.

Ἐὰν ἄρα εὐθεῖα ῥητὴ ὁάκρον καὶ μέσον λόγον τμηθῇ, ἑκάτερον τῶν τμημάτωνὁάλογόξ ἐστιν ἡ καλουμένη ἀποτομή· ὅπερ ἔδει δεῖξαι.

Let *AB* be a rational straight-line cut in extreme and mean ratio at *C*, and let *AC* be the greater piece. I say that *AC* and *CB* is each that irrational (straight-line) called an apotome.

For let *BA* be produced, and let *AD* be made (equal) to half of *BA*. Therefore, since the straight-line *AB* has been cut in extreme and mean ratio at *C*, and *AD*, which is half of *AB*, has been added to the greater piece *AC*, the (square) on *CD* is thus five times the (square) on *DA* [Prop. 13.1]. Thus, the (square) on *CD* has to the (square) on *DA* the ratio which a number (has) to a number. The (square) on *CD* (is) thus commensurable with the (square) on *DA* [Prop. 10.6]. And the (square) on *DA* (is) rational. For *DA* [is] rational, being half of *AB*, which is rational. Thus, the (square) on *CD* (is) also rational [Def. 10.4]. Thus, *CD* is also rational. And since the (square) on *CD* does not have to the (square) on *DA* the ratio which a square number (has) to a square number, *CD* (is) thus incommensurable in length with *DA* [Prop. 10.9]. Thus, *CD* and *DA* are rational (straight-lines which are) commensurable in square only. Thus, *AC* is an apotome [Prop. 10.73]. Again, since *AB* has been cut in extreme and mean ratio, and *AC* is the greater piece, the (rectangle contained) by *AB* and *BC* is thus equal to the (square) on *AC* [Def. 6.3, Prop. 6.17]. Thus, the (square) on the apotome *AC*, applied to the rational (straight-line) *AB*, makes *BC* as width. And the (square) on an apotome, applied to a rational (straight-line), makes a first apotome as width [Prop. 10.97]. Thus, *CB* is a first apotome. And *CA* was also shown (to be) an apotome.

Thus, if a rational straight-line is cut in extreme and mean ratio then each of the pieces is that irrational (straight-line) called an apotome.

7.

Proposition 7

Ἐὰν πενταγώνου ἰσοπλεύρου αἱ τρεῖς γωνίαι ἤτοι αἱ κατὰ τὸ ἑξῆςὁή αἱ μὴ κατὰ τὸ ἑξῆς ἴσαι ὦσιν, ἰσογώνιονὁέσται τὸ πεντάγωνον.

Πενταγώνου γὰρ ἰσοπλεύρου τοῦ ΑΒΓΔΕ αἱ τρεῖς γωνίαι

If three angles, either consecutive or not consecutive, of an equilateral pentagon are equal then the pentagon will be equiangular.

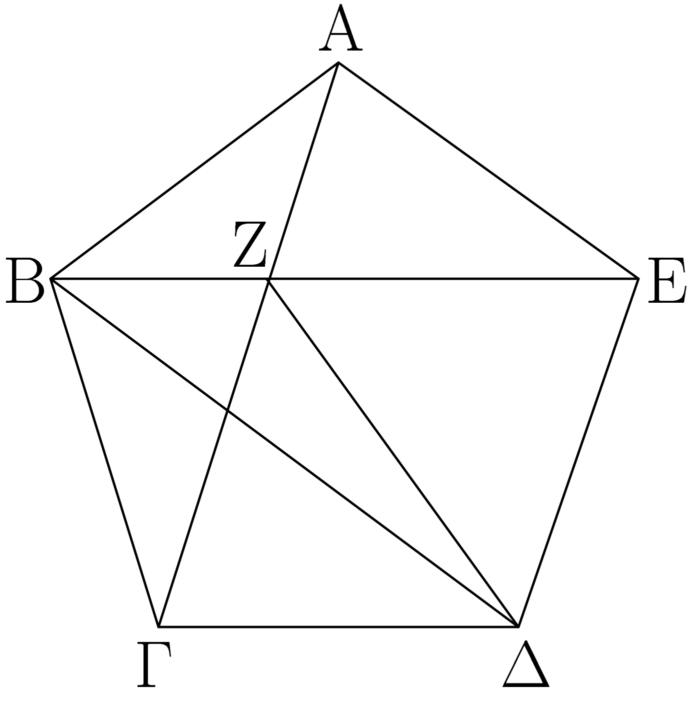
For let three angles of the equilateral pentagon *ABCDE*—

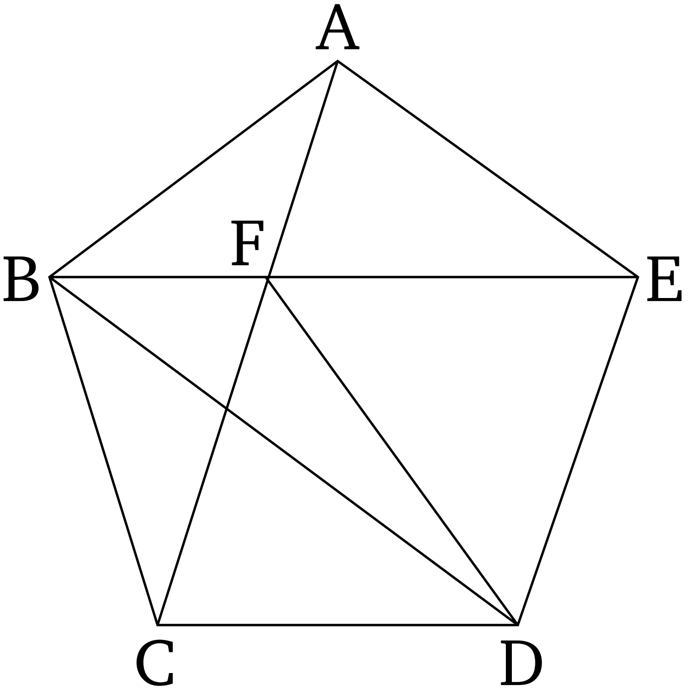
πρότερον αἱ κατὰ τὸ ἑχῆς αἱ πρὸς τοῖς Α, Β, Γ ἴσαι ἀλλήλαιςὀέστωσαν· λέγω, ὅτι ἰσογώνιόν ἐστι τὸ ΑΒΓΔΕ πεντάγωνον.

Ἐπεζεύχθωσαν γὰρ αἱ ΑΓ, ΒΕ, ΖΔ. καὶ ἐπεὶ δύο αἱ ΓΒ, ΒΑ δυσὶ ταῖς ΒΑ, ΑΕ ἴσαι ἐισὶν ἑκατέρα ἑκατέρᾳ, καὶ γωνία ἡ ὑπὸ ΓΒΑ γωνίᾳ τῇ ὑπὸ ΒΑΕ ἐστινὀίση, βάσις ἄρα ἡ ΑΓ βάσει τῇ ΒΕ ἐστινὀίση, καὶ τὸ ΑΒΓ τρίγωνον τῷ ΑΒΕ τριγώνῳ ἴσον, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖξ γωνίαις ἴσαιὀέσονται, ὑφὸ ὁὰξ αἱ ἴσαι πλευραὶ ὑποτείνουσιν, ἡ μὲν ὑπὸ ΒΓΑ τῇ ὑπὸ ΒΕΑ, ἡ δὲ ὑπὸ ΑΒΕ τῇ ὑπὸ ΓΑΒ· ὁώστε καὶ πλευρὰ ἡ ΑΖ πλευρᾷ τῇ ΒΖ ἐστινὀίση. ἐδείχθη δὲ καὶ ὅλη ἡ ΑΓ ὅλῃ τῇ ΒΕὀίση· καὶ λοιπὴ ἄρα ἡ ΖΓ λοιπῇ τῇ ΖΕ ἐστινὀίση. ὀέστι δὲ καὶ ἡ ΓΔ τῇ ΔΕὀίση. δύο δὴ αἱ ΖΓ, ΓΔ δυσὶ ταῖς ΖΕ, ΕΔ ἴσαι εἰσίν· καὶ βάσις αὐτῶν κοινὴ ἡ ΖΔ· γωνία ἄρα ἡ ὑπὸ ΖΓΔ γωνίᾳ τῇ ὑπὸ ΖΕΔ ἐστινὀίση. ἐδείχθη δὲ καὶ ἡ ὑπὸ ΒΓΑ τῇ ὑπὸ ΑΕΒὀίση· καὶ ὅλη ἄρα ἡ ὑπὸ ΒΓΔ ὅλῃ τῇ ὑπὸ ΑΕΔὀίση. ἀλλὸ ἡ ὑπὸ ΒΓΔὀίση ὑπόκειται ταῖς πρὸς τοῖς Α, Β γωνίαις· καὶ ἡ ὑπὸ ΑΕΔ ἄρα ταῖς πρὸς τοῖς Α, Β γωνίαιςὀίση ἐστίν. ὁμοίως δὴ δείχομεν, ὅτι καὶ ἡ ὑπὸ ΓΔΕ γωνίαὀίση ἐστὶ ταῖς πρὸς τοῖς Α, Β, Γ γωνίαις· ἰσογώνιον ἄρα ἐστὶ τὸ ΑΒΓΔΕ πεντάγωνον.

first of all, the consecutive (angles) at *A*, *B*, and *C*—-be equal to one another. I say that pentagon *ABCDE* is equiangular.

For let *AC*, *BE*, and *FD* be joined. And since the two (straight-lines) *CB* and *BA* are equal to the two (straight-lines) *BA* and *AE*, respectively, and angle *CBA* is equal to angle *BAE*, base *AC* is thus equal to base *BE*, and triangle *ABC* equal to triangle *ABE*, and the remaining angles will be equal to the remaining angles which the equal sides subtend [Prop. 1.4], (that is), *BCA* (equal) to *BEA*, and *ABE* to *CAB*. And hence side *AF* is also equal to side *BF* [Prop. 1.6]. And the whole of *AC* was also shown (to be) equal to the whole of *BE*. Thus, the remainder *FC* is also equal to the remainder *FE*. And *CD* is also equal to *DE*. So, the two (straight-lines) *FC* and *CD* are equal to the two (straight-lines) *FE* and *ED* (respectively). And *FD* is their common base. Thus, angle *FCD* is equal to angle *FED* [Prop. 1.8]. And *BCA* was also shown (to be) equal to *AEB*. And thus the whole of *BCD* (is) equal to the whole of *AED*. But, (angle) *BCD* was assumed (to be) equal to the angles at *A* and *B*. Thus, (angle) *AED* is also equal to the angles at *A* and *B*. So, similarly, we can show that angle *CDE* is also equal to the angles at *A*, *B*, *C*. Thus, pentagon *ABCDE* is equiangular.





Ἀλλὰ δὴ μὴ ὀέστωσαν ἴσαι αἱ κατὰ τὸ ἑχῆς γωνίαι, ἀλλὸ ὀέστωσαν ἴσαι αἱ πρὸς τοῖς Α, Γ, Δ σημείοις· λέγω, ὅτι καὶ οὁύτως ἰσογώνιόν ἐστι τὸ ΑΒΓΔΕ πεντάγωνον.

Ἐπεζεύχθω γὰρ ἡ ΒΔ. καὶ ἐπεὶ δύο αἱ ΒΑ, ΑΕ δυσὶ ταῖς ΒΓ, ΓΔ ἴσαι εἰσὶ καὶ γωνίας ἴσας περιέθουσιν, βάσις ἄρα ἡ ΒΕ βάσει τῇ ΒΔὀίση ἐστίν, καὶ τὸ ΑΒΕ τρίγωνον τῷ ΒΓΔ τριγώνῳ ἴσον ἐστίν, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖξ γωνίαις

And so let consecutive angles not be equal, but let the (angles) at points *A*, *C*, and *D* be equal. I say that pentagon *ABCDE* is also equiangular in this case.

For let *BD* be joined. And since the two (straight-lines) *BA* and *AE* are equal to the (straight-lines) *BC* and *CD*, and they contain equal angles, base *BE* is thus equal to base *BD*, and triangle *ABE* is equal to triangle *BCD*, and the remaining

ἴσαιὁέσονται, ὑφὸ ὁὰξ αἱ ἴσαι πλευραὶ ὑποτείνουσιν· ὁίση ἄρα ἐστὶν ἡ ὑπὸ ΑΕΒ γωνία τῇ ὑπὸ ΓΔΒ. ὁέστι δὲ καὶ ἡ ὑπὸ ΒΕΔ γωνία τῇ ὑπὸ ΒΔΕὁίση, ἐπεὶ καὶ πλευρὰ ἡ ΒΕ πλευρᾷ τῇ ΒΔ ἐστινὁίση. καὶ ὅλη ἄρα ἡ ὑπὸ ΑΕΔ γωνία ὅλῃ τῇ ὑπὸ ΓΔΕ ἐστινὁίση. ἀλλὰ ἡ ὑπὸ ΓΔΕ ταῖς πρὸς τοῖς Α, Γ γωνίαις ὑπόκειταιὁίση· καὶ ἡ ὑπὸ ΑΕΔ ἄρα γωνία ταῖς πρὸς τοῖς Α, Γὁίση ἐστίν. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ ΑΒΓὁίση ἐστὶ ταῖς πρὸς τοῖς Α, Γ, Δ γωνίαις. ἰσογώνιον ἄρα ἐστὶ τὸ ΑΒΓΔΕ πεντάγωνον· ὅπερ ἔδει δεῖξαι.

angles will be equal to the remaining angles which the equal sides subtend [Prop. 1.4]. Thus, angle *AEB* is equal to (angle) *CDB*. And angle *BED* is also equal to (angle) *BDE*, since side *BE* is also equal to side *BD* [Prop. 1.5]. Thus, the whole angle *AED* is also equal to the whole (angle) *CDE*. But, (angle) *CDE* was assumed (to be) equal to the angles at *A* and *C*. Thus, angle *AED* is also equal to the (angles) at *A* and *C*. So, for the same (reasons), (angle) *ABC* is also equal to the angles at *A*, *C*, and *D*. Thus, pentagon *ABCDE* is equiangular. (Which is) the very thing it was required to show.

8.

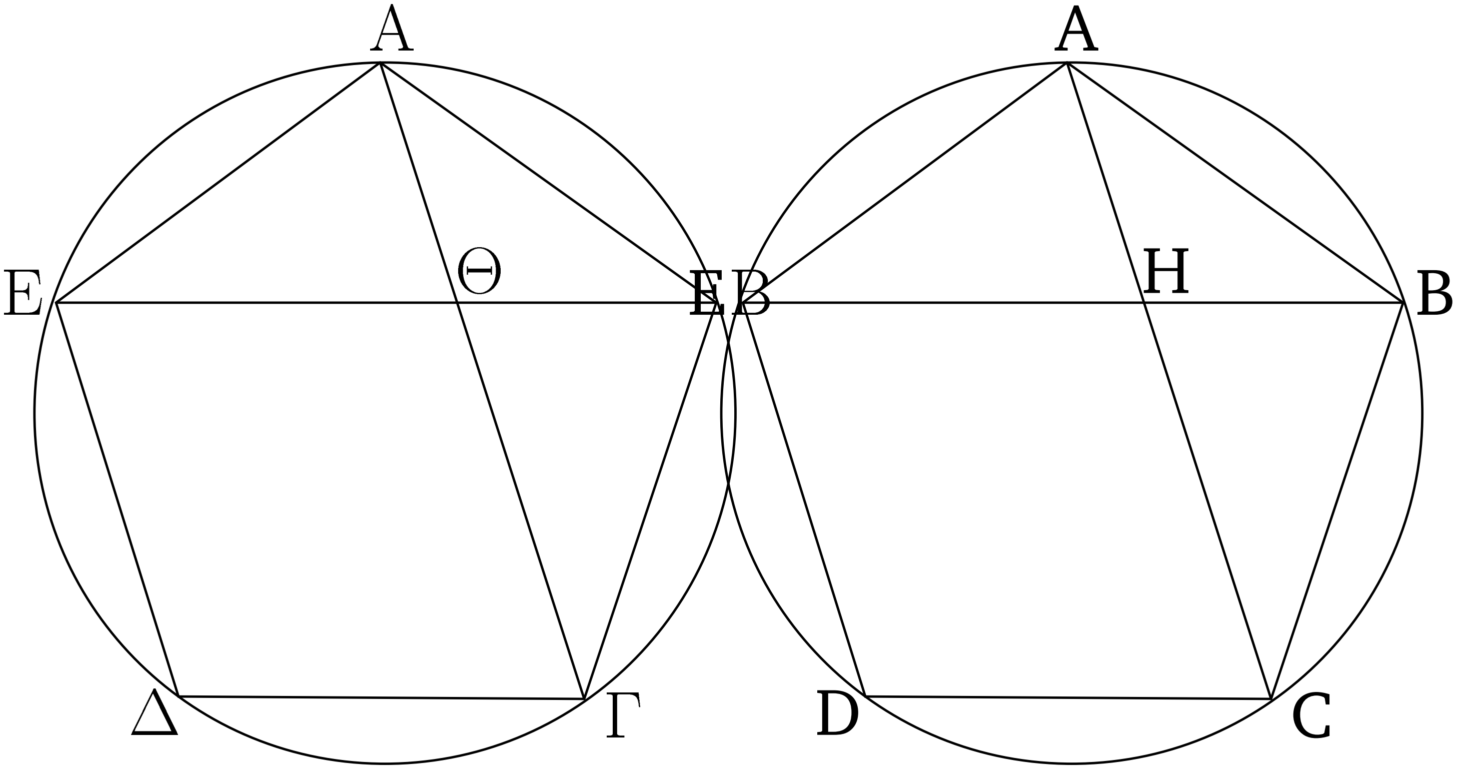
Ἐὰν πενταγώνου ἰσοπλεύρου καὶ ἰσογωνίου τὰξ κατὰ τὸ ἑξῆς δύο γωνίας ὑποτείνωσιν εὐθεῖαι, ὁάκρον καὶ μέσον λόγον τέμνουσιν ἀλλήλας, καὶ τὰ μείζονα αὐτῶν τμήματα ἴσα ἐστὶ τῇ τοῦ πενταγώνου πλευρᾷ.

Πενταγώνου γὰρ ἰσοπλεύρου καὶ ἰσογωνίου τοῦ ΑΒΓΔΕ δύο γωνίας τὰξ κατὰ τὸ ἑξῆς τὰξ πρὸς τοῖς Α, Β ὑποτεινέτωσαν εὐθεῖαι αἱ ΑΓ, ΒΕ τέμνουσαι ἀλλήλας κατὰ τὸ Θ σημεῖον· λέγω, ὅτι ἑκατέρα αὐτῶνὁάκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ Θ σημεῖον, καὶ τὰ μείζονα αὐτῶν τμήματα ἴσα ἐστὶ τῇ τοῦ πενταγώνου πλευρᾷ.

Proposition 8

If straight-lines subtend two consecutive angles of an equilateral and equiangular pentagon then they cut one another in extreme and mean ratio, and their greater pieces are equal to the sides of the pentagon.

For let the two straight-lines, *AC* and *BE*, cutting one another at point *H*, have subtended two consecutive angles, at *A* and *B* (respectively), of the equilateral and equiangular pentagon *ABCDE*. I say that each of them has been cut in extreme and mean ratio at point *H*, and that their greater pieces are equal to the sides of the pentagon.



Περιγεγράφθω γὰρ περὶ τὸ ΑΒΓΔΕ πεντάγωνον κύκλος ὁ ΑΒΓΔΕ. καὶ ἐπεὶ δύο εὐθεῖαι αἱ ΕΑ, ΑΒ δυσὶ ταῖς ΑΒ, ΒΓ ἴσαι εἰσὶ καὶ γωνίας ἴσας περιέθουσιν, βάσις ἄρα ἡ ΒΕ βάσει τῇ ΑΓὁίση ἐστίν, καὶ τὸ ΑΒΕ τρίγωνον τῷ ΑΒΓ τριγώνῳ ἴσον

For let the circle *ABCDE* be circumscribed about pentagon *ABCDE* [Prop. 4.14]. And since the two straight-lines *EA* and *AB* are equal to the two (straight-lines) *AB* and *BC* (respectively), and they contain equal angles, the base *BE* is

ἐστίν, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖξ γωνίαις ἴσαιὀέσονται ἑκατέρα ἑκατέρᾳ, ὑφὸ ὁἀξ αἱ ἴσαι πλευραὶ ὑποτείνουσιν. ὀίση ἄρα ἐστὶν ἡ ὑπὸ ΒΑΓ γωνία τῇ ὑπὸ ΑΒΕ· διπλῆ ἄρα ἡ ὑπὸ ΑΘΕ τῆς ὑπὸ ΒΑΘ. ὀέστι δὲ καὶ ἡ ὑπὸ ΕΑΓ τῆς ὑπὸ ΒΑΓ διπλῆ, ἐπειδήπερ καὶ περιφέρεια ἡ ΕΔΓ περιφερείας τῆς ΓΒ ἐστι διπλῆ· ὀίση ἄρα ἡ ὑπὸ ΘΑΕ γωνία τῇ ὑπὸ ΑΘΕ· ὁώστε καὶ ἡ ΘΕ εὐθεῖα τῇ ΕΑ, τουτέστι τῇ ΑΒ ἐστινὀίση. καὶ ἐπεὶ ὀίση ἐστὶν ἡ ΒΑ εὐθεῖα τῇ ΑΕ, ὀίση ἐστὶ καὶ γωνία ἡ ὑπὸ ΑΒΕ τῇ ὑπὸ ΑΕΒ. ἀλλὰ ἡ ὑπὸ ΑΒΕ τῇ ὑπὸ ΒΑΘ ἐδείϑϑηὀίση· καὶ ἡ ὑπὸ ΒΕΑ ἄρα τῇ ὑπὸ ΒΑΘ ἐστινὀίση. καὶ κοινὴ τῶν δύο τριγώνων τοῦ τε ΑΒΕ καὶ τοῦ ΑΒΘ ἐστιν ἡ ὑπὸ ΑΒΕ· λοιπὴ ἄρα ἡ ὑπὸ ΒΑΕ γωνία λοιπῇ τῇ ὑπὸ ΑΘΒ ἐστινὀίση· ἰσογώνιον ἄρα ἐστὶ τὸ ΑΒΕ τρίγωνον τῷ ΑΒΘ τριγώνῳ· ἀνάλογον ἄρα ἐστὶν ὡξ ἡ ΕΒ πρὸς τὴν ΒΑ, οὑύτως ἡ ΑΒ πρὸς τὴν ΒΘ. ὀίση δὲ ἡ ΒΑ τῇ ΕΘ· ὡξ ἄρα ἡ ΒΕ πρὸς τὴν ΕΘ, οὑύτως ἡ ΕΘ πρὸς τὴν ΘΒ. μείζων δὲ ἡ ΒΕ τῆς ΕΘ· μείζων ἄρα καὶ ἡ ΕΘ τῆς ΘΒ. ἡ ΒΕ ὁάραὀάκρον καὶ μέσον λόγον τέτμ ηται κατὰ τὸ Θ, καὶ τὸ μεῖζον τμῆμα τὸ ΘΕ ἴσον ἐστὶ τῇ τοῦ πενταγώνου πλευρᾷ. ὁμοίως δὴ δείχομεν, ὅτι καὶ ἡ ΑΓὀάκρον καὶ μέσον λόγον τέτμ ηται κατὰ τὸ Θ, καὶ τὸ μεῖζον αὐτῆς τμῆμα ἡ ΓΘ ἴσον ἐστὶ τῇ τοῦ πενταγώνου πλευρᾷ· ὅπερ ἔδει δεῖχαι.

9.

Ἐὰν ἡ τοῦ ἑχαγώνου πλευρὰ καὶ ἡ τοῦ δεκαγώνου τῶν εἰξ τὸν αὐτὸν κύκλον ἐγγραφομένων συντεθῶσιν, ἡ ὅλη εὐθεῖαὀάκρον καὶ μέσον λόγον τέτμ ηται, καὶ τὸ μεῖχον αὐτῆς τμῆμά ἐστιν ἡ τοῦ ἑχαγώνου πλευρά.

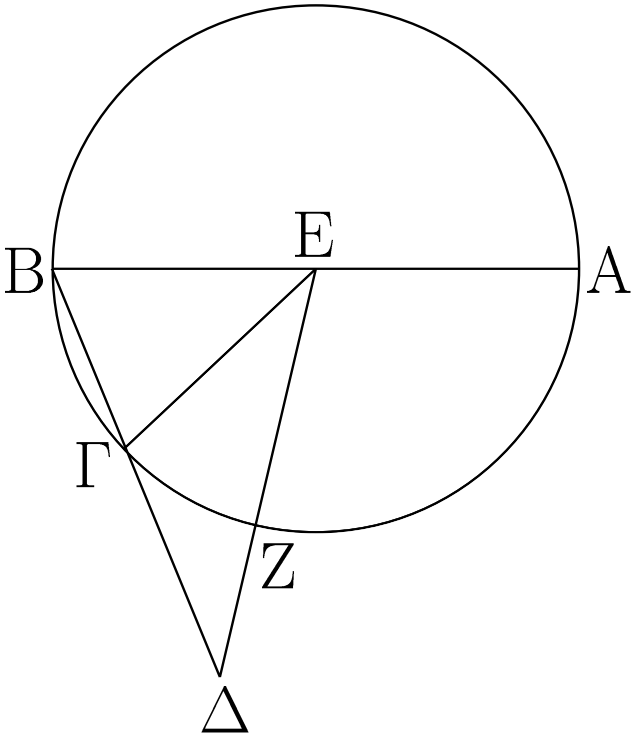
Ἔστω κύκλος ὁ ΑΒΓ, καὶ τῶν εἰξ τὸν ΑΒΓ κύκλον ἐγγραφομένων σχημάτων, δεκαγώνου μὲνὀέστω πλευρὰ ἡ ΒΓ, ἑχαγώνου δὲ ἡ ΓΔ, καὶ ὀέστωσαν ἐπ΄ εὐθείας· λέγω, ὅτι ἡ ὅλη εὐθεῖα ἡ ΒΔὀάκρον καὶ μέσον λόγον τέτμ ηται, καὶ τὸ μεῖζον αὐτῆς τμῆμά ἐστιν ἡ ΓΔ.

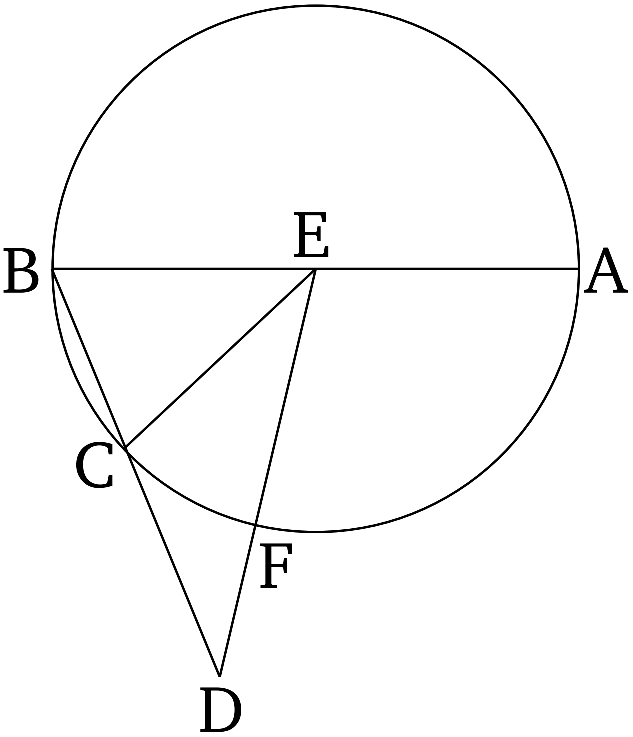
thus equal to the base AC, and triangle ABE is equal to triangle ABC, and the remaining angles will be equal to the remaining angles, respectively, which the equal sides subtend [Prop. 1.4]. Thus, angle BAC is equal to (angle) ABE. Thus, (angle) AHE (is) double (angle) BAH [Prop. 1.32]. And EAC is also double BAC, inasmuch as circumference EDC is also double circumference CB [Props. 3.28, 6.33]. Thus, angle HAE (is) equal to (angle) AHE. Hence, straight-line HE is also equal to (straight-line) EA—that is to say, to (straight-line) AB [Prop. 1.6]. And since straight-line BA is equal to AE, angle ABE is also equal to AEB [Prop. 1.5]. But, ABE was shown (to be) equal to BAH. Thus, BEA is also equal to BAH. And (angle) ABE is common to the two triangles ABE and ABH. Thus, the remaining angle BAE is equal to the remaining (angle) AHB [Prop. 1.32]. Thus, triangle ABE is equiangular to triangle ABH. Thus, proportionally, as EB is to BA, so AB (is) to BH [Prop. 6.4]. And BA (is) equal to EH. Thus, as BE (is) to EH, so EH (is) to HB. And BE (is) greater than EH. EH (is) thus also greater than HB [Prop. 5.14]. Thus, BE has been cut in extreme and mean ratio at H, and the greater piece HE is equal to the side of the pentagon. So, similarly, we can show that AC has also been cut in extreme and mean ratio at H, and that its greater piece CH is equal to the side of the pentagon. (Which is) the very thing it was required to show.

Proposition 9

If the side of a hexagon and of a decagon inscribed in the same circle are added together then the whole straight-line has been cut in extreme and mean ratio (at the junction point), and its greater piece is the side of the hexagon.[†]

Let ABC be a circle. And of the figures inscribed in circle ABC, let BC be the side of a decagon, and CD (the side) of a hexagon. And let them be (laid down) straight-on (to one another). I say that the whole straight-line BD has been cut in extreme and mean ratio (at C), and that CD is its greater piece.





Εἰλήφθω γὰρ τὸ κέντρον τοῦ κύκλου τὸ Ε σημεῖον, καὶ ἐπεζεύθθωσαν αἱ ΕΒ, ΕΓ, ΕΔ, καὶ διήθθω ἡ ΒΕ ἐπὶ τὸ Α. ἐπεὶ δεκαγώνου ἰσοπλεύρον πλευρά ἐστιν ἡ ΒΓ, πενταπλασίων ἄρα ἡ ΑΓΒ περιφέρεια τῆς ΒΓ περιφερείας· τετραπλασίων ἄρα ἡ ΑΓ περιφέρεια τῆς ΓΒ. ὡξ δὲ ἡ ΑΓ περιφέρεια πρὸς τὴν ΓΒ, οὕτως ἡ ὑπὸ ΑΕΓ γωνία πρὸς τὴν ὑπὸ ΓΕΒ· τετραπλασίων ἄρα ἡ ὑπὸ ΑΕΓ τῆς ὑπὸ ΓΕΒ. καὶ ἐπεὶ ὁίση ἡ ὑπὸ ΕΒΓ γωνία τῇ ὑπὸ ΕΓΒ, ἡ ἄρα ὑπὸ ΑΕΓ γωνία διπλασία ἐστὶ τῆς ὑπὸ ΕΓΒ. καὶ ἐπεὶ ὁίση ἐστὶν ἡ ΕΓ εὐθεῖα τῇ ΓΔ· ἑκατέρα γὰρ αὐτῶνὁίση ἐστὶ τῇ τοῦ ἑχαγώνου πλευρᾷ τοῦ εἰξ τὸν ΑΒΓ κύκλον [ἐγγραφομένου]· ὁίση ἐστὶ καὶ ἡ ὑπὸ ΓΕΔ γωνία τῇ ὑπὸ ΓΔΕ γωνίᾳ· διπλασία ἄρα ἡ ὑπὸ ΕΓΒ γωνία τῆς ὑπὸ ΕΔΓ. ἀλλὰ τῆς ὑπὸ ΕΓΒ διπλασία ἐδείθθη ἡ ὑπὸ ΑΕΓ· τετραπλασία ἄρα ἡ ὑπὸ ΑΕΓ τῆς ὑπὸ ΕΔΓ. ἐδείθθη δὲ καὶ τῆς ὑπὸ ΒΕΓ τετραπλασία ἡ ὑπὸ ΑΕΓ· ὁίση ἄρα ἡ ὑπὸ ΕΔΓ τῇ ὑπὸ ΒΕΓ. κοινὴ δὲ τῶν δύο τριγώνων, τοῦ τε ΒΕΓ καὶ τοῦ ΒΕΔ, ἡ ὑπὸ ΕΒΔ γωνία· καὶ λοιπὴ ἄρα ἡ ὑπὸ ΒΕΔ τῇ ὑπὸ ΕΓΒ ἐστινὁίση· ἰσογώνιον ἄρα ἐστὶ τὸ ΕΒΔ τρίγωνον τῷ ΕΒΓ τριγώνῳ. ἀνάλογον ἄρα ἐστὶν ὡξ ἡ ΔΒ πρὸς τὴν ΒΕ, οὕτως ἡ ΕΒ πρὸς τὴν ΒΓ. ὁίση δὲ ἡ ΕΒ τῇ ΓΔ. ὁέστιν ἄρα ὡξ ἡ ΒΔ πρὸς τὴν ΔΓ, οὕτως ἡ ΔΓ πρὸς τὴν ΓΒ. μείζων δὲ ἡ ΒΔ τῆς ΔΓ· μείζων ἄρα καὶ ἡ ΔΓ τῆς ΓΒ. ἡ ΒΔ ἄρα εὐθεῖαὁάκρον καὶ μέσον λόγον τέτμ ηται [κατὰ τὸ Γ], καὶ τὸ μεῖζον τμῆμα αὐτῆς ἐστιν ἡ ΔΓ· ὅπερ ἔδει δεῖχαι.

For let the center of the circle, point *E*, be found [Prop. 3.1], and let *EB*, *EC*, and *ED* be joined, and let *BE* be drawn across to *A*. Since *BC* is a side on an equilateral decagon, circumference *ACB* (is) thus five times circumference *BC*. Thus, circumference *AC* (is) four times *CB*. And as circumference *AC* (is) to *CB*, so angle *AEC* (is) to *CEB* [Prop. 6.33]. Thus, (angle) *AEC* (is) four times *CEB*. And since angle *EBC* (is) equal to *ECB* [Prop. 1.5], angle *AEC* is thus double *ECB* [Prop. 1.32]. And since straight-line *EC* is equal to *CD*—for each of them is equal to the side of the hexagon [inscribed] in circle *ABC* [Prop. 4.15 corr.]— angle *CED* is also equal to angle *CDE* [Prop. 1.5]. Thus, angle *ECB* (is) double *EDC* [Prop. 1.32]. But, *AEC* was shown (to be) double *ECB*. Thus, *AEC* (is) four times *EDC*. And *AEC* was also shown (to be) four times *BEC*. Thus, *EDC* (is) equal to *BEC*. And angle *EBD* (is) common to the two triangles *BEC* and *BED*. Thus, the remaining (angle) *BED* is equal to the (remaining angle) *ECB* [Prop. 1.32]. Thus, triangle *EBD* is equiangular to triangle *EBC*. Thus, proportionally, as *DB* is to *BE*, so *EB* (is) to *BC* [Prop. 6.4]. And *EB* (is) equal to *CD*. Thus, as *BD* is to *DC*, so *DC* (is) to *CB*. And *BD* (is) greater than *DC*. Thus, *DC* (is) also greater than *CB* [Prop. 5.14]. Thus, the straight-line *BD* has been cut in extreme and mean ratio [at *C*], and *DC* is its greater piece. (Which is), the very thing it was required to show.

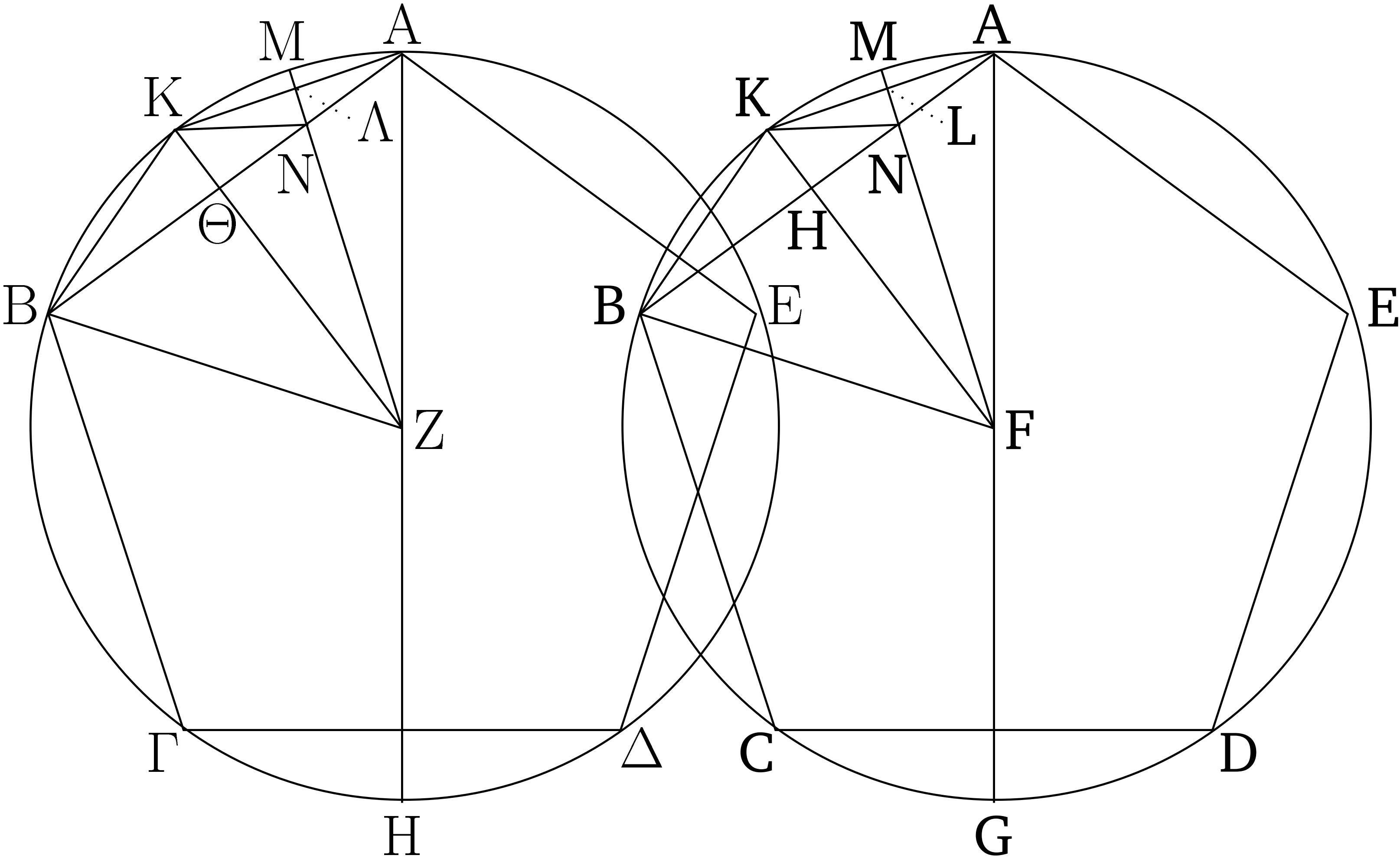
† If the circle is of unit radius then the side of the hexagon is 1, whereas the side of the decagon is $(1/2)(\sqrt{5} - 1)$.

10.

Ἐὰν εἰξ κύκλον πεντάγωνον ἰσόπλευρον ἐγγραφῇ, ἡ τοῦ πενταγώνου πλευρὰ δύναται τήν τε τοῦ ἑξαγώνου καὶ τὴν τοῦ δεκαγώνου τῶν εἰξ τὸν αὐτὸν κύκλον ἐγγραφομένων.

Proposition 10

If an equilateral pentagon is inscribed in a circle then the square on the side of the pentagon is (equal to) the (sum of the squares) on the (sides) of the hexagon and of the decagon inscribed in the same circle.†



Ἔστω κύκλος ὁ ΑΒΓΔΕ, καὶ εἰξ τὸ ΑΒΓΔΕ κύκλον πεντάγωνον ἰσόπλευρον ἐγγεγράφθω τὸ ΑΒΓΔΕ. λέγω, ὅτι ἡ τοῦ ΑΒΓΔΕ πενταγώνου πλευρὰ δύναται τήν τε τοῦ ἑξαγώνου καὶ τὴν τοῦ δεκαγώνου πλευρὰν τῶν εἰξ τὸν ΑΒΓΔΕ κύκλον ἐγγραφομένων.

Εἰλήφθω γὰρ τὸ κέντρον τοῦ κύκλου τὸ Ζ σημεὶον, καὶ ἐπιζευχθεῖσα ἡ ΑΖ διήχθω ἐπὶ τὸ Η σημεῖον, καὶ ἐπεζεύχθω ἡ ΖΒ, καὶ ἀπὸ τοῦ Ζ ἐπὶ τὴν ΑΒ κάθετος ἤχθω ἡ ΖΘ, καὶ διήχθω ἐπὶ τὸ Κ, καὶ ἐπεζεύχθωσαν αἱ ΑΚ, ΚΒ, καὶ πάλιν ἀπὸ τοῦ Ζ ἐπὶ τὴν ΑΚ κάθετος ἤχθω ἡ ΖΛ, καὶ διήχθω ἐπὶ τὸ Μ, καὶ ἐπεζεύχθω ἡ ΚΝ.

Ἐπεὶ ὀίση ἐστὶν ἡ ΑΒΓΗ περιφέρεια τῇ ΑΕΔΗ περιφερείᾳ, ὅῶν ἡ ΑΒΓ τῇ ΑΕΔ ἐστινὀίση, λοιπὴ ἄρα ἡ ΓΗ περιφέρεια λοιπῇ τῇ ΗΔ ἐστινὀίση. πενταγώνου δὲ ἡ ΓΔ·

Let *ABCDE* be a circle. And let the equilateral pentagon *ABCDE* be inscribed in circle *ABCDE*. I say that the square on the side of pentagon *ABCDE* is the (sum of the squares) on the sides of the hexagon and of the decagon inscribed in circle *ABCDE*.

For let the center of the circle, point *F*, be found [Prop. 3.1]. And, *AF* being joined, let it be drawn across to point *G*. And let *FB* be joined. And let *FH* be drawn from *F* perpendicular to *AB*. And let it be drawn across to *K*. And let *AK* and *KB* be joined. And, again, let *FL* have been drawn from *F* perpendicular to *AK*. And let it be drawn across to *M*. And let *KN* be joined.

Since circumference *ABCG* is equal to circumference *AEDG*, of which *ABC* is equal to *AED*, the remaining cir-

δεκαγώνου ἄρα ἡ ΓΗ. καὶ ἐπεὶ ὸίση ἐστὶν ἡ ΖΑ τῇ ΖΒ, καὶ κάθετος ἡ ΖΘ, ὸίση ἄρα καὶ ἡ ὑπὸ ΑΖΚ γωνία τῇ ὑπὸ ΚΖΒ. ὁώστε καὶ περιφέρεια ἡ ΑΚ τῇ ΚΒ ἐστινὸίση· διπλῆ ἄρα ἡ ΑΒ περιφέρεια τῆς ΒΚ περιφερείας· δεκαγώνου ἄρα πλευρά ἐστιν ἡ ΑΚ εὐθεῖα. διὰ τὰ αὐτὰ δὴ καὶ ἡ ΑΚ τῆς ΚΜ ἐστι διπλῆ. καὶ ἐπεὶ διπλῆ ἐστιν ἡ ΑΒ περιφέρεια τῆς ΒΚ περιφερείας, ὸίση δὲ ἡ ΓΔ περιφέρεια τῇ ΑΒ περιφερείᾳ, διπλῆ ἄρα καὶ ἡ ΓΔ περιφέρεια τῆς ΒΚ περιφερείας. ὸέστι δὲ ἡ ΓΔ περιφέρεια καὶ τῆς ΓΗ διπλῆ· ὸίση ἄρα ἡ ΓΗ περιφέρεια τῇ ΒΚ περιφερείᾳ. ἀλλὰ ἡ ΒΚ τῆς ΚΜ ἐστι διπλῆ, ἐπεὶ καὶ ἡ ΚΑ· καὶ ἡ ΓΗ ἄρα τῆς ΚΜ ἐστι διπλῆ. ἀλλὰ μὴν καὶ ἡ ΓΒ περιφέρεια τῆς ΒΚ περιφερείας ἐστὶ διπλῆ· ὸίση γὰρ ἡ ΓΒ περιφέρεια τῇ ΒΑ. καὶ ὅλη ἄρα ἡ ΗΒ περιφέρεια τῆς ΒΜ ἐστι διπλῆ· ὁώστε καὶ γωνία ἡ ὑπὸ ΗΖΒ γωνίας τῆς ὑπὸ ΒΖΜ [ἐστι] διπλῆ. ὸέστι δὲ ἡ ὑπὸ ΗΖΒ καὶ τῆς ὑπὸ ΖΑΒ διπλῆ· ὸίση γὰρ ἡ ὑπὸ ΖΑΒ τῇ ὑπὸ ΑΒΖ. καὶ ἡ ὑπὸ ΒΖΝ ἄρα τῇ ὑπὸ ΖΑΒ ἐστινὸίση. κοινὴ δὲ τῶν δύο τριγώνων, τοῦ τε ΑΒΖ καὶ τοῦ ΒΖΝ, ἡ ὑπὸ ΑΒΖ γωνία· λοιπὴ ἄρα ἡ ὑπὸ ΑΖΒ λοιπῇ τῇ ὑπὸ ΒΝΖ ἐστινὸίση· ὸισογώνιον ἄρα ἐστὶ τὸ ΑΒΖ τρίγωνον τῷ ΒΖΝ τριγώνῳ. ἀνάλογον ἄρα ἐστὶν ὥξ ἡ ΑΒ εὐθεῖα πρὸς τὴν ΒΖ, οὕτως ἡ ΖΒ πρὸς τὴν ΒΝ· τὸ ἄρα ὑπὸ τῶν ΑΒΝ ἴσον ἐστὶ τῷ ἀπὸ ΒΖ. πάλιν ἐπεὶ ὸίση ἐστὶν ἡ ΑΛ τῇ ΛΚ, κοινὴ δὲ καὶ πρὸς ὀρθὰξ ἡ ΛΝ, βάσις ἄρα ἡ ΚΝ βάσει τῇ ΑΝ ἐστινὸίση· καὶ γωνία ἄρα ἡ ὑπὸ ΛΚΝ γωνίᾳ τῇ ὑπὸ ΛΑΝ ἐστινὸίση. ἀλλὰ ἡ ὑπὸ ΛΑΝ τῇ ὑπὸ ΚΒΝ ἐστινὸίση· καὶ ἡ ὑπὸ ΛΚΝ ἄρα τῇ ὑπὸ ΚΒΝ ἐστινὸίση. καὶ κοινὴ τῶν δύο τριγώνων τοῦ τε ΑΚΒ καὶ τοῦ ΑΚΝ ἡ πρὸς τῷ Α. λοιπὴ ἄρα ἡ ὑπὸ ΑΚΒ λοιπῇ τῇ ὑπὸ ΚΝΑ ἐστινὸίση· ἰσογώνιον ἄρα ἐστὶ τὸ ΚΒΑ τρίγωνον τῷ ΚΝΑ τριγώνῳ. ἀνάλογον ἄρα ἐστὶν ὥξ ἡ ΒΑ εὐθεῖα πρὸς τὴν ΑΚ, οὕτως ἡ ΚΑ πρὸς τὴν ΑΝ· τὸ ἄρα ὑπὸ τῶν ΒΑΝ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΑΚ. ἐδείϑθη δὲ καὶ τὸ ὑπὸ τῶν ΑΒΝ ἴσον τῷ ἀπὸ τῆς ΒΖ· τὸ ἄρα ὑπὸ τῶν ΑΒΝ μετὰ τοῦ ὑπὸ ΒΑΝ, ὅπερ ἐστὶ τὸ ἀπὸ τῆς ΒΑ, ἴσον ἐστὶ τῷ ἀπὸ τῆς ΒΖ μετὰ τοῦ ἀπὸ τῆς ΑΚ. καί ἐστιν ἡ μὲν ΒΑ πενταγώνου πλευρά, ἡ δὲ ΒΖ ἑχαγώνου, ἡ δὲ ΑΚ δεκαγώνου.

Ἡ ἄρα τοῦ πενταγώνου πλευρὰ δύναται τήν τε τοῦ ἑχαγώνου καὶ τὴν τοῦ δεκαγώνου τῶν εἴξ τὸν αὐτὸν κύκλον ἐγγραφομένων· ὅπερ ἔδει δεῖχαι.

cumference *CG* is thus equal to the remaining (circumference) *GD*. And *CD* (is the side) of the pentagon. *CG* (is) thus (the side) of the decagon. And since *FA* is equal to *FB*, and *FH* is perpendicular (to *AB*), angle *AFK* (is) thus also equal to *KFB* [Props. 1.5, 1.26]. Hence, circumference *AK* is also equal to *KB* [Prop. 3.26]. Thus, circumference *AB* (is) double circumference *BK*. Thus, straight-line *AK* is the side of the decagon. So, for the same (reasons, circumference) *AK* is also double *KM*. And since circumference *AB* is double circumference *BK*, and circumference *CD* (is) equal to circumference *AB*, circumference *CD* (is) thus also double circumference *BK*. And circumference *CD* is also double *CG*. Thus, circumference *CG* (is) equal to circumference *BK*. But, *BK* is double *KM*, since *KA* (is) also (double *KM*). Thus, (circumference) *CG* is also double *KM*. But, indeed, circumference *CB* is also double circumference *BK*. For circumference *CB* (is) equal to *BA*. Thus, the whole circumference *GB* is also double *BM*. Hence, angle *GFB* [is] also double angle *BFM* [Prop. 6.33]. And *GFB* (is) also double *FAB*. For *FAB* (is) equal to *ABF*. Thus, *BFN* is also equal to *FAB*. And angle *ABF* (is) common to the two triangles *ABF* and *BFN*. Thus, the remaining (angle) *AFB* is equal to the remaining (angle) *BNF* [Prop. 1.32]. Thus, triangle *ABF* is equiangular to triangle *BFN*. Thus, proportionally, as straight-line *AB* (is) to *BF*, so *FB* (is) to *BN* [Prop. 6.4]. Thus, the (rectangle contained) by *ABN* is equal to the (square) on *BF* [Prop. 6.17]. Again, since *AL* is equal to *LK*, and *LN* is common and at right-angles (to *KA*), base *KN* is thus equal to base *AN* [Prop. 1.4]. And, thus, angle *LKN* is equal to angle *LAN*. But, *LAN* is equal to *KBN* [Props. 3.29, 1.5]. Thus, *LKN* is also equal to *KBN*. And the (angle) at *A* (is) common to the two triangles *AKB* and *AKN*. Thus, the remaining (angle) *AKB* is equal to the remaining (angle) *KNA* [Prop. 1.32]. Thus, triangle *KBA* is equiangular to triangle *KNA*. Thus, proportionally, as straight-line *BA* is to *AK*, so *KA* (is) to *AN* [Prop. 6.4]. Thus, the (rectangle contained) by *BAN* is equal to the (square) on *AK* [Prop. 6.17]. And the (rectangle contained) by *ABN* was also shown (to be) equal to the (square) on *BF*. Thus, the (rectangle contained) by *ABN* plus the (rectangle contained) by *BAN*, which is the (square) on *BA* [Prop. 2.2], is equal to the (square) on *BF* plus the (square) on *AK*. And *BA* is the side of the pentagon, and *BF* (the side) of the hexagon [Prop. 4.15 corr.], and *AK* (the side) of the decagon.

Thus, the square on the side of the pentagon (inscribed in a circle) is (equal to) the (sum of the squares) on the (sides) of the hexagon and of the decagon inscribed in the same circle.

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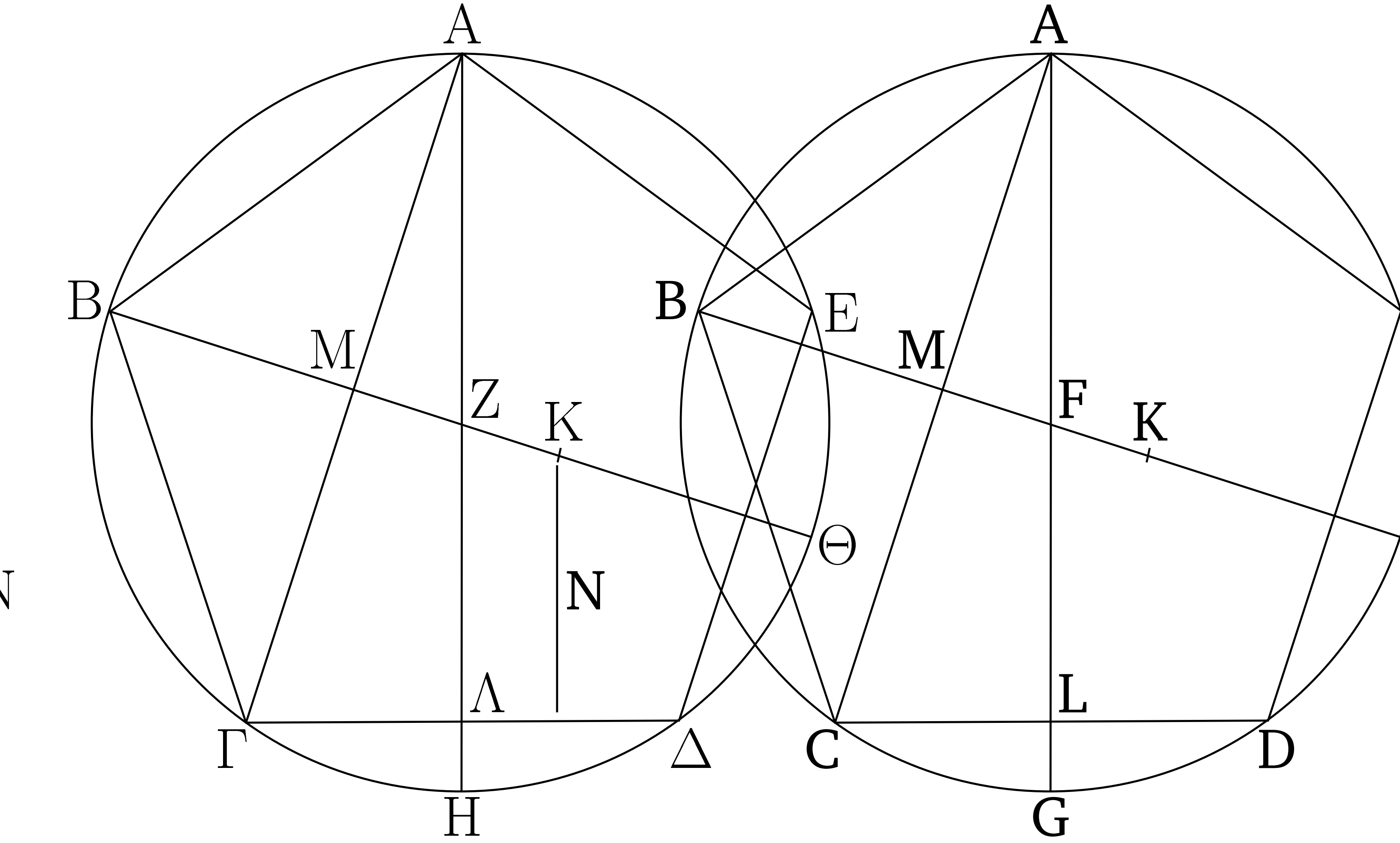
† If the circle is of unit radius then the side of the pentagon is (1/2) $\sqrt{10 - 2\,\sqrt{5}}$.

11.

Proposition 11

Ἐὰν εἰξ κύκλον ὁρητὴν ἔχοντα τὴν διάμετρον πεντάγωνον ἰσόπλευρον ἐγγραφῇ, ἡ τοῦ πενταγώνου πλευρὰ ὀάλογόξ ἐστιν ἡ καλουμένη ἐλάσσων.

If an equilateral pentagon is inscribed in a circle which has a rational diameter then the side of the pentagon is that irrational (straight-line) called minor.



Εἰξ γὰρ κύκλον τὸν ΑΒΓΔΕ ὁρητὴν ἔχοντα τὴν δίαμετρον πεντάγωνον ἰσόπλευρον ἐγγεγράφθω τὸ ΑΒΓΔΕ· λέγω, ὅτι ἡ τοῦ [ΑΒΓΔΕ] πενταγώνου πλευρὰ ὀάλογόξ ἐστιν ἡ καλουμένη ἐλάσσων.

Εἰλήφθω γὰρ τὸ κέντρον τοῦ κύκλου τὸ Ζ σημεῖον, καὶ ἐπεζεύθθωσαν αἱ ΑΖ, ΖΒ καὶ διήθθωσαν ἐπὶ τὰ Η, Θ σημεῖα, καὶ ἐπεζεύθθω ἡ ΑΓ, καὶ κείσθω τῆς ΑΖ τέταρτον μέρος ἡ ΖΚ. ὁρητὴ δὲ ἡ ΑΖ· ὁρητὴ ἄρα καὶ ἡ ΖΚ. ὀέστι δὲ καὶ ἡ ΒΖ ὁρητή· ὅλη ἄρα ἡ ΒΚ ὁρητή ἐστιν. καὶ ἐπεὶ ὀίση ἐστὶν ἡ ΑΓΗ περιφέρεια τῇ ΑΔΗ περιφερείᾳ, ὁῶν ἡ ΑΒΓ τῇ ΑΕΔ ἐστινὀίση, λοιπὴ ἄρα ἡ ΓΗ λοιπῇ τῇ ΗΔ ἐστινὀίση. καὶ ἔαν ἐπιζεύχωμεν τὴν ΑΔ, συνάγονται ὀρθαὶ αἱ πρὸς τῷ Λ γωνίαι, καὶ διπλῆ ἡ ΓΔ τῆς ΓΛ. διὰ τὰ αὐτὰ δὴ καὶ αἱ πρὸς τῷ Μ ὀρθαί εἰσιν, καὶ διπλῆ ἡ ΑΓ τῆς ΓΜ. ἐπεὶ οὀῦνὀίση ἐστὶν ἡ ὑπὸ ΑΛΓ γωνία τῇ ὑπὸ ΑΜΖ, κοινὴ δὲ τῶν δύο τριγώνων τοῦ τε ΑΓΛ καὶ τοῦ ΑΜΖ ἡ ὑπὸ ΛΑΓ, λοιπὴ ἄρα ἡ ὑπὸ

For let the equilateral pentagon *ABCDE* be inscribed in the circle *ABCDE* which has a rational diameter. I say that the side of pentagon [*ABCDE*] is that irrational (straight-line) called minor.

For let the center of the circle, point *F*, be found [Prop. 3.1]. And let *AF* and *FB* be joined. And let them be drawn across to points *G* and *H* (respectively). And let *AC* be joined. And let *FK* made (equal) to the fourth part of *AF*. And *AF* (is) rational. *FK* (is) thus also rational. And *BF* is also rational. Thus, the whole of *BK* is rational. And since circumference *ACG* is equal to circumference *ADG*, of which *ABC* is equal to *AED*, the remainder *CG* is thus equal to the remainder *GD*. And if we join *AD* then the angles at *L* are inferred (to be) right-angles, and *CD* (is inferred to be) double *CL* [Prop. 1.4]. So, for the same (reasons), the (angles) at *M* are also right-angles, and *AC* (is) double *CM*. Therefore, since angle *ALC*

ΑΓΛ λοιπῇ τῇ ὑπὸ ΜΖΑ ἐστινὀίση· ἰσογώνιον ἄρα ἐστὶ τὸ ΑΓΛ τρίγωνον τῷ ΑΜΖ τριγώνῳ· ἀνάλογον ἄρα ἐστὶν ὡξ ἡ ΛΓ πρὸς ΓΑ, οὕτως ἡ ΜΖ πρὸς ΖΑ· καὶ τῶν ἡγουμένων τὰ διπλάσια· ὡξ ἄρα ἡ τῆς ΛΓ διπλῆ πρὸς τὴν ΓΑ, οὕτως ἡ τῆς ΜΖ διπλῆ πρὸς τὴν ΖΑ. ὡξ δὲ ἡ τῆς ΜΖ διπλῆ πρὸς τὴν ΖΑ, οὕτως ἡ ΜΖ πρὸς τὴν ἡμίσειαν τῆς ΖΑ· καὶ ὡξ ἄρα ἡ τῆς ΛΓ διπλῆ πρὸς τὴν ΓΑ, οὕτως ἡ ΜΖ πρὸς τὴν ἡμίσειαν τῆς ΖΑ· καὶ τῶν ἑπομένων τὰ ἡμίσεα· ὡξ ἄρα ἡ τῆς ΛΓ διπλῆ πρὸς τὴν ἡμίσειαν τῆς ΓΑ, οὕτως ἡ ΜΖ πρὸς τὸ τέτατρον τῆς ΖΑ. καί ἐστι τῆς μὲν ΛΓ διπλῆ ἡ ΔΓ, τῆς δὲ ΓΑ ἡμίσεια ἡ ΓΜ, τῆς δὲ ΖΑ τέτατρον μέρος ἡ ΖΚ· ὀέστιν ἄρα ὡξ ἡ ΔΓ πρὸς τὴν ΓΜ, οὕτως ἡ ΜΖ πρὸς τὴν ΖΚ. συνθέντι καὶ ὡξ συναμφότερος ἡ ΔΓΜ πρὸς τὴν ΓΜ, οὕτως ἡ ΜΚ πρὸς ΚΖ· καὶ ὡξ ἄρα τὸ ἀπὸ συναμφοτέρου τῆς ΔΓΜ πρὸς τὸ ἀπὸ ΓΜ, οὕτως τὸ ἀπὸ ΜΚ πρὸς τὸ ἀπὸ ΚΖ. καὶ ἐπεὶ τῆς ὑπὸ δύο πλευρὰξ τοῦ πενταγώνου ὑποτεινούσης, οὀῖον τῆς ΑΓ, ὀάκρον καὶ μέσον λόγον τεμνομένης τὸ μεῖζον τμῆμα ἴσον ἐστὶ τῇ τοῦ πενταγώνου πλευρᾷ, τουτέστι τῇ ΔΓ, τὸ δὲ μεῖζον τμῆμα προσλαβὸν τὴν ἡμίσειαν τῆς ὅλῆς πενταπλάσιον δύναται τοῦ ἀπὸ τῆς ἡμισείας τῆς ὅλης, καί ἐστιν ὅλης τῆς ΑΓ ἡμίσεια ἡ ΓΜ, τὸ ἄρα ἀπὸ τῆς ΔΓΜ ὡξ μιᾶς πενταπλάσιόν ἐστι τοῦ ἀπὸ τῆς ΓΜ. ὡξ δὲ τὸ ἀπὸ τῆς ΔΓΜ ὡξ μιᾶς πρὸς τὸ ἀπὸ τῆς ΓΜ, οὕτως ἐδείχθη τὸ ἀπὸ τῆς ΜΚ πρὸς τὸ ἀπὸ τῆς ΚΖ· πενταπλάσιον ἄρα τὸ ἀπὸ τῆς ΜΚ τοῦ ἀπὸ τῆς ΚΖ. ῥητὸν δὲ τὸ ἀπὸ τῆς ΚΖ· ῥητὴ γὰρ ἡ διάμετρος· ῥητὸν ἄρα καὶ τὸ ἀπὸ τῆς ΜΚ· ῥητὴ ἄρα ἐστὶν ἡ ΜΚ [δυνάμει μόνον]. καὶ ἐπεὶ τετραπλασία ἐστὶν ἡ ΒΖ τῆς ΖΚ, πενταπλασία ἄρα ἐστὶν ἡ ΒΚ τῆς ΚΖ· εἰκοσιπενταπλάσιον ἄρα τὸ ἀπὸ τῆς ΒΚ τοῦ ἀπὸ τῆς ΚΖ. πενταπλάσιον δὲ τὸ ἀπὸ τῆς ΜΚ τοῦ ἀπὸ τῆς ΚΖ· πενταπλάσιον ἄρα τὸ ἀπὸ τῆς ΒΚ τοῦ ἀπὸ τῆς ΚΜ· τὸ ἄρα ἀπὸ τῆς ΒΚ πρὸς τὸ ἀπὸ ΚΜ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ ΒΚ τῇ ΚΜ μήκει. καί ἐστι ῥητὴ ἑκατέρα αὐτῶν. αἱ ΒΚ, ΚΜ ἄρα ῥηταί εἰσι δυνάμει μόνον σύμμετροι. ἐὰν δὲ ἀπὸ ῥητῆς ῥητὴ ἀφαιρεθῇ δυνάμει μόνον σύμμετρος οὀῦσα τῇ ὅλῃ, ἡ λοιπὴ ὀάλογόξ ἐστιν ἀποτομή· ἀποτομὴ ἄρα ἐστὶν ἡ ΜΒ, προσαρμόζουσα δὲ αὐτῇ ἡ ΜΚ. λέγω δή, ὅτι καὶ τετάρτη. ὁῷ δὴ μεῖζόν ἐστι τὸ ἀπὸ τῆς ΒΚ τοῦ ἀπὸ τῆς ΚΜ, ἐκείνῳ ἴσονὀέστω τὸ ἀπὸ τῆς Ν· ἡ ΒΚ ἄρα τῆς ΚΜ μεῖζον δύναται τῇ Ν. καὶ ἐπεὶ σύμμετρόξ ἐστιν ἡ ΚΖ τῇ ΖΒ, καὶ συνθέντι σύμμετρόξ ἐστι ἡ ΚΒ τῇ ΖΒ. ἀλλὰ ἡ ΒΖ τῇ ΒΘ σύμμετρόξ ἐστιν· καὶ ἡ ΒΚ ἄρα τῇ ΒΘ σύμμετρόξ ἐστιν. καὶ ἐπεὶ πενταπλάσιόν ἐστι τὸ ἀπὸ τῆς ΒΚ τοῦ ἀπὸ τῆς ΚΜ, τὸ ἄρα ἀπὸ τῆς ΒΚ πρὸς τὸ ἀπὸ τῆς ΚΜ λόγον ἔχει, ὃν ε πρὸς ἕν. ἀναστρέψαντι ἄρα τὸ ἀπὸ τῆς ΒΚ πρὸς τὸ ἀπὸ τῆς Ν λόγον ἔχει, ὃν $\overline{\epsilon}$ πρὸς $\overline{\delta}$, οὐχ ὃν τετράγωνος πρὸς τετράγωνον· ἀσύμμετρος ἄρα ἐστὶν ἡ ΒΚ τῇ Ν· ἡ ΒΚ ἄρα τῆς ΚΜ μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ. ἐπεὶ ο᾽ὖν ὅλη ἡ ΒΚ τῆς προσαρμοζούσης τῆς ΚΜ μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ, καὶ ὅλη ἡ ΒΚ σύμμετρός ἐστι τῇ ἐκκειμένῃ ῥητῇ τῇ ΒΘ, ἀποτομὴ ἄρα τετάρτη ἐστὶν ἡ ΜΒ. τὸ δὲ ὑπὸ ῥητῆς καὶ

(is) equal to AMF, and (angle) LAC (is) common to the two triangles ACL and AMF, the remaining (angle) ACL is thus equal to the remaining (angle) MFA [Prop. 1.32]. Thus, triangle ACL is equiangular to triangle AMF. Thus, proportionally, as LC (is) to CA, so MF (is) to FA [Prop. 6.4]. And (we can take) the doubles of the leading (magnitudes). Thus, as double LC (is) to CA, so double MF (is) to FA. And as double MF (is) to FA, so MF (is) to half of FA. And, thus, as double LC (is) to CA, so MF (is) to half of FA. And (we can take) the halves of the following (magnitudes). Thus, as double LC (is) to half of CA, so MF (is) to the fourth of FA. And DC is double LC, and CM half of CA, and FK the fourth part of FA. Thus, as DC is to CM, so MF (is) to FK. Via composition, as the sum of DCM (*i.e.*, DC and CM) (is) to CM, so MK (is) to KF [Prop. 5.18]. And, thus, as the (square) on the sum of DCM (is) to the (square) on CM, so the (square) on MK (is) to the (square) on KF. And since the greater piece of a (straight-line) subtending two sides of a pentagon, such as AC, (which is) cut in extreme and mean ratio is equal to the side of the pentagon [Prop. 13.8]—that is to say, to DC—and the square on the greater piece added to half of the whole is five times the (square) on half of the whole [Prop. 13.1], and CM (is) half of the whole, AC, thus the (square) on DCM, (taken) as one, is five times the (square) on CM. And the (square) on DCM, (taken) as one, (is) to the (square) on CM, so the (square) on MK was shown (to be) to the (square) on KF. Thus, the (square) on MK (is) five times the (square) on KF. And the square on KF (is) rational. For the diameter (is) rational. Thus, the (square) on MK (is) also rational. Thus, MK is rational [in square only]. And since BF is four times FK, BK is thus five times KF. Thus, the (square) on BK (is) twenty-five times the (square) on KF. And the (square) on MK (is) five times the square on KF. Thus, the (square) on BK (is) five times the (square) on KM. Thus, the (square) on BK does not have to the (square) on KM the ratio which a square number (has) to a square number. Thus, BK is incommensurable in length with KM [Prop. 10.9]. And each of them is a rational (straight-line). Thus, BK and KM are rational (straight-lines which are) commensurable in square only. And if from a rational (straight-line) a rational (straight-line) is subtracted, which is commensurable in square only with the whole, then the remainder is that irrational (straight-line called) an apotome [Prop. 10.73]. Thus, MB is an apotome, and MK its attachment. So, I say that (it is) also a fourth (apotome). So, let the (square) on N be (made) equal to that (magnitude) by which the (square) on BK is greater than the (square) on KM. Thus, the square on BK is greater than the (square) on KM by the (square) on N. And since KF is commensurable (in length) with FB then, via composition, KB is also commensurable (in length) with FB [Prop. 10.15]. But, BF is commensurable (in length) with BH. Thus, BK is also commensurable

ἀποτομῆς τετάρτης περιεχόμενον ὀρθογώνιον ἄλογόν ἐστιν, καὶ ἡ δυναμένη αὐτὸ ἄλογός ἐστιν, καλεῖται δὲ ἐλάττων. δύναται δὲ τὸ ὑπὸ τῶν ΘΒΜ ἡ ΑΒ διὰ τὸ ἐπιζευγνυμένης τῆς ΑΘ ἰσογώνιον γίνεσθαι τὸ ΑΒΘ τρίγωνον τῷ ΑΒΜ τριγώνῳ καὶ εἶναι ὡς τὴν ΘΒ πρὸς τὴν ΒΑ, οὕτως τὴν ΑΒ πρὸς τὴν ΒΜ.

Ἡ ἄρα ΑΒ τοῦ πενταγώνου πλευρὰ ὁάλογόξ ἐστιν ἡ καλουμένη ἐλάττων· ὅπερ ἔδει δεῖξαι.

(in length) with BH [Prop. 10.12]. And since the (square) on BK is five times the (square) on KM, the (square) on BK thus has to the (square) on KM the ratio which 5 (has) to one. Thus, via conversion, the (square) on BK has to the (square) on N the ratio which 5 (has) to 4 [Prop. 5.19 corr.], which is not (that) of a square (number) to a square (number). BK is thus incommensurable (in length) with N [Prop. 10.9]. Thus, the square on BK is greater than the (square) on KM by the (square) on (some straight-line which is) incommensurable (in length) with (BK). Therefore, since the square on the whole, BK, is greater than the (square) on the attachment, KM, by the (square) on (some straight-line which is) incommensurable (in length) with (BK), and the whole, BK, is commensurable (in length) with the (previously) laid down rational (straight-line) BH, MB is thus a fourth apotome [Def. 10.14]. And the rectangle contained by a rational (straight-line) and a fourth apotome is irrational, and its square-root is that irrational (straight-line) called minor [Prop. 10.94]. And the square on AB is the rectangle contained by HBM, on account of joining AH, (so that) triangle ABH becomes equiangular with triangle ABM [Prop. 6.8], and (proportionally) as HB is to BA, so AB (is) to BM.

Thus, the side AB of the pentagon is that irrational (straight-line) called minor.[†] (Which is) the very thing it was required to show.

[†] If the circle has unit radius then the side of the pentagon is $(1/2)\sqrt{10-2\sqrt{5}}$. However, this length can be written in the “minor” form (see Prop. 10.94) $(\rho/\sqrt{2})\sqrt{1+k/\sqrt{1+k^2}} - (\rho/\sqrt{2})\sqrt{1-k/\sqrt{1+k^2}}$, with $\rho = \sqrt{5/2}$ and $k = 2$.

12.

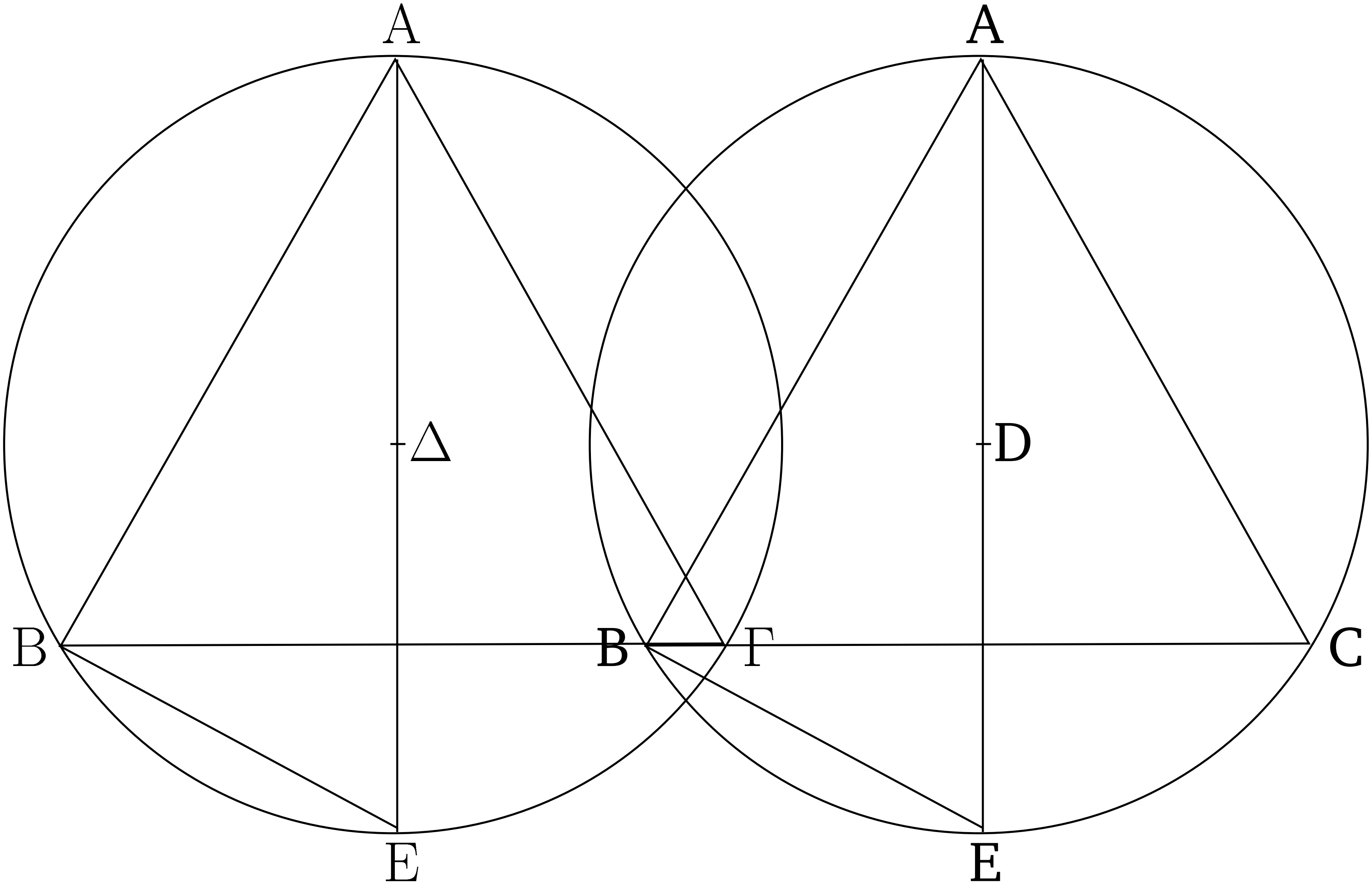
Ἐὰν εἰξ κύκλον τρίγωνον ἰσόπλευρον ἐγγραφῇ, ἡ τοῦ τριγώνου πλευρὰ δυνάμει τριπλασίων ἐστὶ τῆς ἐκ τοῦ κέντρου τοῦ κύκλου.

Ἔστω κύκλος ὁ ΑΒΓ, καὶ εἰξ αὐτὸν τρίγωνον ἰσόπλευρον ἐγγεγράφθω τὸ ΑΒΓ· λέγω, ὅτι τοῦ ΑΒΓ τριγώνου μία πλευρὰ δυνάμει τριπλασίων ἐστὶ τῆς ἐκ τοῦ κέντρου τοῦ ΑΒΓ κύκλου.

Proposition 12

If an equilateral triangle is inscribed in a circle then the square on the side of the triangle is three times the (square) on the radius of the circle.

Let there be a circle ABC, and let the equilateral triangle ABC have been inscribed in it [Prop. 4.2]. I say that the square on one side of triangle ABC is three times the (square) on the radius of circle ABC.



Εἰλήφθω γὰρ τὸ κέντρον τοῦ ΑΒΓ κύκλου τὸ Δ, καὶ ἐπιζευχθεῖσα ἡ ΑΔ διήχθω ἐπὶ τὸ Ε, καὶ ἐπεζεύχθω ἡ ΒΕ.

Καὶ ἐπεὶ ἰσόπλευρόν ἐστι τὸ ΑΒΓ τρίγωνον, ἡ ΒΕΓ ἄρα περιφέρεια τρίτον μέρος ἐστὶ τῆς τοῦ ΑΒΓ κύκλου περιφερείας. ἡ ἄρα ΒΕ περιφέρεια ὁέκτον ἐστὶ μέρος τῆς τοῦ κύκλου περιφερείας· ἑχαγώνου ἄρα ἐστὶν ἡ ΒΕ εὐθεῖα· ὁίση ἄρα ἐστὶ τῇ ἐκ τοῦ κέντρου τῇ ΔΕ. καὶ ἐπεὶ διπλῆ ἐστιν ἡ ΑΕ τῆς ΔΕ, τετραπλάσιον ἐστι τὸ ἀπὸ τῆς ΑΕ τοῦ ἀπὸ τῆς ΕΔ, τουτέστι τοῦ ἀπὸ τῆς ΒΕ. ἴσον δὲ τὸ ἀπὸ τῆς ΑΕ τοῖς ἀπὸ τῶν ΑΒ, ΒΕ· τὰ ἄρα ἀπὸ τῶν ΑΒ, ΒΕ τετραπλάσιά ἐστι τοῦ ἀπὸ τῆς ΒΕ. διελόντι ἄρα τὸ ἀπὸ τῆς ΑΒ τριπλάσιόν ἐστι τοῦ ἀπὸ ΒΕ. ὁίση δὲ ἡ ΒΕ τῇ ΔΕ· τὸ ἄρα ἀπὸ τῆς ΑΒ τριπλάσιόν ἐστι τοῦ ἀπὸ τῆς ΔΕ.

Ἡ ἄρα τοῦ τριγώνου πλευρὰ δυνάμει τριπλασία ἐστὶ τῆς ἐκ τοῦ κέντρου [τοῦ κύκλου]· ὅπερ ἔδει δεῖξαι.

For let the center, *D*, of circle *ABC* be found [Prop. 3.1]. And *AD* (being) joined, let it be drawn across to *E*. And let *BE* be joined.

And since triangle *ABC* is equilateral, circumference *BEC* is thus the third part of the circumference of circle *ABC*. Thus, circumference *BE* is the sixth part of the circumference of the circle. Thus, straight-line *BE* is (the side) of a hexagon. Thus, it is equal to the radius *DE* [Prop. 4.15 corr.]. And since *AE* is double *DE*, the (square) on *AE* is four times the (square) on *ED*—that is to say, of the (square) on *BE*. And the (square) on *AE* (is) equal to the (sum of the squares) on *AB* and *BE* [Props. 3.31, 1.47]. Thus, the (sum of the squares) on *AB* and *BE* is four times the (square) on *BE*. Thus, via separation, the (square) on *AB* is three times the (square) on *BE*. And *BE* (is) equal to *DE*. Thus, the (square) on *AB* is three times the (square) on *DE*.

Thus, the square on the side of the triangle is three times the (square) on the radius [of the circle]. (Which is) the very

13.

Πυραμίδα συστήσασϑαι καὶ σφαίρᾳ περιλαβεῖν τῇ δοϑείσῃ καὶ δεῖξαι, ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει ἡμιολία ἐστὶ τῆς πλευρᾶξ τῆς πυραμίδος.

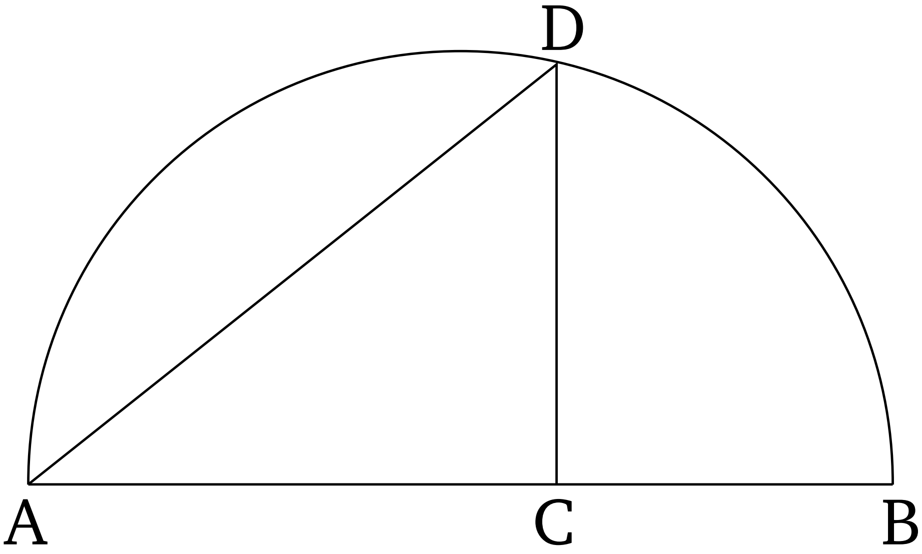
Ἐκκείσϑω ἡ τῆς δοϑείσης σφαίρας διάμετρος ἡ ΑΒ, καὶ τετμήσϑω κατὰ τὸ Γ σημεῖον, ὁώστε διπλασίαν εὁῖναι τὴν ΑΓ τῆς ΓΒ· καὶ γεγράφϑω ἐπὶ τῆς ΑΒ ἡμικύκλιον τὸ ΑΔΒ, καὶ ἤϑϑω ἀπὸ τοῦ Γ σημείου τῇ ΑΒ πρὸς ὀρϑὰξ ἡ ΓΔ, καὶ ἐπεζεύϑϑω ἡ ΔΑ· καὶ ἐκκείσϑω κύκλος ὁ ΕΖΗ ἴσηνὁέϑων τὴν ἐκ τοῦ κέντρου τῇ ΔΓ, καὶ ἐγγεγράφϑω εἰξ τὸν ΕΖΗ κύκλον τρίγωνον ἰσόπλευρον τὸ ΕΖΗ· καὶ εἰλήφϑω τὸ κέντρον τοῦ κύκλου τὸ Θ σημεῖον, καὶ ἐπεζεύϑϑωσαν αἱ ΕΘ, ΘΖ, ΘΗ· καὶ ἀνεστάτω ἀπὸ τοῦ Θ σημείου τῷ τοῦ ΕΖΗ κύκλου ἐπιπέδῳ πρὸς ὀρϑὰξ ἡ ΘΚ, καὶ ἀφῃρήσϑω ἀπὸ τῆς ΘΚ τῇ ΑΓ εὐϑείᾳ ὁίση ἡ ΘΚ, καὶ ἐπεζεύϑϑωσαν αἱ ΚΕ, ΚΖ, ΚΗ. καὶ ἐπεὶ ἡ ΚΘ ὀρϑή ἐστι πρὸς τὸ τοῦ ΕΖΗ κύκλου ἐπίπεδον, καὶ πρὸς πάσας ἄρα τὰξ ἁπτομένας αὐτῆς εὐϑείας καὶ οὁύσας ἐν τῷ τοῦ ΕΖΗ κύκλου ἐπιπέδῳ ὀρϑὰξ ποιήσει γωνίας. ὁάπτεται δὲ αὐτῆς ἑκάστη τῶν ΘΕ, ΘΖ, ΘΗ· ἡ ΘΚ ἄρα πρὸς ἑκάστη τῶν ΘΕ, ΘΖ, ΘΗ ὀρϑή ἐστιν. καὶ ἐπεὶ ὁίση ἐστὶν ἡ μὲν ΑΓ τῇ ΘΚ, ἡ δὲ ΓΔ τῇ ΘΕ, καὶ ὀρϑὰξ γωνίας περιέϑουσιν, βάσις ἄρα ἡ ΔΑ βάσει τῇ ΚΕ ἐστινὁίση. διὰ τὰ αὐτὰ δὴ καὶ ἑκατέρα τῶν ΚΖ, ΚΗ τῇ ΔΑ ἐστινὁίση· αἱ τρεῖς ἄρα αἱ ΚΕ, ΚΖ, ΚΗ ἴσαι ἀλλήλαις εἰσίν. καὶ ἐπεὶ διπλῆ ἐστιν ἡ ΑΓ τῆς ΓΒ, τριπλῆ ἄρα ἡ ΑΒ τῆς ΒΓ. ὡξ δὲ ἡ ΑΒ πρὸς τὴν ΒΓ, οὁύτως τὸ ἀπὸ τῆς ΑΔ πρὸς τὸ ἀπὸ τῆς ΔΓ, ὡξ ἑχῆς δειϑϑήσεται. τριπλάσιον ἄρα τὸ ἀπὸ τῆς ΑΔ τοῦ ἀπὸ τῆς ΔΓ. ὁέστι δὲ καὶ τὸ ἀπὸ τῆς ΖΕ τοῦ ἀπὸ τῆς ΕΘ τριπλάσιον, καί ἐστινὁίση ἡ ΔΓ τῇ ΕΘ· ὁίση ἄρα καὶ ἡ ΔΑ τῇ ΕΖ. ἀλλὰ ἡ ΔΑ ἑκάστῃ τῶν ΚΕ, ΚΖ, ΚΗ ἐδείϑϑηὁίση· καὶ ἑκάστη ἄρα τῶν ΕΖ, ΖΗ, ΗΕ ἑκάστῃ τῶν ΚΕ, ΚΖ, ΚΗ ἐστινὁίση· ἰσόπλευρα ἄρα ἐστὶ τὰ τέσσαρα τρίγωνα τὰ ΕΖΗ, ΚΕΖ, ΚΖΗ, ΚΕΗ. πυραμὶξ ἄρα συνέσταται ἐκ τεσσάρων τριγώνων ἰσοπλέυρων, ἧξ βάσις μέν ἐστι τὸ ΕΖΗ τρίγωνον, κορυφὴ δὲ τὸ Κ σημεῖον.

thing it was required to show.

Proposition 13

To construct a (regular) pyramid (*i.e.*, a tetrahedron), and to enclose (it) in a given sphere, and to show that the square on the diameter of the sphere is one and a half times the (square) on the side of the pyramid.

Let the diameter AB of the given sphere be laid out, and let it be cut at point C such that AC is double CB [Prop. 6.10]. And let the semi-circle ADB be drawn on AB. And let CD be drawn from point C at right-angles to AB. And let DA be joined. And let the circle EFG be laid down having a radius equal to DC, and let the equilateral triangle EFG be inscribed in circle EFG [Prop. 4.2]. And let the center of the circle, point H, be found [Prop. 3.1]. And let EH, HF, and HG be joined. And let HK be set up, at point H, at right-angles to the plane of circle EFG [Prop. 11.12]. And let HK, equal to the straight-line AC, be cut off from HK. And let KE, KF, and KG be joined. And since KH is at right-angles to the plane of circle EFG, it will thus also make right-angles with all of the straight-lines joining it (which are) also in the plane of circle EFG [Def. 11.3]. And HE, HF, and HG each join it. Thus, HK is at right-angles to each of HE, HF, and HG. And since AC is equal to HK, and CD to HE, and they contain right-angles, the base DA is thus equal to the base KE [Prop. 1.4]. So, for the same (reasons), KF and KG is each equal to DA. Thus, the three (straight-lines) KE, KF, and KG are equal to one another. And since AC is double CB, AB (is) thus triple BC. And as AB (is) to BC, so the (square) on AD (is) to the (square) on DC, as will be shown later [see lemma]. Thus, the (square) on AD (is) three times the (square) on DC. And the (square) on FE is also three times the (square) on EH [Prop. 13.12], and DC is equal to EH. Thus, DA (is) also equal to EF. But, DA was shown (to be) equal to each of KE, KF, and KG. Thus, EF, FG, and GE are equal to KE, KF, and KG, respectively. Thus, the four triangles EFG, KEF, KFG, and KEG are equilateral. Thus, a pyramid, whose base is triangle EFG, and apex the point K, has been constructed from four equilateral triangles.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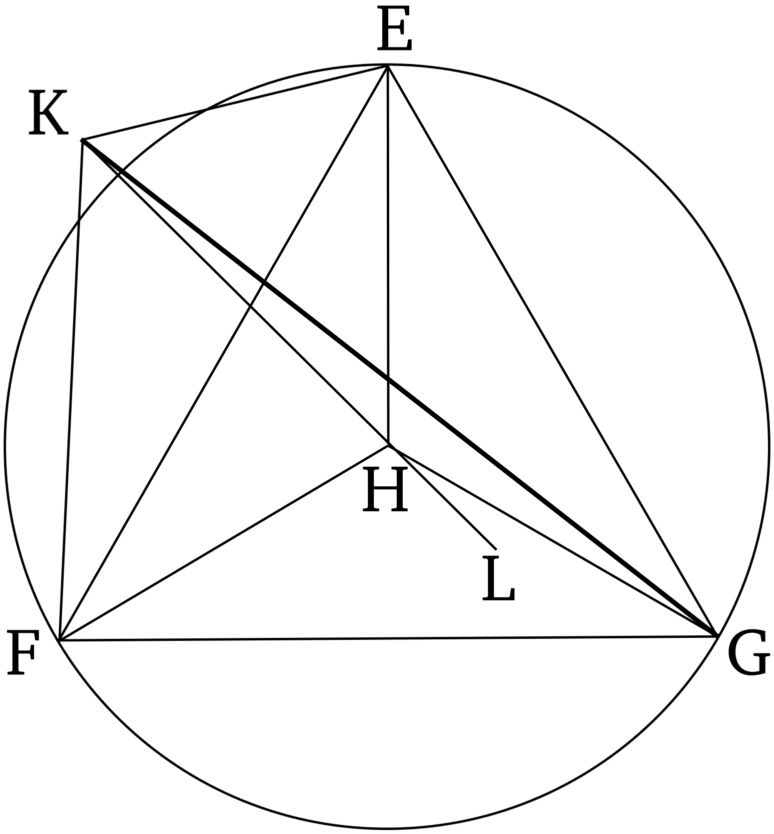
E
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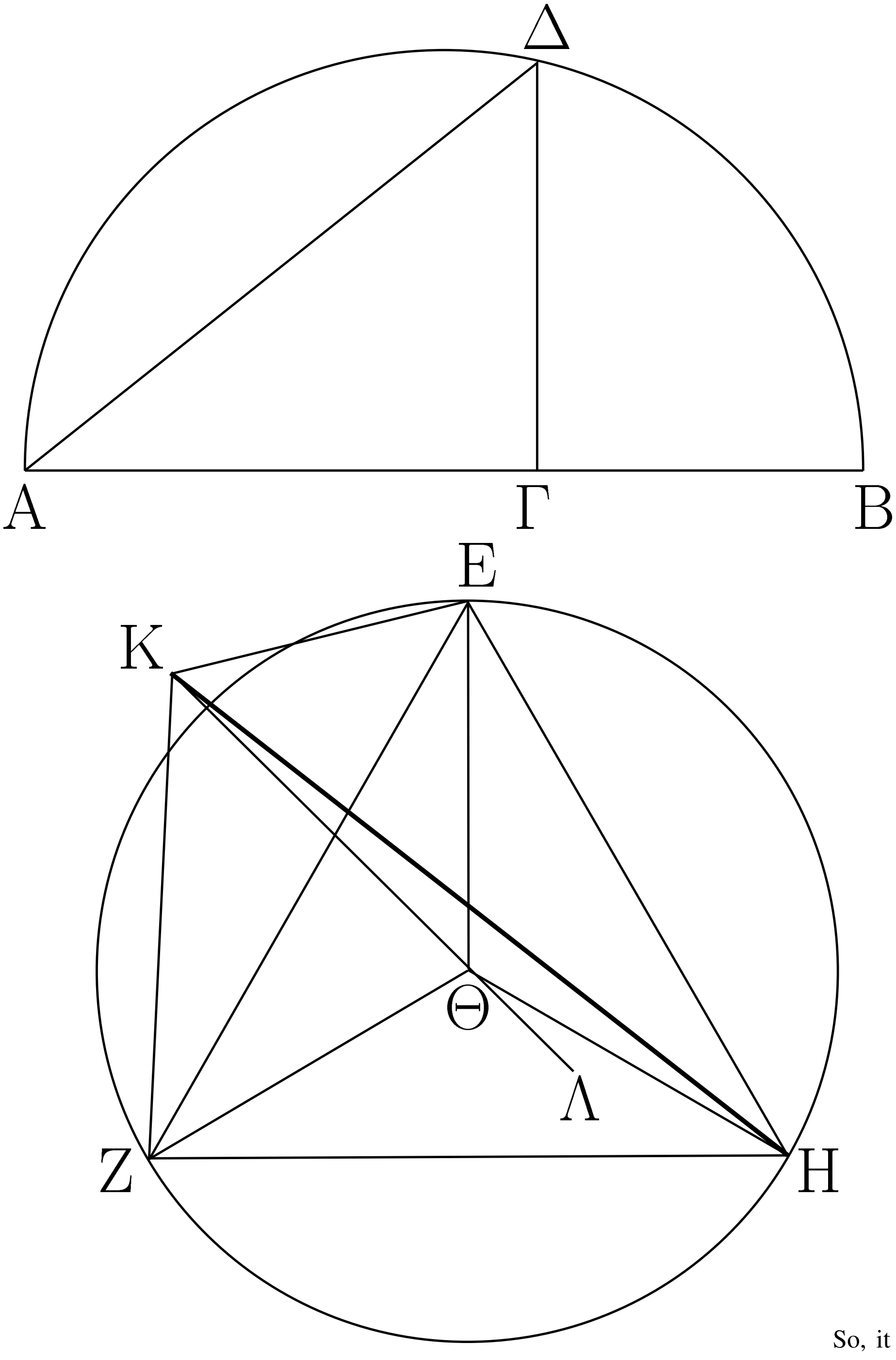


Δεῖ δὴ αὐτὴν καὶ σφαίρᾳ περιλαβεῖν τῇ δοθείσῃ καὶ δεῖχαι, ὅτι ἡ τῆς σφαίρας διάμετρος ἡμιολία ἐστὶ δυνάμει τῆς πλευρᾶξ τῆς πυραμίδος.

Ἐκβεβλήσθω γὰρ ἐπ᾿ εὐθείας τῇ ΚΘ εὐθεῖα ἡ ΘΛ, καὶ κείσθω τῇ ΓΒὀίση ἡ ΘΛ. καὶ ἐπεί ἐστιν ὡξ ἡ ΑΓ πρὸς τὴν ΓΔ, οὕτως ἡ ΓΔ πρὸς τὴν ΓΒ, ὀίση δὲ ἡ μὲν ΑΓ τῇ ΚΘ, ἡ δὲ ΓΔ τῇ ΘΕ, ἡ δὲ ΓΒ τῇ ΘΛ, ὀέστιν ἄρα ὡξ ἡ ΚΘ πρὸς

So, it is also necessary to enclose it in the given sphere, and to show that the square on the diameter of the sphere is one and a half times the (square) on the side of the pyramid.

For let the straight-line HL be produced in a straight-line with KH, and let HL be made equal to CB. And since as AC (is) to CD, so CD (is) to CB [Prop. 6.8 corr.], and AC (is) equal to KH, and CD to HE, and CB to HL, thus as KH is to HE, so EH (is) to HL. Thus, the (rectangle contained) by KH and HL

τὴν ΘΕ, οὕτως ἡ ΕΘ πρὸς τὴν ΘΛ· τὸ ἄρα ὑπὸ τῶν ΚΘ, ΘΛ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΕΘ. καί ἐστιν ὀρθὴ ἑκατέρα τῶν ὑπὸ ΚΘΕ, ΕΘΛ γωνιῶν· τὸ ἄρα ἐπὶ τῆς ΚΛ γραφόμενον ἡμικύκλιον ἥχει καὶ διὰ τοῦ Ε [ἐπειδήπερ ἒαν ἐπιζεύχωμεν τὴν ΕΛ, ὀρθὴ γίνεται ἡ ὑπὸ ΛΕΚ γωνία διὰ τὸ ἰσογώνιον γίνεσθαι τὸ ΕΛΚ τρίγωνον ἑκατέρῳ τῶν ΕΛΘ, ΕΘΚ τριγώνων]. ἒαν δὴ μενούσης τῆς ΚΛ περιενεθθὲν τὸ ἡμικύκλιον εἰξ τὸ αὐτὸ πάλιν ἀποκατασταθῇ, ὅθεν ἤρχατο φέρεσθαι, ἥχει καὶ διὰ τῶν Ζ, Η σημείων ἐπιζευγνυμένων τῶν ΖΛ, ΛΗ καὶ ὀρθῶν ὁμοίως γινομένων τῶν πρὸς τοῖς Ζ, Η γωνιῶν· καὶ ὀέσται ἡ πυραμὶξ σφαίρᾳ περιειλημμένη τῇ δοθείσῃ. ἡ γὰρ ΚΛ τῆς σφαίρας διάμετροςὀίση ἐστὶ τῇ τῆς δοθείσης σφαίρας διαμετρῳ τῇ ΑΒ, ἐπειδήπερ τῇ μὲν ΑΓὀίση κεῖται ἡ ΚΘ, τῇ δὲ ΓΒ ἡ ΘΛ.

Λέγω δή, ὅτι ἡ τῆς σφαίρας διάμετρος ἡμιολία ἐστὶ δυνάμει τῆς πλευρᾶξ τῆς πυραμίδος.

Ἐπεὶ γὰρ διπλῆ ἐστιν ἡ ΑΓ τῆς ΓΒ, τριπλῆ ἄρα ἐστὶν ἡ ΑΒ τῆς ΒΓ· ἀναστρέψαντι ἡμιολία ἄρα ἐστὶν ἡ ΒΑ τῆς ΑΓ. ὥξ δὲ ἡ ΒΑ πρὸς τὴν ΑΓ, οὕτως τὸ ἀπὸ τῆς ΒΑ πρὸς τὸ ἀπὸ τῆς ΑΔ [ἐπειδήπερ ἐπιζευγνμένης τῆς ΔΒ ἐστιν ὥξ ἡ ΒΑ πρὸς τὴν ΑΔ, οὕτως ἡ ΔΑ πρὸς τὴν ΑΓ διὰ τὴν ὁμοιότητα τῶν ΔΑΒ, ΔΑΓ τριγώνων, καὶ εὀῖναι ὥξ τὴν πρώτην πρὸς τὴν τρίτην, οὕτως τὸ ἀπὸ τῆς τρώτης πρὸς τὸ ἀπὸ τῆς δευτέρας]. ἡμιόλιον ἄρα καὶ τὸ ἀπὸ τῆς ΒΑ τοῦ ἀπὸ τῆς ΑΔ. καί ἐστιν ἡ μὲν ΒΑ ἡ τῆς δοθείσης σφαίρας διάμετρος, ἡ δὲ ΑΔὀίση τῇ πλευρᾷ τῆς πυραμίδος.

Ἡ ἄρα τῆς σφαίρας διάμετρος ἡμιολία ἐστὶ τῆς πλευρᾶξ τῆς πυραμίδος· ὅπερ ἔδει δεῖχαι.

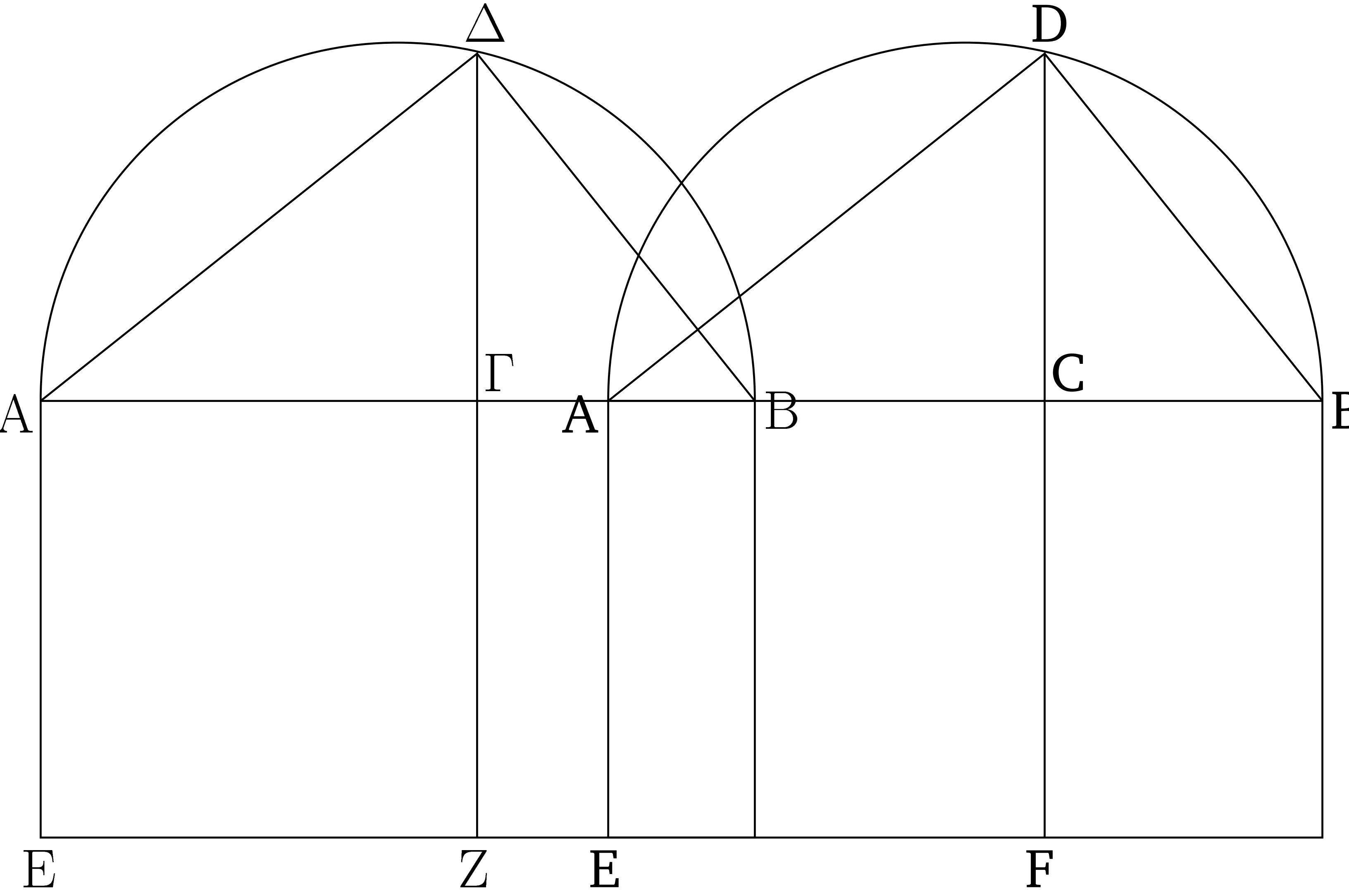
is equal to the (square) on *EH* [Prop. 6.17]. And each of the angles *KHE* and *EHL* is a right-angle. Thus, the semi-circle drawn on *KL* will also pass through *E* [inasmuch as if we join *EL* then the angle *LEK* becomes a right-angle, on account of triangle *ELK* becoming equiangular to each of the triangles *ELH* and *EHK* [Props. 6.8, 3.31]]. So, if *KL* remains (fixed), and the semi-circle is carried around, and again established at the same (position) from which it began to be moved, it will also pass through points *F* and *G*, (because) if *FL* and *LG* are joined, the angles at *F* and *G* will similarly become right-angles. And the pyramid will be enclosed by the given sphere. For the diameter, *KL*, of the sphere is equal to the diameter, *AB*, of the given sphere—inasmuch as *KH* was made equal to *AC*, and *HL* to *CB*.

So, I say that the square on the diameter of the sphere is one and a half times the (square) on the side of the pyramid.

For since *AC* is double *CB*, *AB* is thus triple *BC*. Thus, via conversion, *BA* is one and a half times *AC*. And as *BA* (is) to *AC*, so the (square) on *BA* (is) to the (square) on *AD* [inasmuch as if *DB* is joined then as *BA* is to *AD*, so *DA* (is) to *AC*, on account of the similarity of triangles *DAB* and *DAC*. And as the first is to the third (of four proportional magnitudes), so the (square) on the first (is) to the (square) on the second.] Thus, the (square) on *BA* (is) also one and a half times the (square) on *AD*. And *BA* is the diameter of the given sphere, and *AD* (is) equal to the side of the pyramid.

Thus, the square on the diameter of the sphere is one and a half times the (square) on the side of the pyramid.[†] (Which is) the very thing it was required to show.

[†] If the radius of the sphere is unity then the side of the pyramid (*i.e.*, tetrahedron) is $\sqrt{8/3}$.



Λῆμμα.

Δεικτέον, ὅτι ἐστὶν ὡξ ἡ AB πϱὸς τὴν BΓ, οὕτως τὸ ἀπὸ τῆς AΔ πϱὸς τὸ ἀπὸ τῆς ΔΓ.

Ἐκκείσϑω γὰϱ ἡ τοῦ ἡμικυκλίου καταγϱαφή, καὶ ἐπεζεύϑϑω ἡ ΔB, καὶ ἀναγεγϱάφϑω ἀπὸ τῆς AΓ τετϱάγωνον τὸ EΓ, καὶ συμπεπληϱώσϑω τὸ ZB παϱαλληλόγϱαμμον. ἐπεὶ οὐ̃ν διὰ τὸ ἰσογώνιον εἶναι τὸ ΔAB τϱίγωνον τῷ ΔAΓ τϱιγώνῳ ἐστὶν ὡξ ἡ BA πϱὸς τὴν AΔ, οὕτως ἡ ΔA πϱὸς τὴν AΓ, τὸ ἄϱα ὑπὸ τῶν BA, AΓ ἴσον ἐστὶ τῷ ἀπὸ τῆς AΔ. καὶ ἐπεί ἐστιν ὡξ ἡ AB πϱὸς τὴν BΓ, οὕτως τὸ EB πϱὸς τὸ BZ, καί ἐστι τὸ μὲν EB τὸ ὑπὸ τῶν BA, AΓ· ὁίση γὰϱ ἡ EA τῇ AΓ· τὸ δὲ BZ τὸ ὑπὸ τῶν AΓ, ΓB, ὡξ ἄϱα ἡ AB πϱὸς τὴν BΓ, οὕτως τὸ ὑπὸ τῶν BA, AΓ πϱὸν τὸ ὑπὸ τῶν AΓ, ΓB. καί ἐστι τὸ μὲν ὑπὸ τῶν BA, AΓ ἴσον τῷ ἀπὸ τῆς AΔ, τὸ δὲ ὑπὸ τῶν AΓB ἴσον τῷ ἀπὸ τῆς ΔΓ· ἡ γὰϱ ΔΓ κάϑετος τῶν τῆς βάσεως τμημάτων τῶν AΓ, ΓB μέση ἀνάλογόν ἐστι διὰ τὸ ὀϱϑὴν εἶναι

Lemma

It must be shown that as *AB* is to *BC*, so the (square) on *AD* (is) to the (square) on *DC*.

For, let the figure of the semi-circle be set out, and let *DB* have been joined. And let the square *EC* be described on *AC*. And let the parallelogram *FB* be completed. Therefore, since, on account of triangle *DAB* being equiangular to triangle *DAC* [Props. 6.8, 6.4], (proportionally) as *BA* is to *AD*, so *DA* (is) to *AC*, the (rectangle contained) by *BA* and *AC* is thus equal to the (square) on *AD* [Prop. 6.17]. And since as *AB* is to *BC*, so *EB* (is) to *BF* [Prop. 6.1]. And *EB* is the (rectangle contained) by *BA* and *AC*—for *EA* (is) equal to *AC*. And *BF* the (rectangle contained) by *AC* and *CB*. Thus, as *AB* (is) to *BC*, so the (rectangle contained) by *BA* and *AC* (is) to the (rectangle contained) by *AC* and *CB*. And the (rectangle contained) by *BA* and *AC* is equal to the (square) on *AD*, and the (rectan-

τὴν ὑπὸ ΑΔΒ. ὡξ ἄρα ἡ ΑΒ πρὸς τὴν ΒΓ, οὕτως τὸ ἀπὸ τῆς ΑΔ πρὸς τὸ ἀπὸ τῆς ΔΓ· ὅπερ ἔδει δεῖξαι.

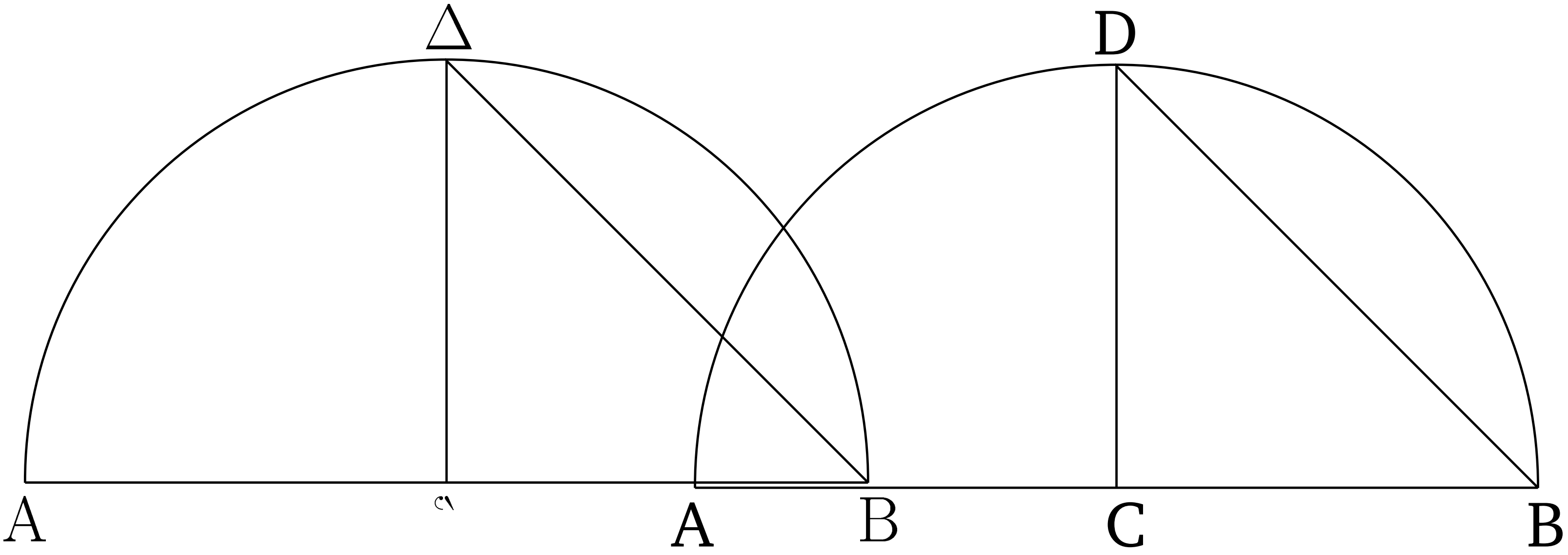
14.

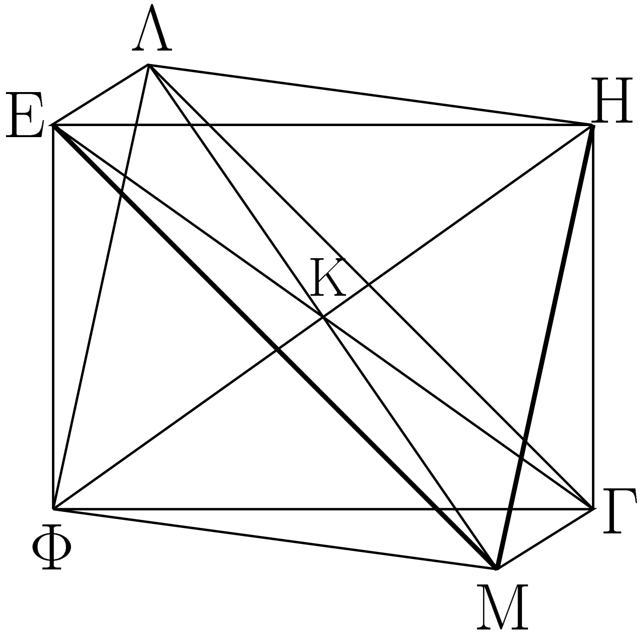
Ὀκτάεδρον συστήσασθαι καὶ σφαίρᾳ περιλαβεῖν, ᾗ καὶ τὰ πρότερα, καὶ δεῖξαι, ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει διπλασία ἐστὶ τῆς πλευρᾶξ τοῦ ὀκταέδρου.

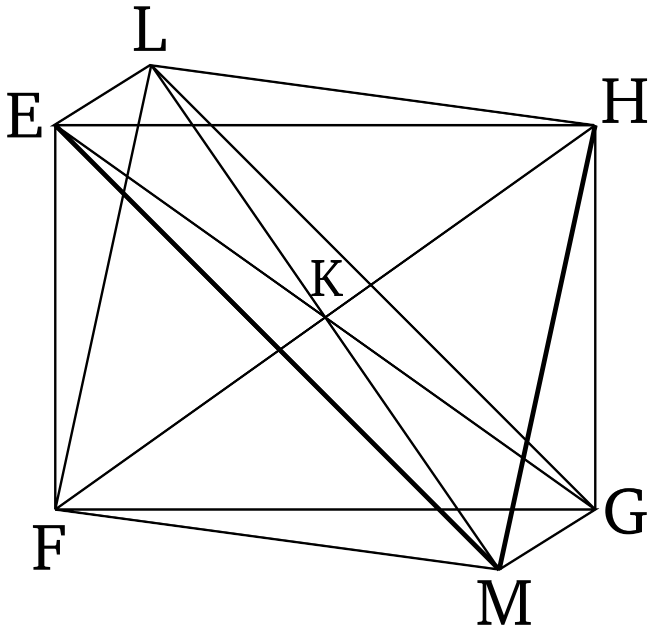
gle contained) by ACB (is) equal to the (square) on DC. For the perpendicular DC is the mean proportional to the pieces of the base, AC and CB, on account of ADB being a right-angle [Prop. 6.8 corr.]. Thus, as AB (is) to BC, so the (square) on AD (is) to the (square) on DC. (Which is) the very thing it was required to show.

Proposition 14

To construct an octahedron, and to enclose (it) in a (given) sphere, like in the preceding (proposition), and to show that the square on the diameter of the sphere is double the (square) on the side of the octahedron.







Ἐκκείσθω ἡ τῆς δοθείσης σφαίρας διάμετρος ἡ ΑΒ, καὶ τετμήσθω δίϑα κατὰ τὸ Γ, καὶ γεγράφθω ἐπὶ τῆς ΑΒ ἡμικύκλιον τὸ ΑΔΒ, καὶ ἤϑϑω ἀπὸ τοῦ Γ τῇ ΑΒ πρὸς ὀρθὰξ ἡ ΓΔ, καὶ ἐπεζεύϑϑω ἡ ΔΒ, καὶ ἐκκείσθω τετράγωνον τὸ ΕΖΗΘ ἴσην ἔχον ἑκάστην τῶν πλευρῶν τῇ ΔΒ, καὶ ἐπεζεύϑϑωσαν αἱ ΘΖ, ΕΗ, καὶ ἀνεστάτω ἀπὸ τοῦ Κ σημείου τῷ τοῦ ΕΖΗΘ τετραγώνου ἐπιπέδῳ πρὸς ὀρθὰξ εὐϑεῖα ἡ ΚΛ καὶ διήϑϑω ἐπὶ τὰ ὁέτερα μέρη τοῦ ἐπιπέδου ὡξ ἡ ΚΜ, καὶ ἀφῃρήσθω ἀφ′ ἑκατέρας τῶν ΚΛ, ΚΜ μιᾷ τῶν ΕΚ, ΖΚ, ΗΚ, ΘΚὀίση ἑκατέρα τῶν ΚΛ, ΚΜ, καὶ ἐπεζεύϑϑωσαν αἱ ΛΕ, ΛΖ, ΛΗ, ΛΘ, ΜΕ, ΜΖ, ΜΗ, ΜΘ.

Καὶ ἐπεὶ ὀίση ἐστὶν ἡ ΚΕ τῇ ΚΘ, καί ἐστιν ὀρθὴ ἡ ὑπὸ

Let the diameter *AB* of the given sphere be laid out, and let it have been cut in half at *C*. And let the semi-circle *ADB* be drawn on *AB*. And let *CD* be drawn from *C* at right-angles to *AB*. And let *DB* be joined. And let the square *EFGH*, having each of its sides equal to *DB*, be laid out. And let *HF* and *EG* be joined. And let the straight-line *KL* be set up, at point *K*, at right-angles to the plane of square *EFGH* [Prop. 11.12]. And let it be drawn across on the other side of the plane, like *KM*. And let *KL* and *KM*, equal to one of *EK*, *FK*, *GK*, and *HK*, be cut off from *KL* and *KM*, respectively. And let *LE*, *LF*, *LG*, *LH*, *ME*, *MF*, *MG*, and *MH* be joined.

And since *KE* is equal to *KH*, and angle *EKH* is a right-

ΕΚΘ γωνία, τὸ ἄρα ἀπὸ τῆς ΘΕ διπλάσιόν ἐστι τοῦ ἀπὸ τῆς ΕΚ. πάλιν, ἐπεὶ ὀίση ἐστὶν ἡ ΛΚ τῇ ΚΕ, καί ἐστιν ὀρϑὴ ἡ ὑπὸ ΛΚΕ γωνία, τὸ ἄρα ἀπὸ τῆς ΕΛ διπλάσιόν ἐστι τοῦ ἀπὸ ΕΚ. ἐδείϑη δὲ καὶ τὸ ἀπὸ τῆς ΘΕ διπλάσιον τοῦ ἀπὸ τῆς ΕΚ· τὸ ἄρα ἀπὸ τῆς ΛΕ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΕΘ· ὀίση ἄρα ἐστὶν ἡ ΛΕ τῇ ΕΘ. διὰ τὰ αὐτὰ δὴ καὶ ἡ ΛΘ τῇ ΘΕ ἐστινὀίση· ἰσόπλευρον ἄρα ἐστὶ τὸ ΛΕΘ τρίγωνον. ὁμοίως δὴ δείχομεν, ὅτι καὶ ὀέκαστον τῶν λοιπῶν τριγώνων, ὁῶν βάσεις μέν εἰσιν αἱ τοῦ ΕΖΗΘ τετραγώνου πλευραί, κορυφαὶ δὲ τὰ Λ, Μ σημεῖα, ἰσόπλευρόν ἐστιν· ὀκτάεδρον ἄρα συνέσταται ὑπὸ ὀκτὼ τριγώνων ἰσοπλεύρων περιεχόμενον.

Δεῖ δὴ αὐτὸ καὶ σφαίρᾳ περιλαβεῖν τῇ δοϑείσῃ καὶ δεῖξαι, ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει διπλασίων ἐστὶ τῆς τοῦ ὀκταέδρου πλευρᾶξ.

Ἐπεὶ γὰρ αἱ τρεῖς αἱ ΛΚ, ΚΜ, ΚΕ ἴσαι ἀλλήλαις εἰσίν, τὸ ἄρα ἐπὶ τῆς ΛΜ γραφόμενον ἡμικύκλιον ἥξει καὶ διὰ τοῦ Ε. καὶ διὰ τὰ αὐτά, ἐὰν μενούσης τῆς ΛΜ περιενεϑὲν τὸ ἡμικύκλιον εἰξ τὸ αὐτὸ ἀποκατασταϑῇ, ὅϑεν ἤρξατο φέρεσϑαι, ἥξει καὶ διὰ τῶν Ζ, Η, Θ σημείων, καὶ ὀέσται σφαίρᾳ περιειλημμένον τὸ ὀκτάεδρον. λέγω δή, ὅτι καὶ τῇ δοϑείσῃ. ἐπεὶ γὰρὀίση ἐστὶν ἡ ΛΚ τῇ ΚΜ, κοινὴ δὲ ἡ ΚΕ, καὶ γωνίας ὀρϑὰξ περιέϑουσιν, βάσις ἄρα ἡ ΛΕ βάσει τῇ ΕΜ ἐστινὀίση. καὶ ἐπεὶ ὀρϑή ἐστιν ἡ ὑπὸ ΛΕΜ γωνία· ἐν ἡμικυκλίῳ γάρ· τὸ ἄρα ἀπὸ τῆς ΛΜ διπλάσιόν ἐστι τοῦ ἀπὸ τῆς ΛΕ. πάλιν, ἐπεὶ ὀίση ἐστὶν ἡ ΑΓ τῇ ΓΒ, διπλασία ἐστὶν ἡ ΑΒ τῆς ΒΓ. ὡξ δὲ ἡ ΑΒ πρὸς τὴν ΒΓ, οὀύτως τὸ ἀπὸ τῆς ΑΒ πρὸς τὸ ἀπὸ τῆς ΒΔ· διπλάσιον ἄρα ἐστὶ τὸ ἀπὸ τῆς ΑΒ τοῦ ἀπὸ τῆς ΒΔ. ἐδείϑη δὲ καὶ τὸ ἀπὸ τῆς ΛΜ διπλάσιον τοῦ ἀπὸ τῆς ΛΕ. καὶ ἐστιν ἴσον τὸ ἀπὸ τῆς ΔΒ τῷ ἀπὸ τῆς ΛΕ· ὀίση γὰρ κεῖται ἡ ΕΘ τῇ ΔΒ. ἴσον ἄρα καὶ τὸ ἀπὸ τῆς ΑΒ τῷ ἀπὸ τῆς ΛΜ· ὀίση ἄρα ἡ ΑΒ τῇ ΛΜ. καί ἐστιν ἡ ΑΒ ἡ τῆς δοϑείσης σφαίρας διάμετρος· ἡ ΛΜ ἄραὀίση ἐστὶ τῇ τῆς δοϑείσης σφαίρας διαμέτρῳ.

Περιείληπται ἄρα τὸ ὀκτάεδρον τῇ δοϑείσῃ σφαίρᾳ. καὶ συναποδέδεικται, ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει διπλασίων ἐστὶ τῆς τοῦ ὀκταέδρου πλευρᾶξ· ὅπερ ἔδει δεῖξαι.

angle, the (square) on the HE is thus double the (square) on EK [Prop. 1.47]. Again, since LK is equal to KE, and angle LKE is a right-angle, the (square) on EL is thus double the (square) on EK [Prop. 1.47]. And the (square) on HE was also shown (to be) double the (square) on EK. Thus, the (square) on LE is equal to the (square) on EH. Thus, LE is equal to EH. So, for the same (reasons), LH is also equal to HE. Triangle LEH is thus equilateral. So, similarly, we can show that each of the remaining triangles, whose bases are the sides of the square $EFGH$, and apexes the points L and M, are equilateral. Thus, an octahedron contained by eight equilateral triangles has been constructed.

So, it is also necessary to enclose it by the given sphere, and to show that the square on the diameter of the sphere is double the (square) on the side of the octahedron.

For since the three (straight-lines) LK, KM, and KE are equal to one another, the semi-circle drawn on LM will thus also pass through E. And, for the same (reasons), if LM remains (fixed), and the semi-circle is carried around, and again established at the same (position) from which it began to be moved, then it will also pass through points F, G, and H, and the octahedron will be enclosed by a sphere. So, I say that (it is) also (enclosed) by the given (sphere). For since LK is equal to KM, and KE (is) common, and they contain right-angles, the base LE is thus equal to the base EM [Prop. 1.4]. And since angle LEM is a right-angle—for (it is) in a semi-circle [Prop. 3.31]—the (square) on LM is thus double the (square) on LE [Prop. 1.47]. Again, since AC is equal to CB, AB is double BC. And as AB (is) to BC, so the (square) on AB (is) to the (square) on BD [Prop. 6.8, Def. 5.9]. Thus, the (square) on AB is double the (square) on BD. And the (square) on LM was also shown (to be) double the (square) on LE. And the (square) on DB is equal to the (square) on LE. For EH was made equal to DB. Thus, the (square) on AB (is) also equal to the (square) on LM. Thus, AB (is) equal to LM. And AB is the diameter of the given sphere. Thus, LM is equal to the diameter of the given sphere.

Thus, the octahedron has been enclosed by the given sphere, and it has been simultaneously proved that the square on the diameter of the sphere is double the (square) on the side of the octahedron.[†] (Which is) the very thing it was required to show.

[†] If the radius of the sphere is unity then the side of octahedron is $\sqrt{2}$.

15.

Κύβον συστήσασϑαι καὶ σφαίρᾳ περιλαβεῖν, ᾗ καὶ τὴν πυραμίδα, καὶ δεῖξαι, ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει τριπλασίων ἐστὶ τῆς τοῦ κύβου πλευρᾶξ.

Ἐκκείσϑω ἡ τῆς δοϑείσης σφαίρας διάμετρος ἡ ΑΒ καὶ

Proposition 15

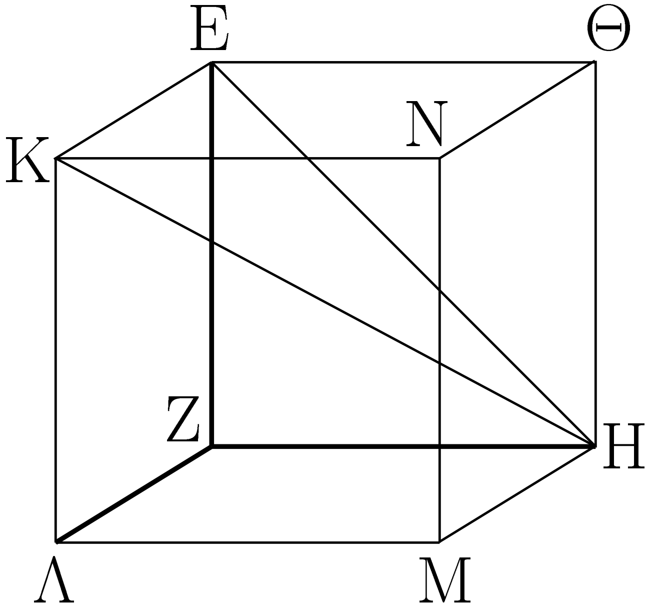
To construct a cube, and to enclose (it) in a sphere, like in the (case of the) pyramid, and to show that the square on the diameter of the sphere is three times the (square) on the side of the cube.

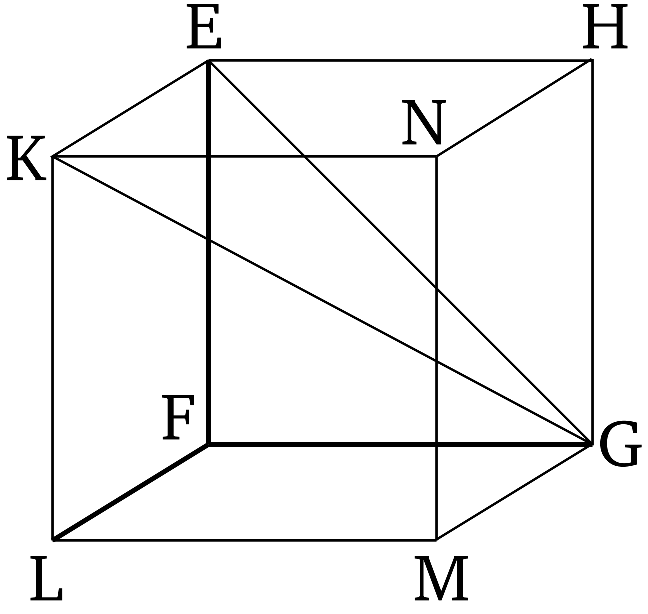
τετμήσθω κατὰ τὸ Γ ὁώστε διπλῆν εὀῖναι τὴν ΑΓ τῆς ΓΒ, καὶ γεγράφθω ἐπὶ τῆς ΑΒ ἡμικύκλιον τὸ ΑΔΒ, καὶ ἀπὸ τοῦ Γ τῇ ΑΒ πρὸς ὀρθὰξ ἤθθω ἡ ΓΔ, καὶ ἐπεζεύχθω ἡ ΔΒ, καὶ ἐκκείσθω τετράγωνον τὸ ΕΖΗΘ ἴσην ἔχον τὴν πλευρὰν τῇ ΔΒ, καὶ ἀπὸ τῶν Ε, Ζ, Η, Θ τῷ τοῦ ΕΖΗΘ τετραγώνου ἐπιπέδῳ πρὸς ὀρθὰξ ἤθθωσαν αἱ ΕΚ, ΖΛ, ΗΜ, ΘΝ, καὶ ἀφῃρήσθω ἀπὸ ἑκάστης τῶν ΕΚ, ΖΛ, ΗΜ, ΘΝ μιᾷ τῶν ΕΖ, ΖΗ, ΗΘ, ΘΕὀίση ἑκάστη τῶν ΕΚ, ΖΛ, ΗΜ, ΘΝ, καὶ ἐπεζεύχθωσαν αἱ ΚΛ, ΛΜ, ΜΝ, ΝΚ· κύβος ἄρα συνέσταται ὁ ΖΝ ὑπὸ δὲχ τετραγώνωνὀίσων περιεχόμενος.

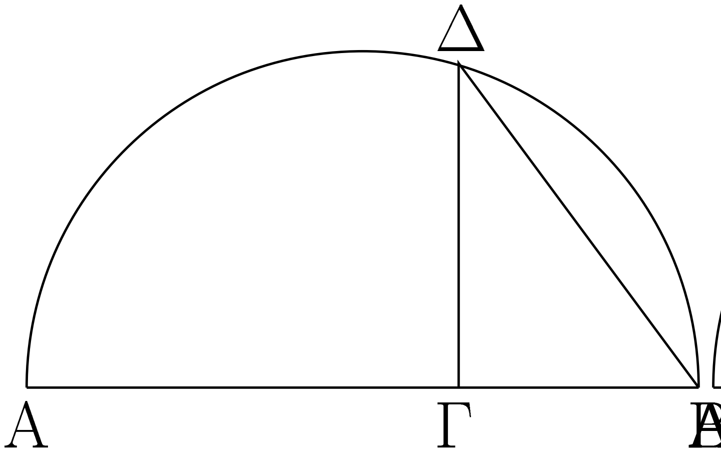
Δεῖ δὴ αὐτὸν καὶ σφαίρᾳ περιλαβεῖν τῇ δοθείσῃ καὶ δεῖξαι, ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει τριπλασία ἐστὶ τῆς πλευρᾶξ τοῦ κύβου.

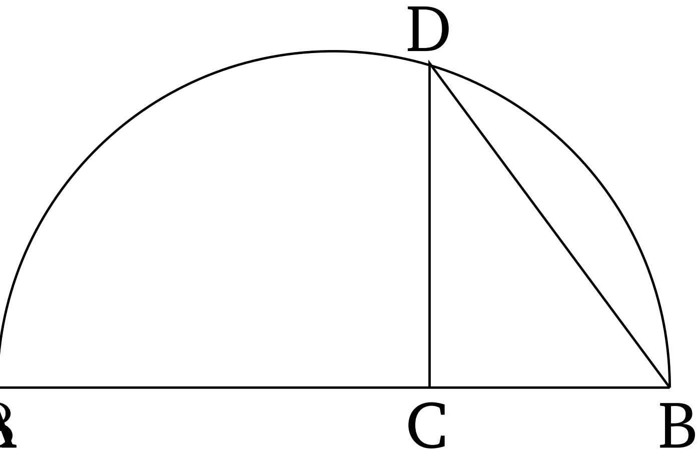
Let the diameter AB of the given sphere be laid out, and let it be cut at C such that AC is double CB. And let the semicircle ADB be drawn on AB. And let CD be drawn from C at right-angles to AB. And let DB be joined. And let the square $EFGH$, having (its) side equal to DB, be laid out. And let EK, FL, GM, and HN be drawn from (points) E, F, G, and H, (respectively), at right-angles to the plane of square $EFGH$. And let EK, FL, GM, and HN, equal to one of EF, FG, GH, and HE, be cut off from EK, FL, GM, and HN, respectively. And let KL, LM, MN, and NK be joined. Thus, a cube contained by six equal squares has been constructed.

So, it is also necessary to enclose it by the given sphere, and to show that the square on the diameter of the sphere is three times the (square) on the side of the cube.









Ἐπεζεύχθωσαν γὰρ αἱ ΚΗ, ΕΗ. καὶ ἐπεὶ ὀρθή ἐστιν ἡ ὑπὸ ΚΕΗ γωνία διὰ τὸ καὶ τὴν ΚΕ ὀρθὴν εὀῖναι πρὸς τὸ ΕΗ ἐπίπεδον δηλαδὴ καὶ πρὸς τὴν ΕΗ εὐθεῖαν, τὸ ἄρα ἐπὶ

For let KG and EG be joined. And since angle KEG is a right-angle—on account of KE also being at right-angles to the plane EG, and manifestly also to the straight-line EG

τῆς ΚΗ γραφόμενον ἡμικύκλιον ἥχει καὶ διὰ τοῦ Ε σημείον. πάλιν, ἐπεὶ ἡ ΗΖ ὀρθή ἐστι πρὸς ἑκατέραν τῶν ΖΛ, ΖΕ, καὶ πρὸς τὸ ΖΚ ἄρα ἐπίπεδον ὀρθή ἐστιν ἡ ΗΖ· ὁώστε καὶ ἐὰν ἐπιζεύχωμεν τὴν ΖΚ, ἡ ΗΖ ὀρθὴ ὁέσται καὶ πρὸς τὴν ΖΚ· καὶ δὶα τοῦτο πάλιν τὸ ἐπὶ τῆς ΗΚ γραφόμενον ἡμικύκλιον ἥχει καὶ διὰ τοῦ Ζ. ὁμοίως καὶ δὶα τῶν λοιπῶν τοῦ κύβου σημείων ἥχει. ἐὰν δὴ μενούσης τῆς ΚΗ περιενεθθὲν τὸ ἡμικύκλιον εἴξ τὸ αὐτὸ ἀποκατασταθῇ, ὅθεν ἤρχατο φέρεσθαι, ὁέσται σφαίρᾳ περιειλημμένος ὁ κύβος. λέγω δή, ὅτι καὶ τῇ δοθείσῃ. ἐπεὶ γὰρὁίση ἐστὶν ἡ ΗΖ τῇ ΖΕ, καί ἐστιν ὀρθὴ ἡ πρὸς τῷ Ζ γωνία, τὸ ἄρα ἀπὸ τῆς ΕΗ διπλάσιόν ἐστι τοῦ ἀπὸ τῆς ΕΖ. ὁίση δὲ ἡ ΕΖ τῇ ΕΚ· τὸ ἄρα ἀπὸ τῆς ΕΗ διπλάσιόν ἐστι τοῦ ἀπὸ τῆς ΕΚ· ὁώστε τὰ ἀπὸ τῶν ΗΕ, ΕΚ, τουτέστι τὸ ἀπὸ τῆς ΗΚ, τριπλάσιόν ἐστι τοῦ ἀπὸ τῆς ΕΚ. καὶ ἐπεὶ τριπλασίων ἐστὶν ἡ ΑΒ τῆς ΒΓ, ὥξ δὲ ἡ ΑΒ πρὸς τὴν ΒΓ, οὁύτως τὸ ἀπὸ τῆς ΑΒ πρὸς τὸ ἀπὸ τῆς ΒΔ, τριπλάσιον ἄρα τὸ ἀπὸ τῆς ΑΒ τοῦ ἀπὸ τῆς ΒΔ. ἐδείθθη δὲ καὶ τὸ ἀπὸ τῆς ΗΚ τοῦ ἀπὸ τῆς ΚΕ τριπλάσιον. καὶ κεῖταιὁίση ἡ ΚΕ τῇ ΔΒ· ὁίση ἄρα καὶ ἡ ΚΗ τῇ ΑΒ. καί ἐστιν ἡ ΑΒ τῆς δοθείσης σφαίρας διάμετρος· καὶ ἡ ΚΗ ἄραὁίση ἐστὶ τῇ τῆς δοθείσης σφαίρας διαμέτρῳ.

Τῇ δοθείσῃ ἄρα σφαίρα περιείληπται ὁ κύβος· καὶ συναποδέδεικται, ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει τριπλασίων ἐστὶ τῆς τοῦ κύβου πλευρᾶξ· ὅπερ ἔδει δεῖχαι.

[Def. 11.3]—the semi-circle drawn on *KG* will thus also pass through point *E*. Again, since *GF* is at right-angles to each of *FL* and *FE*, *GF* is thus also at right-angles to the plane *FK* [Prop. 11.4]. Hence, if we also join *FK* then *GF* will also be at right-angles to *FK*. And, again, on account of this, the semi-circle drawn on *GK* will also pass through point *F*. Similarly, it will also pass through the remaining (angular) points of the cube. So, if *KG* remains (fixed), and the semi-circle is carried around, and again established at the same (position) from which it began to be moved, then the cube will be enclosed by a sphere. So, I say that (it is) also (enclosed) by the given (sphere). For since *GF* is equal to *FE*, and the angle at *F* is a right-angle, the (square) on *EG* is thus double the (square) on *EF* [Prop. 1.47]. And *EF* (is) equal to *EK*. Thus, the (square) on *EG* is double the (square) on *EK*. Hence, the (sum of the squares) on *GE* and *EK*—that is to say, the (square) on *GK* [Prop. 1.47]—is three times the (square) on *EK*. And since *AB* is three times *BC*, and as *AB* (is) to *BC*, so the (square) on *AB* (is) to the (square) on *BD* [Prop. 6.8, Def. 5.9], the (square) on *AB* (is) thus three times the (square) on *BD*. And the (square) on *GK* was also shown (to be) three times the (square) on *KE*. And *KE* was made equal to *DB*. Thus, *KG* (is) also equal to *AB*. And *AB* is the radius of the given sphere. Thus, *KG* is also equal to the diameter of the given sphere.

Thus, the cube has been enclosed by the given sphere. And it has simultaneously been shown that the square on the diameter of the sphere is three times the (square) on the side of the cube.[†] (Which is) the very thing it was required to show.

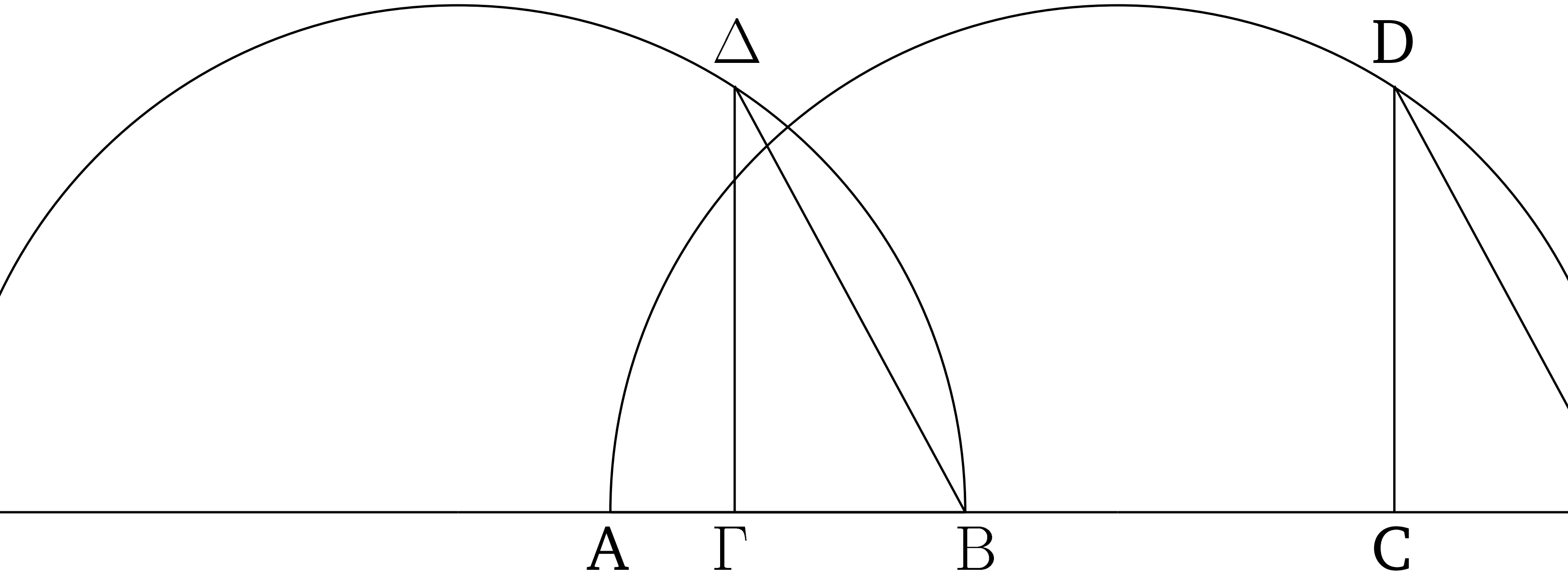
[†] If the radius of the sphere is unity then the side of the cube is $\sqrt{4/3}$.

16.

Εἰκοσάεδρον συστήσασθαι καὶ σφαίρᾳ περιλαβεῖν, ᾗ καὶ τὰ προειρημένα σϑήματα, καὶ δεῖξαι, ὅτι ἡ τοῦ εἰκοσαέδρου πλευρὰ ὁάλογόξ ἐστιν ἡ καλουμένη ἐλάττων.

Proposition 16

To construct an icosahedron, and to enclose (it) in a sphere, like the aforementioned figures, and to show that the side of the icosahedron is that irrational (straight-line) called minor.



Ἐκκείσθω ἡ τῆς δοθείσης σφαίρας διάμετρος ἡ ΑΒ καὶ τετμήσθω κατὰ τὸ Γ ὁώστε τετραπλῆν εὀῖναι τὴν ΑΓ τῆς ΓΒ, καὶ γεγράφθω ἐπὶ τῆς ΑΒ ἡμικύκλιον τὸ ΑΔΒ, καὶ ἤθθω ἀπὸ τοῦ Γ τῇ ΑΒ πρὸς ορθὰξ γωνίας εὐθεῖα γραμμὴ ἡ ΓΔ, καί ἐπεζεύθθω ἡ ΔΒ, καὶ ἐκκείσθω κύκλος ὁ ΕΖΗΘΚ, οὗ ἡ ἐν τοῦ κέντρουὀίσηὀέστω τῇ ΔΒ, καὶ ἐγγεγράφθω εἰξ τὸν ΕΖΗΘΚ κύκλον πεντάγωνον ἰσόπλευρόν τε καὶ ἰσογώνιον τὸ ΕΖΗΘΚ, καὶ τετμήσθωσαν αἱ ΕΖ, ΖΗ, ΗΘ, ΘΚ, ΚΕ περιφέρειαι δίθα κατὰ τὸ Λ, Μ, Ν, Χ, Ο σημεῖα, καὶ ἐπεζεύθθωσαν αἱ ΛΜ, ΜΝ, ΝΧ, ΧΟ, ΟΛ, ΕΟ. ὀίσόπλευρον ἄρα ἐστὶ καὶ τὸ ΛΜΝΧΟ πεντάγωνον, καὶ δεκαγώνου ἡ ΕΟ εὐθεῖα. καὶ ἀνεστάτωσανὀάπὸ τῶν Ε, Ζ, Η, Θ, Κ σημείων τῷ τοῦ κύκλου ἐπιπέδῳ πρὸς ὀρθὰξ γωνίας εὐθεῖαι αἱ ΕΠ, ΖΡ, ΗΣ, ΘΤ, ΚΥ ἴσαι οὀῦσαι τῇ ἐκ τοῦ κέντρου τοῦ ΕΖΗΘΚ κύκλου, καὶ ἐπεζεύθθωσαν αἱ ΠΡ, ΡΣ, ΣΤ, ΤΥ, ΥΠ, ΠΛ, ΛΡ, ΡΜ, ΜΣ, ΣΝ, ΝΤ, ΤΧ, ΧΥ, ΥΟ, ΟΠ.

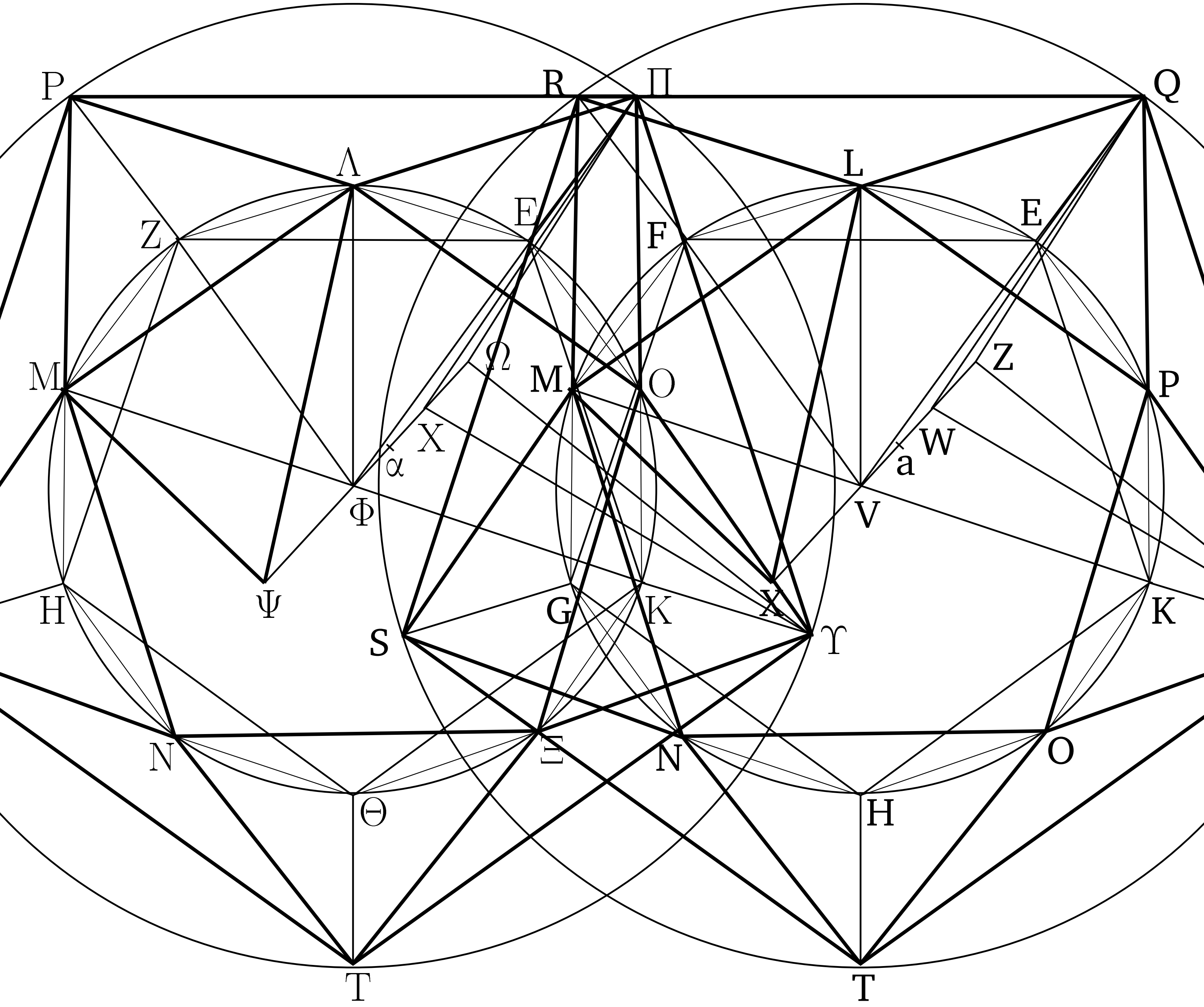
Καὶ ἐπεὶ ἑκατέρα τῶν ΕΠ, ΚΥ τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰξ ἐστιν, παράλληλος ἄρα ἐστὶν ἡ ΕΠ τῇ ΚΥ. ὀέστι δὲ αὐτῇ καὶ ὀίση· αἱ δὲ τὰξ ἴσας τε καὶ παραλλήλους ἐπιζευγνύουσαι ἐπὶ τὰ αὐτὰ μέρη εὐθεῖαι ἴσαι τε καὶ παράλληλοί εἰσιν. ἡ ΠΥ ἄρα τῇ ΕΚὀίση τε καὶ παράλληλόξ ἐστιν. πενταγώνου δὲ ἰσοπλεύρου ἡ ΕΚ· πενταγώνου ἄρα ἰσοπλεύρου καὶ ἡ ΠΥ τοῦ εἰξ τὸν ΕΖΗΘΚ κύκλον ἐγγραφομένου. διὰ τὰ αὐτὰ δὴ καὶ ἑκάστη τῶν ΠΡ, ΡΣ, ΣΤ, ΤΥ πενταγώνου ἐστὶν ἰσοπλεύρου τοῦ εἰξ τὸν ΕΖΗΘΚ κύκλον ἐγγραφομένου· ἰσόπλευρον ἄρα τὸ ΠΡΣΤΥ πεντάγωνον. καὶ ἐπεὶ ἑχαγώνου μέν ἐστιν ἡ ΠΕ, δεκαγώνου δὲ ἡ ΕΟ, καί ἐστιν ὀρθὴ ἡ ὑπὸ ΠΕΟ, πενταγώνου ἄρα ἐστὶν ἡ ΠΟ· ἡ γὰρ τοῦ πενταγώνου πλευρὰ δύναται τήν τε τοῦ ἑχαγώνου καὶ τὴν τοῦ δεκαγώνου τῶν εἰξ τὸν αὐτὸν κύκλον ἐγγραφομένων. διὰ τὰ αὐτὰ δὴ καὶ ἡ ΟΥ πενταγώνου ἐστὶ πλευρά. ὀέστι δὲ καὶ ἡ ΠΥ πενταγώνου· ἰσόπλευρον ἄρα ἐστὶ τὸ ΠΟΥ τρίγωνον. διὰ τὰ αὐτὰ

Let the diameter *AB* of the given sphere be laid out, and let it be cut at *C* such that *AC* is four times *CB* [Prop. 6.10]. And let the semi-circle *ADB* be drawn on *AB*. And let the straight-line *CD* be drawn from *C* at right-angles to *AB*. And let *DB* be joined. And let the circle *EFGHK* be set down, and let its radius be equal to *DB*. And let the equilateral and equiangular pentagon *EFGHK* be inscribed in circle *EFGHK* [Prop. 4.11]. And let the circumferences *EF*, *FG*, *GH*, *HK*, and *KE* be cut in half at points *L*, *M*, *N*, *O*, and *P* (respectively). And let *LM*, *MN*, *NO*, *OP*, *PL*, and *EP* be joined. Thus, pentagon *LMNOP* is also equilateral, and *EP* (is) the side of the decagon (inscribed in the circle). And let the straight-lines *EQ*, *FR*, *GS*, *HT*, and *KU*, which are equal to the radius of circle *EFGHK*, have been set up at right-angles to the plane of the circle, at points *E*, *F*, *G*, *H*, and *K* (respectively). And let *QR*, *RS*, *ST*, *TU*, *UQ*, *QL*, *LR*, *RM*, *MS*, *SN*, *NT*, *TO*, *OU*, *UP*, and *PQ* be joined.

And since *EQ* and *KU* are each at right-angles to the same plane, *EQ* is thus parallel to *KU* [Prop. 11.6]. And it is also equal to it. And straight-lines joining equal and parallel (straight-lines) on the same side are (themselves) equal and parallel [Prop. 1.33]. Thus, *QU* is equal and parallel to *EK*. And *EK* (is the side) of an equilateral pentagon (inscribed in circle *EFGHK*). Thus, *QU* (is) also the side of an equilateral pentagon inscribed in circle *EFGHK*. So, for the same (reasons), *QR*, *RS*, *ST*, and *TU* are also the sides of an equilateral pentagon inscribed in circle *EFGHK*. Pentagon *QRSTU* (is) thus equilateral. And side *QE* is (the side) of a hexagon (inscribed in circle *EFGHK*), and *EP* (the side) of a decagon, and (angle) *QEP* is a right-angle, thus *QP* is (the side) of a pentagon (inscribed in the same circle). For the square on the side of a pentagon is (equal to the sum of) the (squares) on (the

δὴ καὶ ἕκαστον τῶν ΠΛΡ, ΡΜΣ, ΣΝΤ, ΤΧΥ ἰσόπλευρόν ἐστιν. καὶ ἐπεὶ πενταγώνου ἐδείϑϑη ἑκατέρα τῶν ΠΛ, ΠΟ, ἔστι δὲ καὶ ἡ ΛΟ πενταγώνου, ἰσόπλευρον ἄρα ἐστὶ τὸ ΠΛΟ τρίγωνον. διὰ τὰ αὐτὰ δὴ καὶ ἕκαστον τῶν ΛΡΜ, ΜΣΝ, ΝΤΧ, ΧΥΟ τριγώνων ἰσόπλευρόν ἐστιν.

sides of) a hexagon and a decagon inscribed in the same circle [Prop. 13.10]. So, for the same (reasons), PU is also the side of a pentagon. And QU is also (the side) of a pentagon. Thus, triangle QPU is equilateral. So, for the same (reasons), (triangles) QLR, RMS, SNT, and TOU are each also equilateral. And since QL and QP were each shown (to be the sides) of a pentagon, and LP is also (the side) of a pentagon, triangle QLP is thus equilateral. So, for the same (reasons), triangles LRM, MSN, NTO, and OUP are each also equilateral.



Εἰλήφθω τὸ κέντρον τοῦ ΕΖΗΘΚ κύκλου τὸ Φ σημεῖον· καὶ ἀπὸ τοῦ Φ τῷ τοῦ κύκλου ἐπιπέδῳ πρὸς ὀρθὰξ ἀνεστάτω ἡ ΦΩ, καὶ ἐκβεβλήσθω ἐπὶ τὰ ὁέτερα μέρη ὡξ ἡ ΦΨ, καὶ ἀφῃρήσθω ἑχαγώνου μὲν ἡ ΦΘ, δεκαγώνου δὲ ἑκατέρα τῶν ΦΨ, ΘΩ, καὶ ἐπεζεύθθωσαν αἱ ΠΩ, ΠΘ, ΥΩ, ΕΦ, ΛΦ, ΛΨ, ΨΜ.

Καὶ ἐπεὶ ἑκατέρα τῶν ΦΘ, ΠΕ τῷ τοῦ κύκλου ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν, παράλληλος ἄρα ἐστὶν ἡ ΦΘ τῇ ΠΕ. εἰσὶ

Let the center, point V, of circle $EFGHK$ be found [Prop. 3.1]. And let VZ be set up, at (point) V, at right-angles to the plane of the circle. And let it be produced on the other side (of the circle), like VX. And let VW be cut off (from XZ so as to be equal to the side) of a hexagon, and each of VX and WZ (so as to be equal to the side) of a decagon. And let QZ, QW, UZ, EV, LV, LX, and XM be joined.

And since VW and QE are each at right-angles to the plane

δὲ καὶ ἴσαι· καὶ αἱ ΕΦ, ΠΘ ἄρα ἴσαι τε καὶ παράλληλοί εἰσιν. ἑξαγώνου δὲ ἡ ΕΦ· ἑξαγώνου ἄρα καὶ ἡ ΠΘ. καὶ ἐπεὶ ἑξαγώνου μέν ἐστιν ἡ ΠΘ, δεκαγώνου δὲ ἡ ΘΩ, καὶ ὀρθή ἐστιν ἡ ὑπὸ ΠΘΩ γωνία, πενταγώνου ἄρα ἐστὶν ἡ ΠΩ. διὰ τὰ αὐτὰ δὴ καὶ ἡ ΥΩ πενταγώνου ἐστίν, ἐπειδήπερ, ἐὰν ἐπιζεύξωμεν τὰξ ΦΚ, ΘΥ, ἴσαι καὶ ἀπεναντίονὀέσονται, καί ἐστιν ἡ ΦΚ ἐκ τοῦ κέντρου οὀῦσα ἑξαγώνου. ἑξαγώνου ἄρα καὶ ἡ ΘΥ. δεκαγώνου δὲ ἡ ΘΩ, καὶ ὀρθὴ ἡ ὑπὸ ΥΘΩ· πενταγώνου ἄρα ἡ ΥΩ. ὀέστι δὲ καὶ ἡ ΠΥ πενταγώνου· ἰσόπλευρον ἄρα ἐστὶ τὸ ΠΥΩ τρίγωνον. διὰ τὰ αὐτὰ δὴ καὶ ὁέκαστον τῶν λοιπῶν τριγώνων, ὁῶν βάσεις μέν εἰσιν αἱ ΠΡ, ΡΣ, ΣΤ, ΤΥ εὐθεῖαι, κορυφὴ δὲ τὸ Ω σημεῖον, ἰσόπλευρόν ἐστιν. πάλιν, ἐπεὶ ἑξαγώνου μὲν ἡ ΦΛ, δεκαγώνου δὲ ἡ ΦΨ, καὶ ὀρθή ἐστιν ἡ ὑπὸ ΛΦΨ γωνία, πενταγώνου ἄρα ἐστὶν ἡ ΛΨ. διὰ τὰ αὐτὰ δὴ ἐὰν ἐπιζεύξωμεν τὴν ΜΦ οὀῦσαν ἑξαγώνου, συνάγεται καὶ ἡ ΜΨ πενταγώνου. ὀέστι δὲ καὶ ἡ ΛΜ πενταγώνου· ἰσόπλευρον ἄρα ἐστὶ τὸ ΛΜΨ τρίγωνον. ὁμοίως δὴ δειϑϑήσεται, ὅτι καὶ ὁέκαστον τῶν λοιπῶν τριγώνων, ὁῶν βάσεις μέν εἰσιν αἱ ΜΝ, ΝΧ, ΧΟ, ΟΛ, κορυφὴ δὲ τὸ Ψ σημεἰον, ἰσόπλευρόν ἐστιν. συνέσταται ἄρα εἰκοσάεδρον ὑπὸ εὀίκοσι τριγώνων ἰσοπλεύρων περιεχόμενον.

Δεῖ δὴ αὐτὸ καὶ σφαίρᾳ περιλαβεῖν τῇ δοθείσῃ καὶ δεῖξαι, ὅτι ἡ τοῦ εἰκοσαέδρου πλευρὰ ὀάλογόξ ἐστιν ἡ καλουμένη ἐλάσσων.

Ἐπεὶ γὰρ ἑξαγώνου ἐστὶν ἡ ΦΘ, δεκαγώνου δὲ ἡ ΘΩ, ἡ ΦΩ ἄραὀάκρον καὶ μέσον λόγον τέτμ ηται κατὰ τὸ Θ, καὶ τὸ μεῖζον αὐτῆς τμῆμά ἐστιν ἡ ΦΘ· ὀέστιν ἄρα ὡξ ἡ ΩΦ πρὸς τὴν ΦΘ, οὁύτως ἡ ΦΘ πρὸς τὴν ΘΩ. ὀίση δὲ ἡ μὲν ΦΘ τῇ ΦΕ, ἡ δὲ ΘΩ τῇ ΦΨ· ὀέστιν ἄρα ὡξ ἡ ΩΦ πρὸς τὴν ΦΕ, οὁύτως ἡ ΕΦ πρὸς τὴν ΦΨ. καί εἰσιν ὀρθαὶ αἱ ὑπὸ ΩΦΕ, ΕΦΨ γωνίαι· ἐὰν ἄρα ἐπιζεύξωμεν τὴν ΕΩ εὐθεἰαν, ὀρθὴ ὀέσται ἡ ὑπὸ ΨΕΩ γωνία διὰ τὴν ὁμοιότητα τῶν ΨΕΩ, ΦΕΩ τριγώνων. διὰ τὰ αὐτὰ δὴ ἐπεί ἐστιν ὡξ ἡ ΩΦ πρὸς τὴν ΦΘ, οὁύτως ἡ ΦΘ πρὸς τὴν ΘΩ, ὀίση δὲ ἡ μὲν ΩΦ τῇ ΨΘ, ἡ δὲ ΦΘ τῇ ΘΠ, ὀέστιν ἄρα ὡξ ἡ ΨΘ πρὸς τὴν ΘΠ, οὁύτως ἡ ΠΘ πρὸς τὴν ΘΩ. καὶ διὰ τοῦτο πάλιν ἐὰν ἐπιζεύξωμεν τὴν ΠΨ, ὀρθὴ ὀέσται ἡ πρὸς τῷ Π γωνία· τὸ ἄρα ἐπὶ τῆς ΨΩ γραφόμενον ἡμικύκλιον ἥξει καὶ δἰα τοῦ Π. καὶ ἐὰν μενούσης τῆς ΨΩ περιενεϑθὲν τὸ ἡμικύκλιον εἰξ τὸ αὐτὸ πάλιν ἀποκατασταθῇ, ὅθεν ἤρχατο φέρεσθαι, ἥξει καὶ διὰ τοῦ Π καὶ τῶν λοιπῶν σημείων τοῦ εἰκοσαέδρου, καὶ ὀέσται σφαίρᾳ περιειλημμένον τὸ εἰκοσάεδρον. λέγω δή, ὅτι καὶ τῇ δοθείσῃ. τετμήσθω γὰρ ἡ ΦΘ δίϑα κατὰ τὸ α. καὶ ἐπεὶ εὐθεῖα γραμμὴ ἡ ΦΩὀάκρον καὶ μέσον λόγον τέτμ ηται κατὰ τὸ Θ, καὶ τὸ ὀέλασσον αὐτῆς τμῆμά ἐστιν ἡ ΩΘ, ἡ ἄρα ΩΘ προσλαβοῦσα τὴν ἡμίσειαν τοῦ μείζονος τμήματος τὴν Θα πενταπλάσιον δύναται τοῦ ἀπὸ τῆς ἡμισείας τοῦ μείζονος τμήματος· πενταπλάσιον ἄρα ἐστὶ τὸ ἀπὸ τῆς Ωα τοῦ ἀπὸ τῆς αΘ. καί ἐστι τῆς μὲν Ωα διπλῆ ἡ ΩΨ, τὴξ δὲ αΘ διπλῆ ἡ ΦΘ· πενταπλάσιον ἄρα ἐστὶ τὸ ἀπὸ τῆς ΩΨ τοῦ ἀπὸ τῆς ΘΦ. καὶ ἐπεὶ τετραπλῆ ἐστιν ἡ ΑΓ τῆς ΓΒ, πενταπλῆ ἄρα ἐστὶν ἡ ΑΒ τῆς ΒΓ. ὡξ δὲ ἡ ΑΒ πρὸς τὴν ΒΓ, οὁύτως τὸ ἀπὸ τῆς ΑΒ πρὸς τὸ ἀπὸ τῆς ΒΔ· πενταπλάσιον ἄρα ἐστὶ

of the circle, *VW* is thus parallel to *QE* [Prop. 11.6]. And they are also equal. *EV* and *QW* are thus equal and parallel (to one another) [Prop. 1.33]. And *EV* (is the side) of a hexagon. Thus, *QW* (is) also (the side) of a hexagon. And since *QW* is (the side) of a hexagon, and *WZ* (the side) of a decagon, and angle *QWZ* is a right-angle [Def. 11.3, Prop. 1.29], *QZ* is thus (the side) of a pentagon [Prop. 13.10]. So, for the same (reasons), *UZ* is also (the side) of a pentagon—inasmuch as, if we join *VK* and *WU* then they will be equal and opposite. And *VK*, being (equal) to the radius (of the circle), is (the side) of a hexagon [Prop. 4.15 corr.]. Thus, *WU* (is) also the side of a hexagon. And *WZ* (is the side) of a decagon, and (angle) *UWZ* (is) a right-angle. Thus, *UZ* (is the side) of a pentagon [Prop. 13.10]. And *QU* is also (the side) of a pentagon. Triangle *QUZ* is thus equilateral. So, for the same (reasons), each of the remaining triangles, whose bases are the straight-lines *QR*, *RS*, *ST*, and *TU*, and apexes the point *Z*, are also equilateral. Again, since *VL* (is the side) of a hexagon, and *VX* (the side) of a decagon, and angle *LVX* is a right-angle, *LX* is thus (the side) of a pentagon [Prop. 13.10]. So, for the same (reasons), if we join *MV*, which is (the side) of a hexagon, *MX* is also inferred (to be the side) of a pentagon. And *LM* is also (the side) of a pentagon. Thus, triangle *LMX* is equilateral. So, similarly, it can be shown that each of the remaining triangles, whose bases are the (straight-lines) *MN*, *NO*, *OP*, and *PL*, and apexes the point *X*, are also equilateral. Thus, an icosahedron contained by twenty equilateral triangles has been constructed.

So, it is also necessary to enclose it in the given sphere, and to show that the side of the icosahedron is that irrational (straight-line) called minor.

For, since *VW* is (the side) of a hexagon, and *WZ* (the side) of a decagon, *VZ* has thus been cut in extreme and mean ratio at *W*, and *VW* is its greater piece [Prop. 13.9]. Thus, as *ZV* is to *VW*, so *VW* (is) to *WZ*. And *VW* (is) equal to *VE*, and *WZ* to *VX*. Thus, as *ZV* is to *VE*, so *EV* (is) to *VX*. And angles *ZVE* and *EVX* are right-angles. Thus, if we join straight-line *EZ* then angle *XEZ* will be a right-angle, on account of the similarity of triangles *XEZ* and *VEZ*. [Prop. 6.8]. So, for the same (reasons), since as *ZV* is to *VW*, so *VW* (is) to *WZ*, and *ZV* (is) equal to *XW*, and *VW* to *WQ*, thus as *XW* is to *WQ*, so *QW* (is) to *WZ*. And, again, on account of this, if we join *QX* then the angle at *Q* will be a right-angle [Prop. 6.8]. Thus, the semi-circle drawn on *XZ* will also pass through *Q* [Prop. 3.31]. And if *XZ* remains fixed, and the semi-circle is carried around, and again established at the same (position) from which it began to be moved, then it will also pass through (point) *Q*, and (through) the remaining (angular) points of the icosahedron. And the icosahedron will be enclosed by a sphere. So, I say that (it is) also (enclosed) by the given (sphere). For let *VW* be cut in half at *a*. And since the straight-line *VZ* has been cut in extreme and mean ratio at *W*, and *ZW* is its lesser piece, then

τὸ ἀπὸ τῆς ΑΒ τοῦ ἀπὸ τῆς ΒΔ. ἐδείϑϑη δὲ καὶ τὸ ἀπὸ τῆς ΩΨ πενταπλάσιον τοῦ ἀπὸ τῆς ΦΘ. καί ἐστινὀίση ἡ ΔΒ τῄ ΦΘ· ἑκατέρα γὰρ αὐτῶνὀίση ἐστὶ τῄ ἐκ τοῦ κέντρου τοῦ ΕΖΗΘΚ κύκλου· ὀίση ἄρα καὶ ἡ ΑΒ τῄ ΨΩ. καί ἐστιν ἡ ΑΒ ἡ τῆς δοθείσης σφαίρας διάμετρος· καὶ ἡ ΨΩ ἄραὀίση ἐστὶ τῄ τῆς δοθείσης σφαίρας διαμέτρῳ· τῄ ἄρα δοθείσῃ σφαίρᾳ περιείληπται τὸ εἰκοσάεδρον.

Λέγω δή, ὅτι ἡ τοῦ εἰκοσαέδρου πλευρὰ ὀάλογόξ ἐστιν ἡ καλουμένη ἐλάττων. ἐπεὶ γὰρ ὁρητή ἐστιν ἡ τῆς σφαίρας διάμετρος, καί ἐστι δυνάμει πενταπλασίων τῆς ἐκ τοῦ κέντρου τοῦ ΕΖΗΘΚ κύκλου, ὁρητὴ ἄρα ἐστὶ καὶ ἡ ἐκ τοῦ κέντρου τοῦ ΕΖΗΘΚ κύκλου· ὁώστε καὶ ἡ διάμετρος αὐτοῦ ὁρητή ἐστιν. ἐὰν δὲ εἰξ κύκλον ὁρητὴν ἔχοντα τὴν διάμετρον πεντάγωνον ἰσόπλευρον ἐγγραφῇ, ἡ τοῦ πενταγώνου πλευρὰ ὀάλογόξ ἐστιν ἡ καλουμένη ἐλάττων. ἡ δὲ τοῦ ΕΖΗΘΚ πενταγώνου πλευρὰ ἡ τοῦ εἰκοσαέδρου ἐστίν. ἡ ἄρα τοῦ είκοσαέδρου πλευρὰ ὀάλογόξ ἐστιν ἡ καλουμένη ἐλάττων.

the square on ZW added to half of the greater piece, Wa, is five times the (square) on half of the greater piece [Prop. 13.3]. Thus, the (square) on Za is five times the (square) on aW. And ZX is double Za, and VW double aW. Thus, the (square) on ZX is five times the (square) on WV. And since AC is four times CB, AB is thus five times BC. And as AB (is) to BC, so the (square) on AB (is) to the (square) on BD [Prop. 6.8, Def. 5.9]. Thus, the (square) on AB is five times the (square) on BD. And the (square) on ZX was also shown (to be) five times the (square) on VW. And DB is equal to VW. For each of them is equal to the radius of circle $EFGHK$. Thus, AB (is) also equal to XZ. And AB is the diameter of the given sphere. Thus, XZ is equal to the diameter of the given sphere. Thus, the icosahedron has been enclosed by the given sphere.

So, I say that the side of the icosahedron is that irrational (straight-line) called minor. For since the diameter of the sphere is rational, and the square on it is five times the (square) on the radius of circle $EFGHK$, the radius of circle $EFGHK$ is thus also rational. Hence, its diameter is also rational. And if an equilateral pentagon is inscribed in a circle having a rational diameter then the side of the pentagon is that irrational (straight-line) called minor [Prop. 13.11]. And the side of pentagon $EFGHK$ is (the side) of the icosahedron. Thus, the side of the icosahedron is that irrational (straight-line) called minor.

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει πενταπλασίων ἐστὶ τῆς ἐκ τοῦ κέντρου τοῦ κύκλου, ἀφ᾿ οὗ τὸ εἰκοσάεδρον ἀναγέγραπται, καὶ ὅτι ἡ τῆς σφαίρας διάμετρος σύγκειταιὀέκ τε τῆς τοῦ ἑχαγώνου καὶ δύο τῶν τοῦ δεκαγώνου τῶν εἰξ τὸν αὐτὸν κύκλον ἐγγραφομένων. ὅπερ ἔδει δεῖξαι.

Corollary

So, (it is) clear, from this, that the square on the diameter of the sphere is five times the square on the radius of the circle from which the icosahedron has been described, and that the the diameter of the sphere is the sum of (the side) of the hexagon, and two of (the sides) of the decagon, inscribed in the same circle.[†]

17.

Δωδεκάεδρον συστήσασθαι καὶ σφαίρᾳ περιλαβεῖν, ᾗ καὶ τὰ προειρημένα σϑήματα, καὶ δεῖξαι, ὅτι ἡ τοῦ δωδεκαέδρου πλευρὰ ὀάλογόξ ἐστιν ἡ καλουμένη ἀποτομή.

Ἐκκείσθωσαν τοῦ προειρημένου κύβου δύο ἐπίπεδα πρὸς ὀρθὰξ ἀλλήλοις τὰ ΑΒΓΔ, ΓΒΕΖ, καὶ τετμήσθω ἑκάστη τῶν ΑΒ, ΒΓ, ΓΔ, ΔΑ, ΕΖ, ΕΒ, ΖΓ πλευρῶν δίϑα κατὰ τὰ Η, Θ, Κ, Λ, Μ, Ν, Χ, καὶ ἐπεζεύϑθωσαν αἱ ΗΚ, ΘΛ, ΜΘ, ΝΧ, καὶ τετηήσθω ἑκάστη τῶν ΝΟ, ΟΧ, ΘΠὀάκρον καὶ μέσον λόγον κατὰ τὰ Ρ, Σ, Τ σημεῖα, καὶ ὀέστω αὐτῶν μείζονα τμήματα τὰ ΡΟ, ΟΣ, ΤΠ, καὶ ἀνεστάτωσαν ἀπὸ τῶν Ρ, Σ, Τ σημείων τοῖς τοῦ κύβου ἐπιπέδοις πρὸς ὀρθὰξ ἐπὶ τὰ ἐκτὸξ μέρη τοῦ κύβου αἱ ΡΥ, ΣΦ, ΤΘ, καὶ κείσθωσαν ἴσαι ταῖς ΡΟ, ΟΣ, ΤΠ, καὶ ἐπεζεύϑθωσαν αἱ ΥΒ, ΒΘ, ΘΓ, ΓΦ, ΦΥ.

Proposition 17

To construct a dodecahedron, and to enclose (it) in a sphere, like the aforementioned figures, and to show that the side of the dodecahedron is that irrational (straight-line) called an apotome.

Let two planes of the aforementioned cube [Prop. 13.15], $ABCD$ and $CBEF$, (which are) at right-angles to one another, be laid out. And let the sides AB, BC, CD, DA, EF, EB, and FC have each been cut in half at points G, H, K, L, M, N, and O (respectively). And let GK, HL, MH, and NO be joined. And let NP, PO, and HQ have each been cut in extreme and mean ratio at points R, S, and T (respectively). And let their greater pieces be RP, PS, and TQ (respectively). And let RU, SV, and TW be set up on the exterior side of the cube, at points

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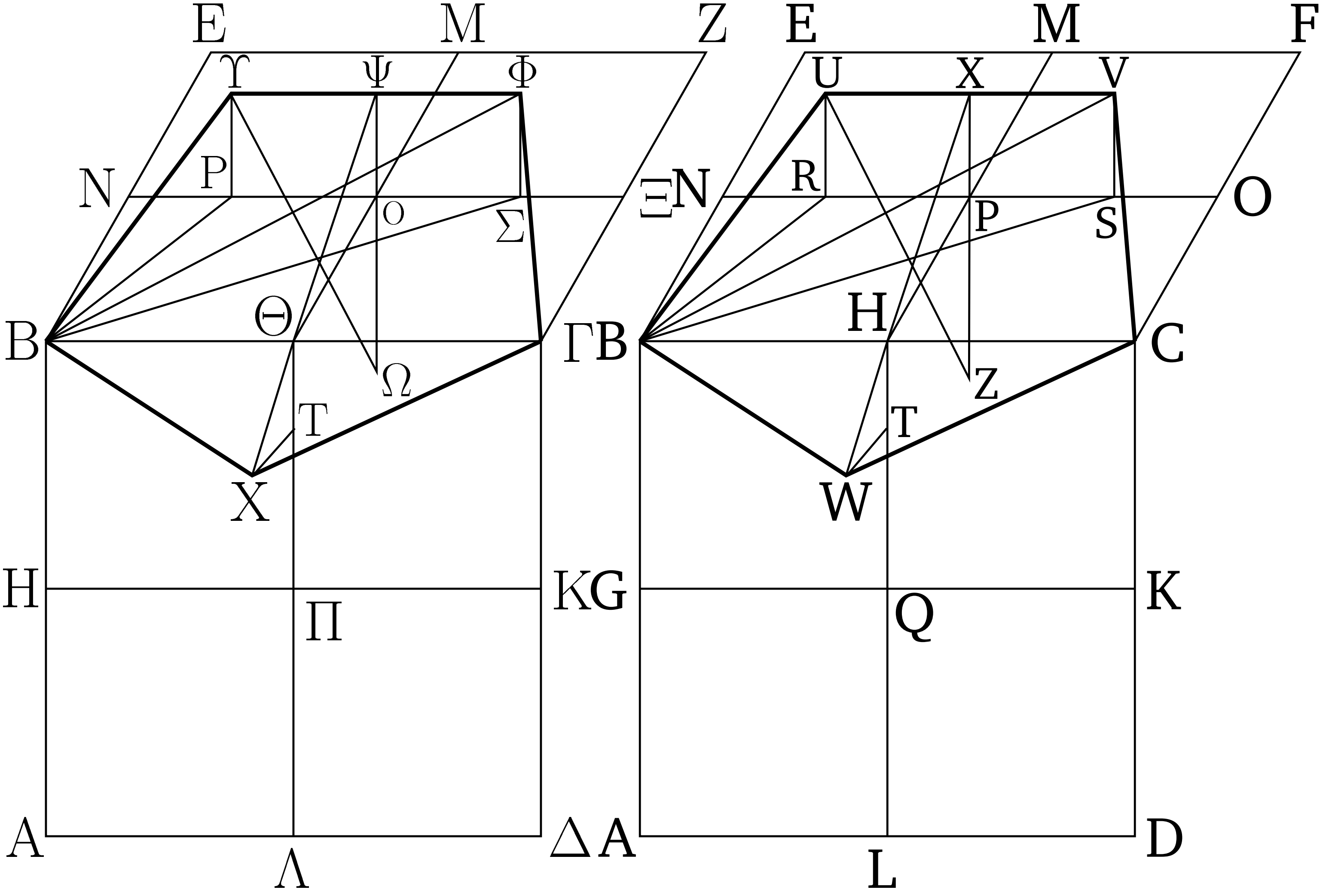
[†] If the radius of the sphere is unity then the radius of the circle is $2/\sqrt{5}$, and the sides of the hexagon, decagon, and pentagon/icosahedron are $2/\sqrt{5}$, $1-1/\sqrt{5}$, and $(1/\sqrt{5})\sqrt{10-2\sqrt{5}}$, respectively.

R, *S*, and *T* (respectively), at right-angles to the planes of the cube. And let them be made equal to *RP*, *PS*, and *TQ*. And let *UB*, *BW*, *WC*, *CV*, and *VU* be joined.



Λέγω, ὅτι τὸ ΥΒΘΓΦ πεντάγωνον ἰσόπλευρόν τε καὶ ἐν ἑνὶ ἐπιπέδῳ καὶ ὁέτι ἰσογώνιόν ἐστιν. ἐπεζεύχθωσαν γὰρ αἱ ΡΒ, ΣΒ, ΦΒ. καὶ ἐπεὶ εὐθεῖα ἡ ΝΟὀάκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ Ρ, καὶ τὸ μεῖζον τμῆμά ἐστιν ἡ ΡΟ, τὰ ἄρα ἀπὸ τῶν ΟΝ, ΝΡ τριπλάσιά ἐστι τοῦ ἀπὸ τῆς ΡΟ. ὀίση δὲ ἡ μὲν ΟΝ τῇ ΝΒ, ἡ δὲ ΟΡ τῇ ΡΥ· τὰ ἄρα ἀπὸ τῶν ΒΝ, ΝΡ τριπλάσιά ἐστι τοῦ ἀπὸ τῆς ΡΥ. τοῖς δὲ ἀπὸ τῶν ΒΝ, ΝΡ τὸ ἀπὸ τῆς ΒΡ ἐστιν ἴσον· τὸ ἄρα ἀπὸ τῆς ΒΡ τριπλάσιόν ἐστι τοῦ ἀπὸ τῆς ΡΥ· ὁώστε τὰ ἀπὸ τῶν ΒΡ, ΡΥ τετραπλάσιά ἐστι τοῦ ἀπὸ τῆς ΡΥ. τοῖς δὲ ἀπὸ τῶν ΒΡ, ΡΥ ἴσον ἐστι τὸ ἀπὸ τῆς ΒΥ· τὸ ἄραὀάπὸ τῆς ΒΥ τετραπλάσιόν ἐστι τοῦ ἀπὸ τῆς ΥΡ· διπλῆ ἄρα ἐστὶν ἡ ΒΥ τῆς ΡΥ. ὀέστι δὲ καὶ ἡ ΦΥ τῆς ΥΡ διπλῆ, ἐπειδήπερ καὶ ἡ ΣΡ τῆς ΟΡ, τουτέστι τῆς ΡΥ, ἐστι διπλῆ· ὀίση ἄρα ἡ ΒΥ τῇ ΥΦ. ὁμοίως δὴ δειχθήσεται, ὅτι καὶ ἑκάστη τῶν ΒΘ, ΘΓ, ΓΦ ἑκατέρᾳ τῶν ΒΥ, ΥΦ ἐστινὀίση.

I say that the pentagon *UBWCV* is equilateral, and in one plane, and, further, equiangular. For let *RB*, *SB*, and *VB* be joined. And since the straight-line *NP* has been cut in extreme and mean ratio at *R*, and *RP* is the greater piece, the (sum of the squares) on *PN* and *NR* is thus three times the (square) on *RP* [Prop. 13.4]. And *PN* (is) equal to *NB*, and *PR* to *RU*. Thus, the (sum of the squares) on *BN* and *NR* is three times the (square) on *RU*. And the (square) on *BR* is equal to the (sum of the squares) on *BN* and *NR* [Prop. 1.47]. Thus, the (square) on *BR* is three times the (square) on *RU*. Hence, the (sum of the squares) on *BR* and *RU* is four times the (square) on *RU*. And the (square) on *BU* is equal to the (sum of the squares) on *BR* and *RU* [Prop. 1.47]. Thus, the (square) on *BU* is four times the (square) on *UR*. Thus, *BU* is double *RU*. And *VU* is also double *UR*, inasmuch as *SR* is also double *PR*—that is

ἰσόπλευρον ἄρα ἐστὶ τὸ ΒΥΦΓΘ πεντάγωνον. λέγω δή, ὅτι καὶ ἐν ἑνί ἐστιν ἐπιπέδῳ. ἤθθω γὰρ ἀπὸ τοῦ Ο ἑκατέρᾳ τῶν ΡΥ, ΣΦ παράλληλος ἐπὶ τὰ ἐκτὸξ τοῦ κύβου μέρη ἡ ΟΨ, καὶ ἐπεζεύθθωσαν αἱ ΨΘ, ΘΘ· λέγω, ὅτι ἡ ΨΘΘ εὐθεῖά ἐστιν. ἐπεὶ γὰρ ἡ ΘΠὀάκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ Τ, καὶ τὸ μεῖζον αὐτῆς τμῆμά ἐστιν ἡ ΠΤ, ὀέστιν ἄρα ὥξ ἡ ΘΠ πρὸς τὴν ΠΤ, οὁύτως ἡ ΠΤ πρὸς τὴν ΤΘ. ὀίση δὲ ἡ μὲν ΘΠ τῇ ΘΟ, ἡ δὲ ΠΤ ἑκατέρᾳ τῶν ΤΘ, ΟΨ· ὀέστιν ἄρα ὥξ ἡ ΘΟ πρὸς τὴν ΟΨ, οὁύτως ἡ ΘΤ πρὸς τὴν ΤΘ. καί ἐστι παράλληλος ἡ μὲν ΘΟ τῇ ΤΘ· ἑκατέρα γὰρ αὐτῶν τῷ ΒΔ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν· ἡ δὲ ΤΘ τῇ ΟΨ· ἑκατέρα γὰρ αὐτῶν τῷ ΒΖ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν. ἔαν δὲ δύο τρίγωνα συντεθῇ κατὰ μίαν γωνίαν, ὥξ τὰ ΨΟΘ, ΘΤΘ, τὰξ δύο πλευρὰξ ταῖς δυνὶν ἀνάλογον ἔχοντα, ὁώστε τὰξ ὁμολόγους αὐτῶν πλευρὰξ καὶ παραλλήλους εὀῖναι, αἱ λοιπαὶ εὐθεῖαι ἐπ′ εὐθείαςὀέσονται· ἐπ′ εὐθείας ἄρα ἐστὶν ἡ ΨΘ τῇ ΘΘ. πᾶσα δὲ εὐθεῖα ἐν ἑνί ἐστιν ἐπιπέδῳ· ἐν ἑνὶ ἄρα ἐπιπέδῳ ἐστὶ τὸ ΥΒΘΓΦ πεντάγωνον.

Λέγω δή, ὅτι καὶ ἰσογώνιόν ἐστιν.

Ἐπεὶ γὰρ εὐθεῖα γραμμὴ ἡ ΝΟὀάκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ Ρ, καὶ τὸ μεῖζον τμῆμά ἐστιν ἡ ΟΡ [ὀέστιν ἄρα ὥξ συναμφότερος ἡ ΝΟ, ΟΡ πρὸς τὴν ΟΝ, οὁύτως ἡ ΝΟ πρὸς τὴν ΟΡ], ὀίση δὲ ἡ ΟΡ τῇ ΟΣ [ὀέστιν ἄρα ὥξ ἡ ΣΝ πρὸς τὴν ΝΟ, οὁύτως ἡ ΝΟ πρὸς τὴν ΟΣ], ἡ ΝΣ ἄραὀάκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ Ο, καὶ τὸ μεῖζον τμῆμά ἐστιν ἡ ΝΟ· τὰ ἄρα ἀπὸ τῶν ΝΣ, ΣΟ τριπλάσιά ἐστι τοῦ ἀπὸ τῆς ΝΟ. ὀίση δὲ ἡ μὲν ΝΟ τῇ ΝΒ, ἡ δὲ ΟΣ τῇ ΣΦ· τὰ ἄρα ἀπὸ τῶν ΝΣ, ΣΦ τετράγωνα τριπλάσιά ἐστι τοῦ ἀπὸ τῆς ΝΒ· ὁώστε τὰ ἀπὸ τῶν ΦΣ, ΣΝ, ΝΒ τετραπλάσιά ἐστι τοῦ ἀπὸ τῆς ΝΒ. τοῖς δὲ ἀπὸ τῶν ΣΝ, ΝΒ ἴσον ἐστὶ τὸ ἀπὸ τῆς ΣΒ· τὰ ἄρα ἀπὸ τῶν ΒΣ, ΣΦ, τουτέστι τὸ ἀπὸ τῆς ΒΦ [ὀρθὴ γὰρ ἡ ὑπὸ ΦΣΒ γωνία], τετραπλάσιόν ἐστι τοῦ ἀπὸ τῆς ΝΒ· διπλῆ ἄρα ἐστὶν ἡ ΦΒ τῆς ΒΝ. ὀέστι δὲ καὶ ἡ ΒΓ τῆς ΒΝ διπλῆ· ὀίση ἄρα ἐστὶν ἡ ΒΦ τῇ ΒΓ. καὶ ἐπεὶ δύο αἱ ΒΥ, ΥΦ δυσὶ ταῖς ΒΘ, ΘΓ ἴσαι εἰσίν, καὶ βάσις ἡ ΒΦ βάσει τῇ ΒΓὀίση, γωνία ἄρα ἡ ὑπὸ ΒΥΦ γωνίᾳ τῇ ὑπὸ ΒΘΓ ἐστινὀίση. ὁμοίως δὴ δείχομεν, ὅτι καὶ ἡ ὑπὸ ΥΦΓ γωνίαὀίση ἐστὶ τῇ ὑπὸ ΒΘΓ· αἱ ἄρα ὑπὸ ΒΘΓ, ΒΥΦ, ΥΦΓ τρεῖς γωνίαι ἴσαι ἀλλήλαις εἰσίν. ἔαν δὲ πενταγώνου ἰσοπλεύρου αἱ τρεῖς γωνίαι ἴσαι ἀλλήλαις ὦσιν, ἰσογώνιονὀέσται τὸ πεντάγωνον· ἰσογώνιον ἄρα ἐστὶ τὸ ΒΥΦΓΘ πεντάγωνον. ἐδείθθη δὲ καὶ ἰσόπλευρον· τὸ ἄρα ΒΥΦΓΘ πεντάγωνον ἰσόπλευρόν ἐστι καὶ ἰσογώνιον, καί ἐστιν ἐπὶ μιᾶς τοῦ κύβου πλευρᾶξ τῆς ΒΓ. ἔαν ἄρα ἐφ′ ἑκάστης τῶν τοῦ κύβου δώδεκα πλευρῶν τὰ αὐτὰ κατασκευάσωμεν, συσταθήσεταί τι σχῆμα στερεὸν ὑπὸ δώδεκα πενταγώνων ἰσοπλεύρων τε καὶ ἰσογωνίων περιεχόμενον, ὅ καλεῖται δωδεκάεδρον.

Δεῖ δὴ αὐτὸ καὶ σφαίρᾳ περιλαβεῖν τῇ δοθείσῃ καὶ δεῖχαι, ὅτι ἡ τοῦ δωδεκαέδρου πλευρὰ ὀάλογόξ ἐστιν ἡ καλουμένη ἀποτομή.

Ἐκβεβλήσθω γὰρ ἡ ΨΟ, καὶ ὀέστω ἡ ΨΩ· συμβάλλει ἄρα

to say, RU. Thus, BU (is) equal to UV. So, similarly, it can be shown that each of BW, WC, CV is equal to each of BU and UV. Thus, pentagon $BUVCW$ is equilateral. So, I say that it is also in one plane. For let PX be drawn from P, parallel to each of RU and SV, on the exterior side of the cube. And let XH and HW be joined. I say that XHW is a straight-line. For since HQ has been cut in extreme and mean ratio at T, and QT is its greater piece, thus as HQ is to QT, so QT (is) to TH. And HQ (is) equal to HP, and QT to each of TW and PX. Thus, as HP is to PX, so WT (is) to TH. And HP is parallel to TW. For of each of them is at right-angles to the plane BD [Prop. 11.6]. And TH (is parallel) to PX. For each of them is at right-angles to the plane BF [Prop. 11.6]. And if two triangles, like XPH and HTW, having two sides proportional to two sides, are placed together at a single angle such that their corresponding sides are also parallel then the remaining sides will be straight-on (to one another) [Prop. 6.32]. Thus, XH is straight-on to HW. And every straight-line is in one plane [Prop. 11.1]. Thus, pentagon $UBWCV$ is in one plane.

So, I say that it is also equiangular.

For since the straight-line NP has been cut in extreme and mean ratio at R, and PR is the greater piece [thus as the sum of NP and PR is to PN, so NP (is) to PR], and PR (is) equal to PS [thus as SN is to NP, so NP (is) to PS], NS has thus also been cut in extreme and mean ratio at P, and NP is the greater piece [Prop. 13.5]. Thus, the (sum of the squares) on NS and SP is three times the (square) on NP [Prop. 13.4]. And NP (is) equal to NB, and PS to SV. Thus, the (sum of the) squares on NS and SV is three times the (square) on NB. Hence, the (sum of the squares) on VS, SN, and NB is four times the (square) on NB. And the (square) on SB is equal to the (sum of the squares) on SN and NB [Prop. 1.47]. Thus, the (sum of the squares) on BS and SV—that is to say, the (square) on BV [for angle VSB (is) a right-angle]—is four times the (square) on NB [Def. 11.3, Prop. 1.47]. Thus, VB is double BN. And BC (is) also double BN. Thus, BV is equal to BC. And since the two (straight-lines) BU and UV are equal to the two (straight-lines) BW and WC (respectively), and the base BV (is) equal to the base BC, angle BUV is thus equal to angle BWC [Prop. 1.8]. So, similarly, we can show that angle UVC is equal to angle BWC. Thus, the three angles BWC, BUV, and UVC are equal to one another. And if three angles of an equilateral pentagon are equal to one another then the pentagon is equiangular [Prop. 13.7]. Thus, pentagon $BUVCW$ is equiangular. And it was also shown (to be) equilateral. Thus, pentagon $BUVCW$ is equilateral and equiangular, and it is on one of the sides, BC, of the cube. Thus, if we make the same construction on each of the twelve sides of the cube then some solid figure contained by twelve equilateral and equiangular pentagons will be constructed, which is called a dodecahedron.

So, it is necessary to enclose it in the given sphere, and

ἡ ΟΩ τῇ τοῦ κύβου διαμέτρῳ, καὶ δίθα τέμνουσιν ἀλλήλας· τοῦτο γὰρ δέδεικται ἐν τῷ παρατελεύτῳ θεωρήματι τοῦ ἑνδεκάτου βιβλίου. τεμνέτωσαν κατὰ τὸ Ω· τὸ Ω ἄρα κέντρον ἐστὶ τῆς σφαίρας τῆς περιλαμβανούσης τὸν κύβον, καὶ ἡ ΩΟ ἡμίσεια τῆς πλευρᾶξ τοῦ κύβου. ἐπεζεύθθω δὴ ἡ ΥΩ. καὶ ἐπεὶ εὐθεῖα γραμμὴ ἡ ΝΣὀάκρον καὶ μέσον λόγον τέτμ ηται κατὰ τὸ Ο, καὶ τὸ μεῖζον αὐτῆς τμῆμά ἐστιν ἡ ΝΟ, τὰ ἄρα ἀπὸ τῶν ΝΣ, ΣΟ τριπλάσιά ἐστι τοῦ ἀπὸ τῆς ΝΟ. ὀίση δὲ ἡ μὲν ΝΣ τῇ ΨΩ, ἐπειδήπερ καὶ ἡ μὲν ΝΟ τῇ ΟΩ ἐστινὀίση, ἡ δὲ ΨΟ τῇ ΟΣ. ἀλλὰ μὴν καὶ ἡ ΟΣ τῇ ΨΥ, ἐπεὶ καὶ τῇ ΡΟ· τὰ ἄρα ἀπὸ τῶν ΩΨ, ΨΥ τριπλάσιά ἐστι τοῦ ἀπὸ τῆς ΝΟ. τοῖς δὲ ἀπὸ τῶν ΩΨ, ΨΥ ἴσον ἐστὶ τὸ ἀπὸ τῆς ΥΩ· τὸ ἄρα ἀπὸ τῆς ΥΩ τριπλάσιόν ἐστι τοῦ ἀπὸ τῆς ΝΟ. ὀέστι δὲ καὶ ἡ ἐκ τοῦ κέντρου τῆς σφαίρας τῆς περιλαμβανούσης τὸν κύβον δυνάμει τριπλασίων τῆς ἡμισείας τῆς τοῦ κύβου πλευρᾶξ· προδέδεικται γὰρ κύβον συστήσασθαι καὶ σφαίρᾳ περιλαβεῖν καὶ δεῖξαι, ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει τριπλασίων ἐστὶ τῆς πλευρᾶξ τοῦ κύβου. εἰ δὲ ὅλη τῆς ὅλης, καὶ [ἡ] ἡμίσεια τῆς ἡμισείας· καί ἐστιν ἡ ΝΟ ἡμίσεια τῆς τοῦ κύβου πλευρᾶξ· ἡ ἄρα ΥΩὀίση ἐστὶ τῇ ἐκ τοῦ κέντρου τῆς σφαίρας τῆς περιλαμβανούσης τὸν κύβον. καί ἐστι τὸ Ω κέντρον τῆς σφαίρας τῆς περιλαμβανούσης τὸν κύβον· τὸ Υ ἄρα σημεῖον πρὸς τῇ ἐπιφανείᾳ ἐστι τῆς σφαίρας. ὁμοίως δὴ δείξομεν, ὅτι καὶ ἑκάστη τῶν λοιπῶν γωνιῶν τοῦ δωδεκαέδρου πρὸς τῇ ἐπιφανείᾳ ἐστὶ τῆς σφαίρας· περιείληπται ἄρα τὸ δωδεκαέδρον τῇ δοθείσῃ σφαίρᾳ.

Λέγω δή, ὅτι ἡ τοῦ δωδεκαέδρου πλευρὰ ὀάλογόξ ἐστιν ἡ καλουμένη ἀποτομή.

Ἐπεὶ γὰρ τῆς ΝΟὀάκρον καὶ μέσον λόγον τετμ ημένης τὸ μεῖζον τμῆμά ἐστιν ὁ ΡΟ, τῆς δὲ ΟΧὀάκρον καὶ μέσον λόγον τετμ ημένης τὸ μεῖζον τμῆμά ἐστιν ἡ ΟΣ, ὅλης ἄρα τῆς ΝΧὀάκρον καὶ μέσον λόγον τεμνομένης τὸ μεῖζον τμῆμά ἐστιν ἡ ΡΣ. [οὁῖον ἐπεί ἐστιν ὡ̔ξ ἡ ΝΟ πρὸς τὴν ΟΡ, ἡ ΟΡ πρὸς τὴν ΡΝ, καὶ τὰ διπλάσια· τὰ γὰρ μέρη τοῖς ἰσάκις πολλαπλασίοις τὸν αὐτὸν ἔχει λόγον· ὡ̔ξ ἄρα ἡ ΝΧ πρὸς τὴν ΡΣ, οὁύτως ἡ ΡΣ πρὸς συναμφότερον τὴν ΝΡ, ΣΧ. μείζων δὲ ἡ ΝΧ τῆς ΡΣ· μείζων ἄρα καὶ ἡ ΡΣ συναμφοτέρου τῆς ΝΡ, ΣΧ· ἡ ΝΧ ἄραὀάκρον καὶ μέσον λόγον τέτμ ηται, καὶ τὸ μεῖζον αὐτῆς τμῆμά ἐστιν ἡ ΡΣ.] ὀίση δὲ ἡ ΡΣ τῇ ΥΦ· τῆς ἄρα ΝΧὀάκρον καὶ μέσον λόγον τεμνομένης τὸ μεῖζον τμῆμά ἐστιν ἡ ΥΦ. καὶ ἐπεὶ ὁρητή ἐστιν τῆς σφαίρας διάμετρος καί ἐστι δυνάμει τριπλασίων τῆς τοῦ κύβου πλευρᾶξ, ὁρητὴ ἄρα ἐστὶν ἡ ΝΧ πλευρὰ οὀῦσα τοῦ κύβου. ἔαν δὲ ὁρητὴ γραμμὴ ὀάκρον καὶ μέσον λόγον τμ ηθῇ, ἑκάτερον τῶν τμ ημάτωνὀάλογόξ ἐστιν ἀποτομή.

Ἡ ΥΦ ἄρα πλευρὰ οὀῦσα τοῦ δωδεκαέδρουὀάλογόξ ἐστιν ἀποτομή.

to show that the side of the dodecahedron is that irrational (straight-line) called an apotome.

For let *XP* be produced, and let (the produced straight-line) be *XZ*. Thus, *PZ* meets the diameter of the cube, and they cut one another in half. For, this has been proved in the penultimate theorem of the eleventh book [Prop. 11.38]. Let them cut (one another) at *Z*. Thus, *Z* is the center of the sphere enclosing the cube, and *ZP* (is) half the side of the cube. So, let *UZ* be joined. And since the straight-line *NS* has been cut in extreme and mean ratio at *P*, and its greater piece is *NP*, the (sum of the squares) on *NS* and *SP* is thus three times the (square) on *NP* [Prop. 13.4]. And *NS* (is) equal to *XZ*, inasmuch as *NP* is also equal to *PZ*, and *XP* to *PS*. But, indeed, *PS* (is) also (equal) to *XU*, since (it is) also (equal) to *RP*. Thus, the (sum of the squares) on *ZX* and *XU* is three times the (square) on *NP*. And the (square) on *UZ* is equal to the (sum of the squares) on *ZX* and *XU* [Prop. 1.47]. Thus, the (square) on *UZ* is three times the (square) on *NP*. And the square on the radius of the sphere enclosing the cube is also three times the (square) on half the side of the cube. For it has previously been demonstrated (how to) construct the cube, and to enclose (it) in a sphere, and to show that the square on the diameter of the sphere is three times the (square) on the side of the cube [Prop. 13.15]. And if the (square on the) whole (is three times) the (square on the) whole, then the (square on the) half (is) also (three times) the (square on the) half. And *NP* is half of the side of the cube. Thus, *UZ* is equal to the radius of the sphere enclosing the cube. And *Z* is the center of the sphere enclosing the cube. Thus, point *U* is on the surface of the sphere. So, similarly, we can show that each of the remaining angles of the dodecahedron is also on the surface of the sphere. Thus, the dodecahedron has been enclosed by the given sphere.

So, I say that the side of the dodecahedron is that irrational straight-line called an apotome.

For since *RP* is the greater piece of *NP*, which has been cut in extreme and mean ratio, and *PS* is the greater piece of *PO*, which has been cut in extreme and mean ratio, *RS* is thus the greater piece of the whole of *NO*, which has been cut in extreme and mean ratio. [Thus, since as *NP* is to *PR*, (so) *PR* (is) to *RN*, and (the same is also true) of the doubles. For parts have the same ratio as similar multiples (taken in corresponding order) [Prop. 5.15]. Thus, as *NO* (is) to *RS*, so *RS* (is) to the sum of *NR* and *SO*. And *NO* (is) greater than *RS*. Thus, *RS* (is) also greater than the sum of *NR* and *SO* [Prop. 5.14]. Thus, *NO* has been cut in extreme and mean ratio, and *RS* is its greater piece.] And *RS* (is) equal to *UV*. Thus, *UV* is the greater piece of *NO*, which has been cut in extreme and mean ratio. And since the diameter of the sphere is rational, and the square on it is three times the (square) on the side of the cube, *NO*, which is the side of the cube, is thus rational. And if a rational (straight)-line is cut in extreme and

mean ratio then each of the pieces is the irrational (straight-line called) an apotome.

Thus, UV, which is the side of the dodecahedron, is the irrational (straight-line called) an apotome [Prop. 13.6].

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι τῆς τοῦ κύβου πλευρᾶξὀάκρον καὶ μέσον λόγον τεμνομένης τὸ μεῖζον τμῆμά ἐστιν ἡ τοῦ δωδεκαέδρου πλευρά. ὅπερ ἔδει δεῖχαι.

Corollary

So, (it is) clear, from this, that the side of the dodecahedron is the greater piece of the side of the cube, when it is cut in extreme and mean ratio.[†] (Which is) the very thing it was required to show.

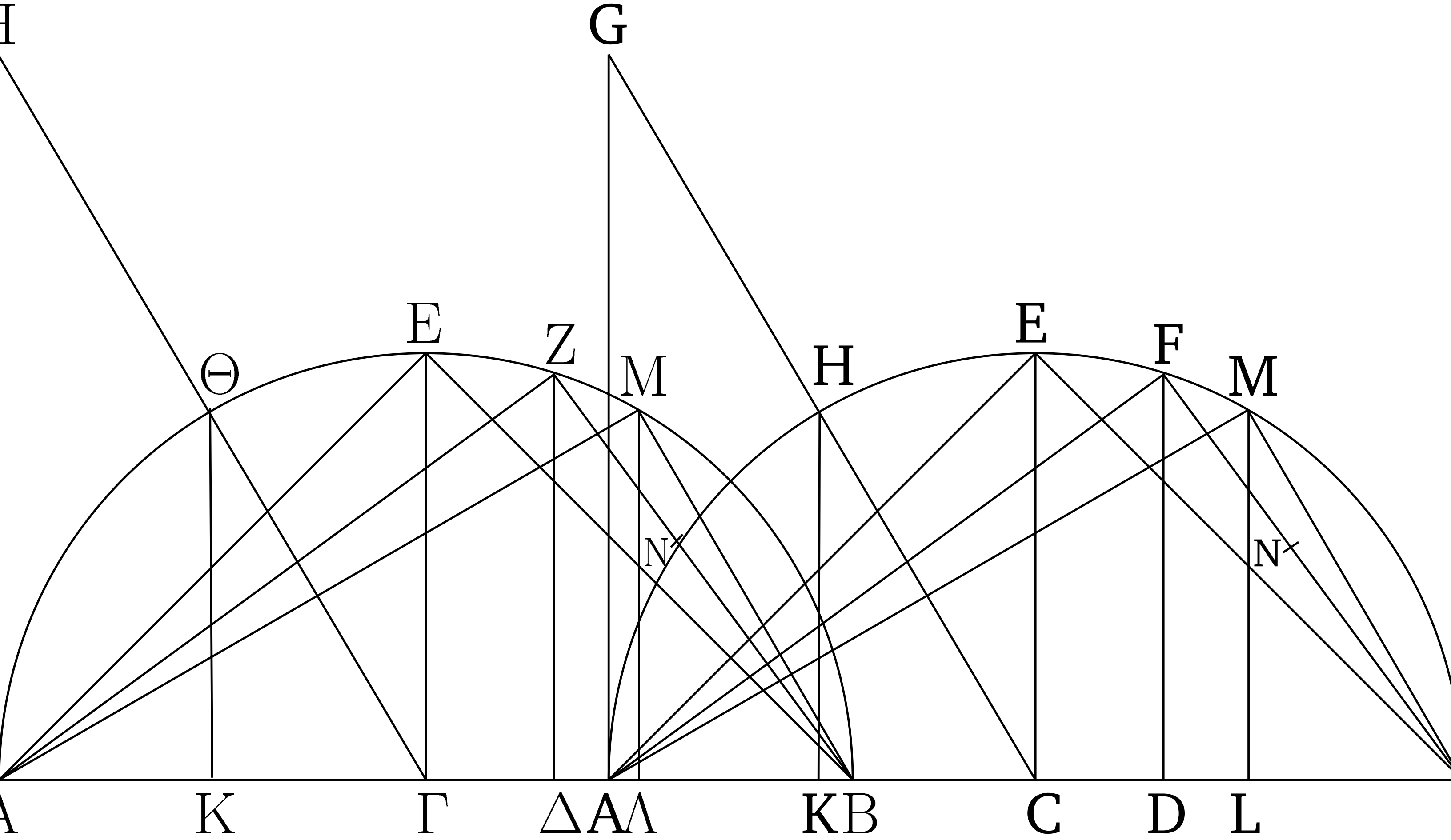
[†] If the radius of the circumscribed sphere is unity then the side of the cube is $\sqrt{4/3}$, and the side of the dodecahedron is $(1/3)(\sqrt{15}-\sqrt{3})$.

18.

Τὰξ πλευρὰξ τῶν πέντε σχημάτων ἐκθέσθαι καὶ συγκρῖναι πρὸς ἀλλήλας.

Proposition 18

To set out the sides of the five (aforementioned) figures, and to compare (them) with one another.[†]



Ἐκκείσθω ἡ τῆς δοθείσης σφαίρας διάμετρος ἡ AB, καὶ τετμήσθω κατὰ τὸ Γ ὁώστε ἴσην εὀῖναι τὴν AΓ τῇ ΓB, κατὰ δὲ τὸ Δ ὁώστε διπλασίονα εὀῖναι τὴν AΔ τῆς ΔB, καὶ γεγράφθω ἐπὶ τῆς AB ἡμικύκλιον τὸ AEB, καὶ ἀπὸ τῶν Γ, Δ τῇ AB

Let the diameter, AB, of the given sphere be laid out. And let it be cut at C, such that AC is equal to CB, and at D, such that AD is double DB. And let the semi-circle AEB be drawn on AB. And let CE and DF be drawn from C and D (respec-

πρὸς ὀρθὰξ ἤθθωσαν αἵ ΓΕ, ΔΖ, καὶ ἐπεζεύθθωσαν αἵ ΑΖ, ΖΒ, ΕΒ. καὶ ἐπεὶ διπλῆ ἐστιν ἡ ΑΔ τῆς ΔΒ, τριπλῆ ἄρα ἐστὶν ἡ ΑΒ τῆς ΒΔ. ἀναστρέψαντι ἡμιολία ἄρα ἐστὶν ἡ ΒΑ τῆς ΑΔ. ὡξ δὲ ἡ ΒΑ πρὸς τὴν ΑΔ, οὑύτως τὸ ἀπὸ τῆς ΒΑ πρὸς τὸ ἀπὸ τῆς ΑΖ· ἰσογώνιον γάρ ἐστι τὸ ΑΖΒ τρίγωνον τῷ ΑΖΔ τριγώνῳ· ἡμιόλιον ἄρα ἐστὶ τὸ ἀπὸ τῆς ΒΑ τοῦ ἀπὸ τῆς ΑΖ. ὁέστι δὲ καὶ ἡ τῆς σφαίρας διάμετρος δυνάμει ἡμιολία τῆς πλευρᾶξ τῆς πυραμίδος. καί ἐστιν ἡ ΑΒ ἡ τῆς σφαίρας διάμετρος· ἡ ΑΖ ἄραὀίση ἐστὶ τῇ πλευρᾷ τῆς πυραμίδος.

Πάλιν, ἐπεὶ διπλασίων ἐστὶν ἡ ΑΔ τῆς ΔΒ, τριπλῆ ἄρα ἐστὶν ἡ ΑΒ τῆς ΒΔ. ὡξ δὲ ἡ ΑΒ πρὸς τὴν ΒΔ, οὑύτως τὸ ἀπὸ τῆς ΑΒ πρὸς τὸ ἀπὸ τῆς ΒΖ· τριπλάσιον ἄρα ἐστὶ τὸ ἀπὸ τῆς ΑΒ τοῦ ἀπὸ τῆς ΒΖ. ὁέστι δὲ καὶ ἡ τῆς σφαίρας διάμετρος δυνάμει τριπλασίων τῆς τοῦ κύβου πλευρᾶξ. καί ἐστιν ἡ ΑΒ ἡ τῆς σφαίρας διάμετρος· ἡ ΒΖ ἄρα τοῦ κύβου ἐστὶ πλευρά.

Καὶ ἐπεὶ ὀίση ἐστὶν ἡ ΑΓ τῇ ΓΒ, διπλῆ ἄρα ἐστὶν ἡ ΑΒ τῆς ΒΓ. ὡξ δὲ ἡ ΑΒ πρὸς τὴν ΒΓ, οὑύτως τὸ ἀπὸ τῆς ΑΒ πρὸς τὸ ἀπὸ τῆς ΒΕ· διπλάσιον ἄρα ἐστὶ τὸ ἀπὸ τῆς ΑΒ τοῦ ἀπὸ τῆς ΒΕ. ὁέστι δὲ καὶ ἡ τῆς σφαίρας διάμετρος δυνάμει διπλασίων τῆς τοῦ ὀκταέδρου πλευρᾶξ. καὶ ἐστιν ἡ ΑΒ ἡ τῆς δοθείσης σφαίρας διάμετρος· ἡ ΒΕ ἄρα τοῦ ὀκταέδρου ἐστὶ πλευρά.

ὀΉθθω δὴ ἀπὸ τοῦ Α σημείου τῇ ΑΒ εὐθείᾳ πρὸς ὀρθὰξ ἡ ΑΗ, καὶ κείσθω ἡ ΑΗὀίση τῇ ΑΒ, καὶ ἐπεζεύθθω ἡ ΗΓ, καὶ ἀπὸ τοῦ Θ ἐπὶ τὴν ΑΒ κάθετος ἤθθω ἡ ΘΚ. καὶ ἐπεὶ διπλῆ ἐστιν ἡ ΗΑ τῆς ΑΓ· ὀίση γὰρ ἡ ΗΑ τῇ ΑΒ· ὡξ δὲ ἡ ΗΑ πρὸς τὴν ΑΓ, οὑύτως ἡ ΘΚ πρὸς τὴν ΚΓ, διπλῆ ἄρα καὶ ἡ ΘΚ τῆς ΚΓ. τετραπλάσιον ἄρα ἐστὶ τὸ ἀπὸ τῆς ΘΚ τοῦ ἀπὸ τῆς ΚΓ· τὰ ἄρα ἀπὸ τῶν ΘΚ, ΚΓ, ὅπερ ἐστὶ τὸ ἀπὸ τῆς ΘΓ, πενταπλάσιόν ἐστι τοῦ ἀπὸ τῆς ΚΓ. ὀίση δὲ ἡ ΘΓ τῇ ΓΒ· πενταπλάσιον ἄρα ἐστὶ τὸ ἀπὸ τῆς ΒΓ τοῦ ἀπὸ τῆς ΓΚ. καὶ ἐπεὶ διπλῆ ἐστιν ἡ ΑΒ τῆς ΓΒ, ὁῶν ἡ ΑΔ τῆς ΔΒ ἐστι διπλῆ, λοιπὴ ἄρα ἡ ΒΔ λοιπῆς τῆς ΔΓ ἐστι διπλῆ. τριπλῆ ἄρα ἡ ΒΓ τῆς ΓΔ· ἐνναπλάσιον ἄρα τὸ ἀπὸ τῆς ΒΓ τοῦ ἀπὸ τῆς ΓΔ. πενταπλάσιον δὲ τὸ ἀπὸ τῆς ΒΓ τοῦ ἀπὸ τῆς ΓΚ· μεῖζον ἄρα τὸ ἀπὸ τῆς ΓΚ τοῦ ἀπὸ τῆς ΓΔ. μείζων ἄρα ἐστὶν ἡ ΓΚ τῆς ΓΔ. κείσθω τῇ ΓΚὀίση ἡ ΓΛ, καὶ ἀπὸ τοῦ Λ τῇ ΑΒ πρὸς ὀρθὰξ ἤθθω ἡ ΛΜ, καὶ ἐπεζεύθθω ἡ ΜΒ. καὶ ἐπεὶ πενταπλάσιόν ἐστι τὸ ἀπὸ τῆς ΒΓ τοῦ ἀπὸ τῆς ΓΚ, καί ἐστι τῆς μὲν ΒΓ διπλῆ ἡ ΑΒ, τῆς δὲ ΓΚ διπλῆ ἡ ΚΛ, πενταπλάσιον ἄρα ἐστὶ τὸ ἀπὸ τῆς ΑΒ τοῦ ἀπὸ τῆς ΚΛ. ὁέστι δὲ καὶ ἡ τῆς σφαίρας διάμετρος δυνάμει πενταπλασίων τῆς ἐκ τοῦ κέντρου τοῦ κύκλου, ἀφ᾽ οὗ τὸ εἰκοσάεδρον ἀναγέγραπται. καί ἐστιν ἡ ΑΒ ἡ τῆς σφαίρας διάμετρος· ἡ ΚΛ ἄρα ἐκ τοῦ κέντρου ἐστὶ τοῦ κύκλου, ἀφ᾽ οὗ τὸ εἰκοσάεδρον ἀναγέγραπται· ἡ ΚΛ ἄρα ἑχαγώνου ἐστὶ πλευρὰ τοῦ εἰρημένου κύκλου. καὶ ἐπεὶ ἡ τῆς σφαίρας διάμετρος σύγκειταιὀέκ τε τῆς τοῦ ἑχαγώνου καὶ δύο τῶν τοῦ δεκαγώνου τῶν εἰξ τὸν εἰρημένον κύκλον ἐγγραφομένων, καί ἐστιν ἡ μὲν ΑΒ ἡ τῆς σφαίρας διάμετρος, ἡ δὲ ΚΛ ἑχαγώνου πλευρά, καὶ ὀίση ἡ ΑΚ τῇ ΛΒ, ἑκατέρα ἄρα

tively), at right-angles to AB. And let AF, FB, and EB be joined. And since AD is double DB, AB is thus triple BD. Thus, via conversion, BA is one and a half times AD. And as BA (is) to AD, so the (square) on BA (is) to the (square) on AF [Def. 5.9]. For triangle AFB is equiangular to triangle AFD [Prop. 6.8]. Thus, the (square) on BA is one and a half times the (square) on AF. And the square on the diameter of the sphere is also one and a half times the (square) on the side of the pyramid [Prop. 13.13]. And AB is the diameter of the sphere. Thus, AF is equal to the side of the pyramid.

Again, since AD is double DB, AB is thus triple BD. And as AB (is) to BD, so the (square) on AB (is) to the (square) on BF [Prop. 6.8, Def. 5.9]. Thus, the (square) on AB is three times the (square) on BF. And the square on the diameter of the sphere is also three times the (square) on the side of the cube [Prop. 13.15]. And AB is the diameter of the sphere. Thus, BF is the side of the cube.

And since AC is equal to CB, AB is thus double BC. And as AB (is) to BC, so the (square) on AB (is) to the (square) on BE [Prop. 6.8, Def. 5.9]. Thus, the (square) on AB is double the (square) on BE. And the square on the diameter of the sphere is also double the (square) on the side of the octagon [Prop. 13.14]. And AB is the diameter of the given sphere. Thus, BE is the side of the octagon.

So let AG be drawn from point A at right-angles to the straight-line AB. And let AG be made equal to AB. And let GC be joined. And let HK be drawn from H, perpendicular to AB. And since GA is double AC. For GA (is) equal to AB. And as GA (is) to AC, so HK (is) to KC [Prop. 6.4]. HK (is) thus also double KC. Thus, the (square) on HK is four times the (square) on KC. Thus, the (sum of the squares) on HK and KC, which is the (square) on HC [Prop. 1.47], is five times the (square) on KC. And HC (is) equal to CB. Thus, the (square) on BC (is) five times the (square) on CK. And since AB is double CB, of which AD is double DB, the remainder BD is thus double the remainder DC. BC (is) thus triple CD. The (square) on BC (is) thus nine times the (square) on CD. And the (square) on BC (is) five times the (square) on CK. Thus, the (square) on CK (is) greater than the (square) on CD. CK is thus greater than CD. Let CL be made equal to CK. And let LM be drawn from L at right-angles to AB. And let MB be joined. And since the (square) on BC is five times the (square) on CK, and AB is double BC, and KL double CK, the (square) on AB is thus five times the (square) on KL. And the square on the diameter of the sphere is also five times the (square) on the radius of the circle from which the icosahedron has been described [Prop. 13.16 corr.]. And AB is the diameter of the sphere. Thus, KL is the radius of the circle from which the icosahedron has been described. Thus, KL is (the side) of the hexagon (inscribed) in the aforementioned circle [Prop. 4.15 corr.]. And since the diameter of the

τῶν ΑΚ, ΛΒ δεκαγώνου ἐστὶ πλευρὰ τοῦ ἐγγραφομένου εἴξ τὸν κύκλον, ἀφ᾿ οὗ τὸ εἰκοσάεδρον ἀναγέγραπται. καὶ ἐπεὶ δεκαγώνου μὲν ἡ ΛΒ, ἑχαγώνου δὲ ἡ ΜΛ· ὁίση γάρ ἐστι τῇ ΚΛ, ἐπεὶ καὶ τῇ ΘΚ· ἴσον γὰρ ἀπέθουσιν ἀπὸ τοῦ κέντρου· καί ἐστιν ἑκατέρα τῶν ΘΚ, ΚΛ διπλασίων τῆς ΚΓ· πενταγώνου ἄρα ἐστὶν ἡ ΜΒ. ἡ δὲ τοῦ πενταγώνου ἐστὶν ἡ τοῦ εἰκοσαέδρου· εἰκοσαέδρου ἄρα ἐστὶν ἡ ΜΒ.

Καὶ ἐπεὶ ἡ ΖΒ κύβου ἐστὶ πλευρά, τετμήσθωὁάκρον καὶ μέσον λόγον κατὰ τὸ Ν, καὶ ὁέστω μεῖζον τμῆμα τὸ ΝΒ· ἡ ΝΒ ἄρα δωδεκαέδρου ἐστὶ πλευρά.

Καὶ ἐπεὶ ἡ τῆς σφαίρας διάμετρος ἐδείθθη τῆς μὲν ΑΖ πλευρᾶξ τῆς πυραμίδος δυνάμει ἡμιολία, τῆς δὲ τοῦ ὀκταέδρου τῆς ΒΕ δυνάμει διπλασίων, τῆς δὲ τοῦ κύβου τῆς ΖΒ δυνάμει τριπλασίων, οὁίων ἄρα ἡ τῆς σφαίρας διάμετρος δυνάμει ὁέχ, τοιούτων ἡ μὲν τῆς πυραμίδος τεσσάρων, ἡ δὲ τοῦ ὀκταέδρου τριῶν, ἡ δὲ τοῦ κύβου δύο. ἡ μὲν ἄρα τῆς πυραμίδος πλευρὰ τῆς μὲν τοῦ ὀκταέδρου πλευρᾶξ δυνάμει ἐστὶν ἐπίτριτος, τῆς δὲ τοῦ κύβου δυνάμει διπλῆ, ἡ δὲ τοῦ ὀκταέδρου τῆς τοῦ κύβου δυνάμει ἡμιολία. αἱ μὲν οὀῦν εἰρημέναι τῶν τριῶν σχημάτων πλευραί, λέγω δὴ πυραμίδος καὶ ὀκταέδρου καὶ κύβου, πρὸς ἀλλήλας εἰσὶν ἐν λόγοις ὁρητοῖς. αἱ δὲ λοιπαὶ δύο, λέγω δὴ ἥ τε τοῦ εἰκοσαέδρου καὶ ἡ τοῦ δωδεκαέδρου, οοὔτε πρὸς ἀλλήλας οοὔτε πρὸς τάξ προειρημένας εἰσὶν ἐν λόγοις ὁρητοῖς· ὀάλογοι γάρ εἰσιν, ἡ μὲν ἐλάττων, ἡ δὲ ἀποτομή.

῞ΟΟτι μείζων ἐστὶν ἡ τοῦ εἰκοσαέδρου πλευρὰ ἡ ΜΒ τῆς τοῦ δωδεκαέδρου τῆς ΝΒ, δείχομεν οὁύτως.

Ἐπεὶ γὰρ ἰσογώνιόν ἐστι τὸ ΖΔΒ τρίγωνον τῷ ΖΑΒ τριγώνῳ, ἀνάλογόν ἐστιν ὡξ ἡ ΔΒ πρὸς τὴν ΒΖ, οὁύτως ἡ ΒΖ πρὸς τὴν ΒΑ. καὶ ἐπεὶ τρεῖς εὐθεῖαι ἀνάλογόν εἰσιν, ὁέστιν ὡξ ἡ πρώτη πρὸς τὴν τρίτην, οὁύτως τὸ ἀπὸ τῆς πρώτης πρὸς τὸ ἀπὸ τῆς δευτέρας· ὁέστιν ἄρα ὡξ ἡ ΔΒ πρὸς τὴν ΒΑ, οὁύτως τὸ ἀπὸ τῆς ΔΒ πρὸς τὸ ἀπὸ τῆς ΒΖ· ἀνάπαλιν ἄρα ὡξ ἡ ΑΒ πρὸς τὴν ΒΔ, οὁύτως τὸ ἀπὸ τῆς ΖΒ πρὸς τὸ ἀπὸ τῆς ΒΔ. τριπλῆ δὲ ἡ ΑΒ τῆς ΒΔ· τριπλάσιον ἄρα τὸ ἀπὸ τῆς ΖΒ τοῦ ἀπὸ τῆς ΒΔ. ὁέστι δὲ καὶ τὸ ἀπὸ τῆς ΑΔ τοῦ ἀπὸ τῆς ΔΒ τετραπλάσιον· διπλῆ γὰρ ἡ ΑΔ τῆς ΔΒ· μεῖζον ἄρα τὸ ἀπὸ τῆς ΑΔ τοῦ ἀπὸ τῆς ΖΒ· μείζων ἄρα ἡ ΑΔ τῆς ΖΒ· πολλῷ ἄρα ἡ ΑΛ τῆς ΖΒ μείζων ἐστίν. καὶ τῆς μὲν ΑΛὁάκρον καὶ μέσον λόγον τεμνομένης τὸ μεῖζον τμῆμά ἐστιν ἡ ΚΛ, ἐπειδήπερ ἡ μὲν ΛΚ ἑχαγώνου ἐστίν, ἡ δὲ ΚΑ δεκαγώνου· τῆς δὲ ΖΒὁάκρον καὶ μέσον λόγον τεμνομένης τὸ μεῖζον τμῆμά ἐστιν ἡ ΝΒ· μείζων ἄρα ἡ ΚΛ τῆς ΝΒ. ὁίση δὲ ἡ ΚΛ τῇ ΛΜ· μείζων ἄρα ἡ ΛΜ τῆς ΝΒ [τῆς δὲ ΛΜ μείζων ἐστὶν ἡ ΜΒ]. πολλῷ ἄρα ἡ ΜΒ πλευρὰ οὀῦσα τοῦ εἰκοσαέδρου μείζων ἐστὶ τῆς ΝΒ πλευρᾶξ οὀύσης τοῦ δωδεκαέδρου· ὅπερ ἔδει δεῖχαι.

sphere is composed of (the side) of the hexagon, and two of (the sides) of the decagon, inscribed in the aforementioned circle, and AB is the diameter of the sphere, and KL the side of the hexagon, and AK (is) equal to LB, thus AK and LB are each sides of the decagon inscribed in the circle from which the icosahedron has been described. And since LB is (the side) of the decagon. And ML (is the side) of the hexagon—for (it is) equal to KL, since (it is) also (equal) to HK, for they are equally far from the center. And HK and KL are each double KC. MB is thus (the side) of the pentagon (inscribed in the circle) [Props. 13.10, 1.47]. And (the side) of the pentagon is (the side) of the icosahedron [Prop. 13.16]. Thus, MB is (the side) of the icosahedron.

And since FB is the side of the cube, let it be cut in extreme and mean ratio at N, and let NB be the greater piece. Thus, NB is the side of the dodecahedron [Prop. 13.17 corr.].

And since the (square) on the diameter of the sphere was shown (to be) one and a half times the square on the side, AF, of the pyramid, and twice the square on (the side), BE, of the octagon, and three times the square on (the side), FB, of the cube, thus, of whatever (parts) the (square) on the diameter of the sphere (makes) six, of such (parts) the (square) on (the side) of the pyramid (makes) four, and (the square) on (the side) of the octagon three, and (the square) on (the side) of the cube two. Thus, the (square) on the side of the pyramid is one and a third times the square on the side of the octagon, and double the square on (the side) of the cube. And the (square) on (the side) of the octahedron is one and a half times the square on (the side) of the cube. Therefore, the aforementioned sides of the three figures—I mean, of the pyramid, and of the octahedron, and of the cube—are in rational ratios to one another. And (the sides of) the remaining two (figures)—I mean, of the icosahedron, and of the dodecahedron—are neither in rational ratios to one another, nor to the (sides) of the aforementioned (three figures). For they are irrational (straight-lines): (namely), a minor [Prop. 13.16], and an apotome [Prop. 13.17].

(And), we can show that the side, MB, of the icosahedron is greater that the (side), NB, or the dodecahedron, as follows.

For, since triangle FDB is equiangular to triangle FAB [Prop. 6.8], proportionally, as DB is to BF, so BF (is) to BA [Prop. 6.4]. And since three straight-lines are (continually) proportional, as the first (is) to the third, so the (square) on the first (is) to the (square) on the second [Def. 5.9, Prop. 6.20 corr.]. Thus, as DB is to BA, so the (square) on DB (is) to the (square) on BF. Thus, inversely, as AB (is) to BD, so the (square) on FB (is) to the (square) on BD. And AB (is) triple BD. Thus, the (square) on FB (is) three times the (square) on BD. And the (square) on AD is also four times the (square) on DB. For AD (is) double DB. Thus, the (square) on AD (is) greater than the (square) on FB. Thus, AD (is)

greater than FB. Thus, AL is much greater than FB. And KL is the greater piece of AL, which is cut in extreme and mean ratio—inasmuch as LK is (the side) of the hexagon, and KA (the side) of the decagon [Prop. 13.9]. And NB is the greater piece of FB, which is cut in extreme and mean ratio. Thus, KL (is) greater than NB. And KL (is) equal to LM. Thus, LM (is) greater than NB [and MB is greater than LM]. Thus, MB, which is (the side) of the icosahedron, is much greater than NB, which is (the side) of the dodecahedron. (Which is) the very thing it was required to show.

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† If the radius of the given sphere is unity then the sides of the pyramid (*i.e.*, tetrahedron), octahedron, cube, icosahedron, and dodecahedron, respectively, satisfy the following inequality: $\sqrt{8/3} > \sqrt{2} > \sqrt{4/3} > (1/\sqrt{5})\sqrt{10-2\sqrt{5}} > (1/3)(\sqrt{15}-\sqrt{3})$.

Λέγω δή, ὅτι παρὰ τὰ εἰρημένα πέντε σϑήματα οὐ συσταϑήσεται ὁέτερον σχῆμα περιεχόμενον ὑπὸ ἰσοπλεύρων τε καὶ ἰσογωνίωνὀίσων ἀλλήλοις.

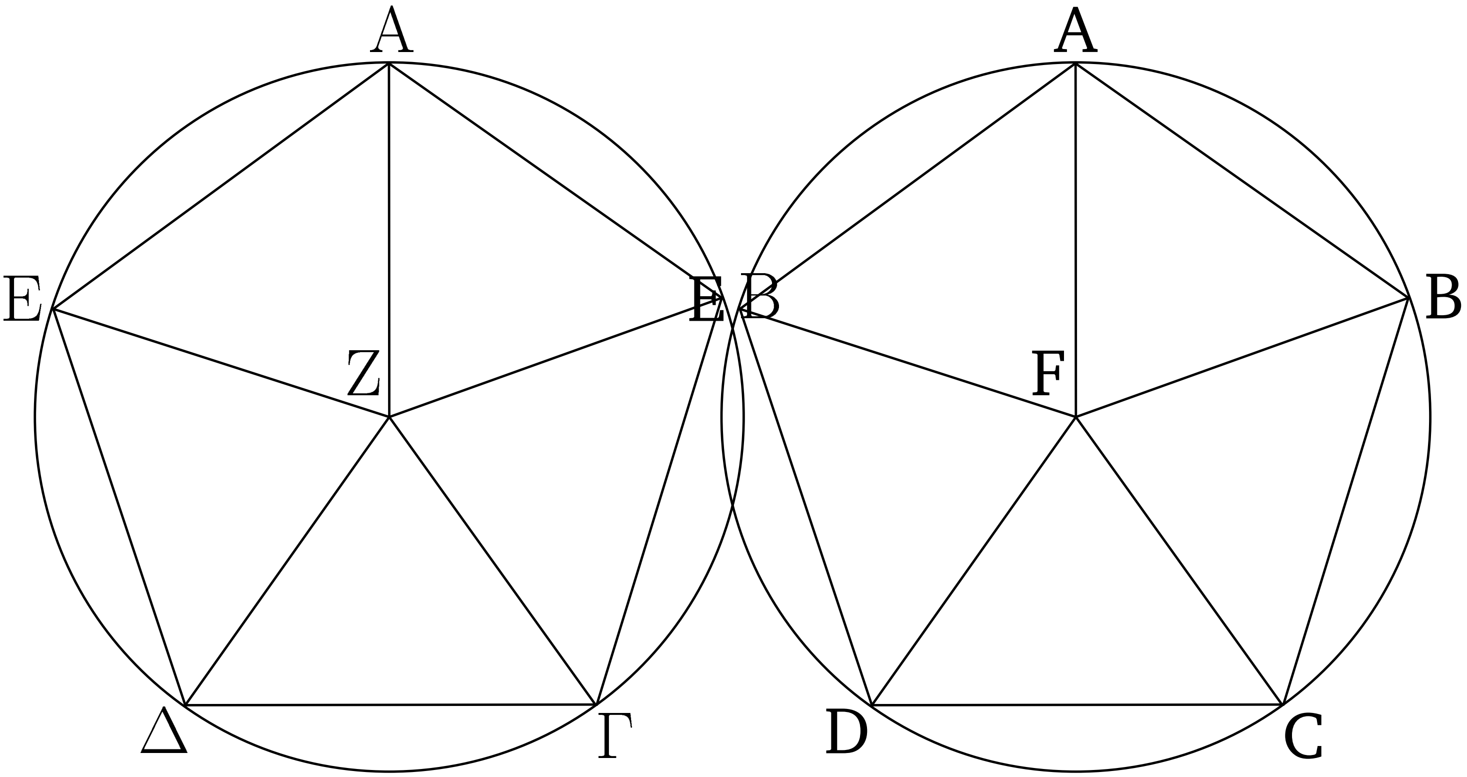
Ὑπὸ μὲν γὰρ δύο τριγώνωνὀὴ ὅλως ἐπιπέδων στερεὰ γωνία οὐ συνίσταται. ὑπὸ δὲ τριῶν τριγώνων ἡ τῆς πυραμίδος, ὑπὸ δὲ τεσσάρων ἡ τοῦ ὀκταέδρου, ὑπὸ δὲ πέντε ἡ τοῦ εἰκοσαέδρου· ὑπὸ δὲ ὁὲχ τριγώνων ἰσοπλεύρων τε καὶ ἰσογωνίων πρὸς ἑνὶ σημείῳ συνισταμένων οὐκὀέσται στερεὰ γωνία· οὀύσης γὰρ τῆς τοῦ ἰσοπλεύρου τριγώνου γωνίας διμοίρου ὀρϑῆςὀέσονται αἱ ὁὲχ τέσσαρσιν ὀρϑαῖξ ἴσαι· ὅπερ ἀδύνατον· ὁάπασα γὰρ στερεὰ γωνία ὑπὸ ἐλασσόνωνὀὴ τεσσάρων ὀρϑῶν περιέϑεται. διὰ τὰ αὐτὰ δὴ οὐδὲ ὑπὸ πλειόνωνὀὴ ὁὲχ γωνιῶν ἐπιπέδων στερεὰ γωνία συνίσταται. ὑπὸ δὲ τετραγώνων τριῶν ἡ τοῦ κύβου γωνία περιέϑεται· ὑπὸ δὲ τεσσάρων ἀδύνατον· ὀέσονται γὰρ πάλιν τέσσαρες ὀρϑαί. ὑπὸ δὲ πενταγώνων ἰσοπλεύρων καὶ ἰσογωνίων, ὑπὸ μὲν τριῶν ἡ τοῦ δωδεκαέδρου· ὑπὸ δὲ τεσσάρων ἀδύνατον· οὀύσης γὰρ τῆς τοῦ πενταγώνου ἰσοπλεύρου γωνίας ὀρϑῆς καὶ πέμπτου, ὀέσονται αἱ τέσσαρες γωνίαι τεσσάρων ὀρϑῶν μείζους· ὅπερ ἀδύνατον. οὐδὲ μὴν ὑπὸ πολυγώνων ἑτέρων σχημάτων περισϑεϑήσεται στερεὰ γωνία διὰ τὸ αὐτὸ ὀάτοπον.

Οὐκ ἄρα παρὰ τὰ εἰρημένα πέντε σϑήματα ὁέτερον σχῆμα στερεὸν συσταϑήσεται ὑπὸ ἰσοπλεύρων τε καὶ ἰσογωνίων περιεχόμενον· ὅπερ ἔδει δεῖξαι.

So, I say that, beside the five aforementioned figures, no other (solid) figure can be constructed (which is) contained by equilateral and equiangular (planes), equal to one another.

For a solid angle cannot be constructed from two triangles, or indeed (two) planes (of any sort) [Def. 11.11]. And (the solid angle) of the pyramid (is constructed) from three (equiangular) triangles, and (that) of the octahedron from four (triangles), and (that) of the icosahedron from (five) triangles. And a solid angle cannot be (made) from six equilateral and equiangular triangles set up together at one point. For, since the angles of a equilateral triangle are (each) two-thirds of a right-angle, the (sum of the) six (plane) angles (containing the solid angle) will be four right-angles. The very thing (is) impossible. For every solid angle is contained by (plane angles whose sum is) less than four right-angles [Prop. 11.21]. So, for the same (reasons), a solid angle cannot be constructed from more than six plane angles (equal to two-thirds of a right-angle) either. And the (solid) angle of a cube is contained by three squares. And (a solid angle contained) by four (squares is) impossible. For, again, the (sum of the plane angles containing the solid angle) will be four right-angles. And (the solid angle) of a dodecahedron (is contained) by three equilateral and equiangular pentagons. And (a solid angle contained) by four (equiangular pentagons is) impossible. For, the angle of an equilateral pentagon being one and one-fifth of right-angle, four (such) angles will be greater (in sum) than four right-angles. The very thing (is) impossible. And, on account of the same absurdity, a solid angle cannot be constructed from any other (equiangular) polygonal figures either.

Thus, beside the five aforementioned figures, no other solid figure can be constructed (which is) contained by equilateral and equiangular (planes). (Which is) the very thing it was required to show.



Λῆμμα.

῞ΟΟτι δὲ ἡ τοῦ ἰσοπλεύρου καὶ ἰσογωνίου πενταγώνου γωνία ὀρθή ἐστι καὶ πέμπτον, οὑύτω δεικτέον.

῎Εστω γὰρ πεντάγωνον ἰσόπλευρον καὶ ἰσογώνιον τὸ ΑΒΓΔΕ, καὶ περιγεγράφθω περὶ αὐτὸ κύκλος ὁ ΑΒΓΔΕ, καὶ εἰλήφθω αὐτοῦ τὸ κέντρον τὸ Ζ, καὶ ἐπεζεύθθωσαν αἱ ΖΑ, ΖΒ, ΖΓ, ΖΔ, ΖΕ. δίθα ἄρα τέμνουσι τὰξ πρὸς τοῖς Α, Β, Γ, Δ, Ε τοῦ πενταγώνου γωνίας. καὶ ἐπεὶ αἱ πρὸς τῷ Ζ πέντε γωνίαι τέσσαρσιν ὀρθαῖξ ἴσαι εἰσὶ καί εἰσιν ἴσαι, μία ἄρα αὐτῶν, ὡξ ἡ ὑπὸ ΑΖΒ, μιᾶς ὀρθῆς ἐστι παρὰ πέμπτον· λοιπαὶ ἄρα αἱ ὑπὸ ΖΑΒ, ΑΒΖ μιᾶς εἰσιν ὀρθῆς καὶ πέμπτον. οἰση δὲ ἡ ὑπὸ ΖΑΒ τῇ ὑπὸ ΖΒΓ· καὶ ὅλη ἄρα ἡ ὑπὸ ΑΒΓ τοῦ πενταγώνου γωνία μιᾶς ἐστιν ὀρθῆς καὶ πέμπτον· ὅπερ ἔδει δεῖχαι.

Lemma

It can be shown that the angle of an equilateral and equiangular pentagon is one and one-fifth of a right-angle, as follows.

For let $ABCDE$ be an equilateral and equiangular pentagon, and let the circle $ABCDE$ be circumscribed about it [Prop. 4.14]. And let its center, F, be found [Prop. 3.1]. And let FA, FB, FC, FD, and FE be joined. Thus, they cut the angles of the pentagon in half at (points) A, B, C, D, and E [Prop. 1.4]. And since the five angles at F are equal (in sum) to four right-angles, and are also equal (to one another), (any) one of them, like AFB, is thus one less a fifth of a right-angle. Thus, the (sum of the) remaining (angles in triangle ABF), FAB and ABF, is one plus a fifth of a right-angle [Prop. 1.32]. And FAB (is) equal to FBC. Thus, the whole angle, ABC, of the pentagon is also one and one-fifth of a right-angle. (Which is) the very thing it was required to show.

GREEK-ENGLISH LEXICON

ABBREVIATIONS: *act* - active; *adj* - adjective; *adv* - adverb; *conj* - conjunction; *fut* - future; *gen* - genitive; *imperat* - imperative; *impf* - imperfect; *ind* - indeclinable; *indic* - indicative; *intr* - intransitive; *mid* - middle; *neut* - neuter; *no* - noun; *par* - particle; *part* - participle; *pass* - passive; *perf* - perfect; *pre* - preposition; *pres* - present; *pro* - pronoun; *sg* - singular; *tr* - transitive; *vb* - verb.

ἄγω, *ἄχω, ἤγαγον, -ἦθα, ἦγμαι, ἤθθην* : *vb*, lead, draw (a line).

ἀδύνατοξ -ον : *adj*, impossible.

ἀεί : *adv*, always, for ever.

αἱρέω, αἱρήσω, ε[˜]λον, ἥρηκα, ἥρημαι, ἡρέθην : *vb*, grasp.

ἀιτέω, αἰτήσω, ἤτησα, ἤτηκα, ἤτημαι, ἠτήθη : *vb*, postulate.

α ἴτημα -ατοξ, τό : *no*, postulate.

ἀκόλουθοξ -ον : *adj*, analogous, consequent on, in conformity with.

ἄκροξ -α -ον : *adj*, outermost, end, extreme.

ἀλλά : *conj*, but, otherwise.

ἄλογοξ -ον : *adj*, irrational.

ἅμα : *adv*, at once, at the same time, together.

ἀμβλυγώνιοξ -ον : *adj*, obtuse-angled; *τὸ ἀμβλυγώνιον*, *no*, obtuse angle.

ἀμβλύξ -εῖα -ύ : *adj*, obtuse.

ἀμφότεροξ -α -ον : *pro*, both.

ἀναγράφω : *vb*, describe (a figure); see *γράφω*.

ἀναλογία, ἡ : *no*, proportion, (geometric) progression.

ἀνάλογοξ -ον : *adj*, proportional.

ἀνάπαλιν : *adv*, inverse(ly).

αναπληρόω : *vb*, fill up.

ἀναστρέφω : *vb*, turn upside down, convert (ratio); see *στρέφω*.

ἀναστροφή, ἡ : *no*, turning upside down, conversion (of ratio).

ἀνθυφαιρέω : *vb*, take away in turn; see *αἱρέω*.

ἀνίστημι : *vb*, set up; see *ἵστημι*.

ἄνισοξ -ον : *adj*, unequal, uneven.

ἀντιπάσθω : *vb*, be reciprocally proportional; see *πάσθω*.

ἄχων -ονοξ, ὁ : *vb*, axis.

ἅπαχ : *adv*, once.

ἅπαξ, ἅπασα, ἅπαν : *adj*, quite all, the whole.

ἄπειροξ -ον : *adj*, infinite.

ἀπεναντίον : *ind*, opposite.

ἀπέθω : *vb*, be far from, be away from; see *ἔθω*.

ἀπλατήξ -έξ : *adj*, without breadth.

ἀπόδειχιξ -εωξ, ἡ : *no*, proof.

ἀποκαθίστημι : *vb*, re-establish, restore; see *ἵστημι*.

ἀπολαμβάνω : *vb*, take from, subtract from, cut off from; see *λαμβάνω*.

ἀποτέμνω : *vb*, cut off, subtend.

ἀπότμημα -ατος, τό : *no*, piece cut off, segment.

ἀποτομή, ἡ : *vb*, piece cut off, apotome.

ἅπτω, ἅψω, ἧψα, **—,** *ἧμμαι*, **—** : *vb*, touch, join, meet.

ἀπώτεροξ -α -ον : *adj*, further off.

ἄρα : *par*, thus, as it seems (inferential).

ἀριθμόξ, ὁ : *no*, number.

ἀρτιάκιξ : *adv*, an even number of times.

αρτιόπλευροξ -ον : *adj*, having a even number of sides.

ἄρθω, ἄρχω, ἦρχα, ἦρθα, ἦργμαι, ἦρθθην : *vb*, rule; *mid.*, begin.

ἀσύμμετροξ -ον : *adj*, incommensurable.

ἀσύμπτωτοξ -ον : *adj*, not touching, not meeting.

ἄρτιοξ -α -ον : *adj*, even, perfect.

ἄτμητοξ -ον : *adj*, uncut.

ἀτόποξ -ον : *adj*, absurd, paradoxical.

αὐτόθεν : *adv*, immediately, obviously.

ἀφαίρεω : *vb*, take from, subtract from, cut off from; see *αἱρέω*.

ἁφή, ἡ : *no*, point of contact.

βάθοξ -εοξ, τό : *no*, depth, height.

βαίνω, -βήσομαι, -έβην, βέβηκα, **—, —** : *vb*, walk; *perf*, stand (of angle).

βάλλω, βαλῶ, ἔβαλον, βέβληκα, βέβλημαι, ἐβλήθην : *vb*, throw.

βάσιξ -εωξ, ἡ : *no*, base (of a triangle).

γάρ : *conj*, for (explanatory).

γί[γ]νομαι, γενήσομαι, ἐγενόμην, γέγονα, γεγένημαι, **—** : *vb*, happen, become.

γνώμων -ονοξ, ἡ : *no*, gnomon.

γραμμή, ἡ : *no*, line.

γράφω, γράψω, ἔγρα[ψ/φ]α, γέγραφα, γέγραμμαι, ἐραψάμην : *vb*, draw (a figure).

γωνία, ἡ : *no*, angle.

δεῖ : *vb*, be necessary; *δεῖ*, it is necessary; *ἔδει*, it was necassary; *δέον*, being necessary.

δείκνυμι, δείχω, ἔδειχα, δέδειθα, δέδειγμαι, ἐδείθθην : *vb*, show, demonstrate.

δεικτέον : *ind*, one must show.

δεῖχιξ -εωξ, ἡ : *no*, proof.

δεκαγώνοξ -ον : *adj*, ten-sided; *τὸ δεκαγώνον*, *no*, decagon.

δέθομαι, δέχομαι, ἐδεχάμην, **—,** *δέδεγμαι, ἐδέθθην* : *vb*, receive, accept.

δή : *conj*, so (explanatory).

δηλαδή : *ind*, quite clear, manifest.

δῆλοξ -η -ον : *adj*, clear.

δηλονότι : *adv*, manifestly.

διάγω : *vb*, carry over, draw through, draw across; see *ἄγω*.

διαγώνιοξ -ον : *adj*, diagonal.

διαλείπω : *vb*, leave an interval between.

διάμετροξ -ον : *adj*, diametrical; *ἡ διάμετροξ*, *no*, diameter, diagonal.

διαίρεσιξ -εωξ, ἡ : *no*, division, separation.

διαιρέω : *vb*, divide (in two); *διαρεθέντοξ -η -ον*, *adj*, separated (ratio); see *αἱρέω*.

διάστημα -ατοξ, τό : *no*, radius.

διαφέρω : *vb*, differ; see *φέρω*.

δίδωμι, δώσω, ἔδωκα, δέδωκα, δέδομαι, ἐδόθην : *vb*, give.

διμοίροξ -ον : *adj*, two-thirds.

διπλασιάζω : *vb*, double.

διπλάσιοξ -α -ον : *adj*, double, twofold.

διπλασίων -ον : *adj*, double, twofold.

διπλοῦξ -ῆ -οῦν : *adj*, double.

δίξ : *adv*, twice.

δίϑα : *adv*, in two, in half.

διϑοϱομία, ἡ : *no*, point of bisection.

δυάξ -άδοξ, ἡ : *no*, the number two, dyad.

δύναμαι : *vb*, be able, be capable, generate, square, be when squared; *δυναμένη, ἡ*, *no*, square-root (of area)—*i.e.*, straight-line whose square is equal to a given area.

δύναμιξ -εωξ, ἡ : *no*, power (usually 2nd power when used in mathematical sence, hence), square.

δυνατόξ -ή -όν : *adj*, possible.

δωδεκάεδϱοξ -ον : *adj*, twelve-sided.

ἑαυτοῦ -ῆξ -οῦ : *adj*, of him/her/it/self, his/her/its/own.

ἐγγίων -ον : *adj*, nearer, nearest.

ἐγγϱάφω : *vb*, inscribe; see *γϱάφω.*

εἶδοξ -εοξ, τό : *no*, figure, form, shape.

εἰκοσάεδϱοξ -ον : *adj*, twenty-sided.

εἴϱω/λέγω, ἐϱῶ/εϱέω, εἶπον, εἴϱηκα, εἴϱημαι, ἐϱϱήϑην : *vb*, say, speak; *per pass part*, *ειϱημένοξ -η -ον*, *adj*, said, aforementioned.

εἴτε **…** *εἴτε* : *ind*, either … or.

ἕκαστοξ -η -ον : *pro*, each, every one.

ἑκατέϱοξ -α -ον : *pro*, each (of two).

ἐκβάλλω : *vb*, produce (a line); see *βάλλω.*

ἐκϑέω : *vb*, set out.

ἔκκειμαι : *vb*, be set out, be taken; see *κεῖμαι.*

ἐκτίϑημι : *vb*, set out; see *τίϑημι.*

ἐκτόξ : *pre + gen*, outside, external.

ἐλάσσων/ἐλάττων -ον : *adj*, less, lesser.

ἐλάϑιστοξ -η -ον : *adj*, least.

ἐλλείπω : *vb*, be less than, fall short of.

ἐμπίπτω : *vb*, meet (of lines), fall on; see *πίπτω.*

ἔμπϱοσϑεν : *adv*, in front.

ἐναλλάχ : *adv*, alternate(ly).

ἐναϱμόζω : *vb*, insert; *perf indic pass 3rd sg*, *ἐνήϱμοσται.*

ἐνδέϑομαι : *vb*, admit, allow.

ἕνεκεν : *ind*, on account of, for the sake of.

ἐνναπλάσιοξ -α -ον : *adj*, nine-fold, nine-times.

ἔννοια, ἡ : *no*, notion.

ενπεϱιέϑω : *vb*, encompass; see *ἔϑω.*

ἐνπίπτω : see *ἐμπίπτω.*

ἐντόξ : *pre + gen*, inside, interior, within, internal.

ἑχάγωνοξ -ον : *adj*, hexagonal; *τὸ ἑχάγωνον*, *no*, hexagon.

ἑχαπλάσιοξ -α -ον : *adj*, sixfold.

ἑχῆξ : *adv*, in order, successively, consecutively.

ἔχωϑεν : *adv*, outside, extrinsic.

ἐπάνω : *adv*, above.

ἐπαφή, ἡ : *no*, point of contact.

ἐπεί : *conj*, since (causal).

ἐπειδήπεϱ : *ind*, inasmuch as, seeing that.

ἐπιζεύγνῦμι, ἐπιζεύχω, ἐπέζευχα, **—**, *ἐπέζευγμαι, ἐπέζεύϑϑην* : *vb*, join (by a line).

ἐπιλογίζομαι : *vb*, conclude.

ἐπινοέω : *vb*, think of, contrive.

ἐπιπέδοξ -ον : *adj*, level, flat, plane; *τὸ ἐπιπέδον*, *no*, plane.

ἐπισκέπτομαι : *vb*, investigate.

ἐπίσκεψιξ -εωξ, ἡ : *no*, inspection, investigation.

ἐπιτάσσω : *vb*, put upon, enjoin; *τὸ ἐπιταϑϑέν*, *no*, the (thing) prescribed; see *τάσσω.*

ἐπίτϱιτοξ -ον : *adj*, one and a third times.

ἐπιφάνεια, ἡ : *no*, surface.

ἔπομαι : *vb*, follow.

ἔϱϑομαι, ἐλεύσομαι, ἦλϑον, ἐλήλυϑα, **—**, **—** : *vb*, come, go.

ἔσϑατοξ -η -ον : *adj*, outermost, uttermost, last.

ἑτεϱόμηκηξ -εξ : *adj*, oblong; *τὸ ἑτεϱόμηκεξ*, *no*, rectangle.

ἕτεϱοξ -α -ον : *adj*, other (of two).

ἔτι : *par*, yet, still, besides.

εὐϑύγϱαμμοξ -ον : *adj*, rectilinear; *τὸ εὐϑύγϱαμμον*, *no*, rectilinear figure.

εὐϑύξ -εῖα -ύ : *adj*, straight; *ἡ εὐϑεῖα*, *no*, straight-line; *ἐπ᾽ εὐϑεῖαξ*, in a straight-line, straight-on.

εὑϱίσκω, εὑϱήσκω, ηὗϱον, εὕϱεκα, εὕϱημαι, εὑϱέϑην : *vb*, find.

ἐφάπτω : *vb*, bind to; *mid*, touch; *ἡ ἐφαπτομένη*, *no*, tangent; see *ἅπτω.*

ἐφαϱμόζω, ἐφαϱμόσω, ἐφήϱμοσα, ἐφήμοκα, ἐφήμοσμαι, ἐφήμόσϑην : *vb*, coincide; *pass*, be applied.

ἐφεχῆξ : *adv*, in order, adjacent.

ἐφίστημι : *vb*, set, stand, place upon; see *ἵστημι.*

ἔϑω, ἕχω, ἔσϑον, ἔσϑηκα, -έσϑημαι, **—** : *vb*, have.

ἡγέομαι, ἡγήσομαι, ἡγησάμην, ἥγημαι, **—**, *ἡγήϑην* : *vb*, lead.

ἤδη : *ind*, already, now.

ἥκω, ἥχω, **—**, **—**, **—**, **—** : *vb*, have come, be present.

ἡμικύκλιον, τό : *no*, semi-circle.

ἡμιόλιοξ -α -ον : *adj*, containing one and a half, one and a half times.

ἥμισυξ -εια -υ : *adj*, half.

ἤπεϱ **=** *ἤ* **+** *πεϱ* : *conj*, than, than indeed.

ἤτοι **…** *ἤ* : *par*, surely, either … or; in fact, either … or.

ϑέσις -εωξ, ἡ : *no*, placing, setting, position.

ϑεωϱημα -ατοξ, τό : *no*, theorem.

ἴδιοξ -α -ον : *adj*, one’s own.

ἰσάκιξ : *adv*, the same number of times; *ἰσάκιξ πολλαπλάσια*, the same multiples, equal multiples.

ἰσογώνιοξ -ον : *adj*, equiangular.

ἰσόπλευϱοξ -ον : *adj*, equilateral.

ἰσοπληϑήξ -έξ : *adj*, equal in number.

ἴσοξ -η -ον : *adj*, equal; *ἐχ ἴσου*, equally, evenly.

ἰσοσκελής -ές : *adj*, isosceles.

ἵστημι, στήσω, ἔστησα, **—, —**, *ἐστάθην* : *vb tr*, stand (something).

ἵστημι, στήσω, ἔστην, ἕστηκα, ἕσταμαι, ἐσταθην : *vb intr*, stand up (oneself); Note: perfect *I have stood up* can be taken to mean present *I am standing*.

ἰσοϋψής -ές : *adj*, of equal height.

καθάπερ : *ind*, according as, just as.

κάθετος -ον : *adj*, perpendicular.

καθόλου : *adv*, on the whole, in general.

καλέω : *vb*, call.

κἀκεινος = *καὶ ἐκεῖνος* .

κἄν = *καὶ ἄν* : *ind*, even if, and if.

καταγραφή, ἡ : *no*, diagram, figure.

καταγράφω : *vb*, describe/draw, inscribe (a figure); see *γράφω.*

κατακολουθέω : *vb*, follow after.

καταλείπω : *vb*, leave behind; see *λείπω*; *τὰ καταλειπόμενα*, *no*, remainder.

κατάλληλος -ον : *adj*, in succession, in corresponding order.

καταμετρέω : *vb*, measure (exactly).

καταντάω : *vb*, come to, arrive at.

κατασκευάζω : *vb*, furnish, construct.

κεῖμαι, κεῖσομαι, **—, —, —, —** : *vb*, have been placed, lie, be made; see *τίθημι.*

κέντρον, τό : *no*, center.

κλάω : *vb*, break off, inflect.

κλίνω, κλίνω, ἔκλινα, κέκλικα, κέκλιμαι, ἐκλίθην : *vb*, lean, incline.

κλίσις -εως, ἡ : *no*, inclination, bending.

κοῖλος -η -ον : *adj*, hollow, concave.

κορυφή, ἡ : *no*, top, summit, apex; *κατὰ κορυφήν*, vertically opposite (of angles).

κρίνω, κρινῶ, ἔκρῖνα, κέκρικα, κέκριμαι, ἐκρίθην : *vb*, judge.

κύβος, ὁ : *no*, cube.

κύκλος, ὁ : *no*, circle.

κύλινδρος, ὁ : *no*, cylinder.

κυρτός -ή -όν : *adj*, convex.

κῶνος, ὁ : *no*, cone.

λαμβάνω, λήψομαι, ἔλαβον, εἴληφα εἴλημμαι, ἐλήφθην : *vb*, take.

λέγω : *vb*, say; *pres pass part*, *λεγόμενος -η -ον*, *adj*, so-called; see *ἔιρω.*

λείπω, λείψω, ἔλιπον, λέλοιπα, λέλειμμαι, ἐλείφθην : *vb*, leave, leave behind.

λημμάτιον, τό : *no*, diminutive of *λῆμμα.*

λῆμμα -ατος, τό : *no*, lemma.

λῆψις -εως, ἡ : *no*, taking, catching.

λόγος, ὁ : *no*, ratio, proportion, argument.

λοιπός -ή -όν : *adj*, remaining.

μανθάνω, μαθήσομαι, ἔμαθον, μεμάθηκα, **—, —** : *vb*, learn.

μέγεθος -εος, τό : *no*, magnitude, size.

μείζων -ον : *adj*, greater.

μένω, μενῶ, ἔμεινα, μεμένηκα, **—, —** : *vb*, stay, remain.

μέρος -ους, τό : *no*, part, direction, side.

μέσος -η -ον : *adj*, middle, mean, medial; *ἐκ δύο μέσων*, bimedial.

μεταλαμβάνω : *vb*, take up.

μεταχύ : *adv*, between.

μετέωρος -ον : *adj*, raised off the ground.

μετρέω : *vb*, measure.

μέτρον, τό : *no*, measure.

μηδείς, μηδεμία, μηδέν : *adj*, not even one, (neut.) nothing.

μηδέποτε : *adv*, never.

μηδέτερος -α -ον : *pro*, neither (of two).

μῆκος -εος, τό : *no*, length.

μήν : *par*, truely, indeed.

μονάς -άδος, ἡ : *no*, unit, unity.

μοναθός -ή -όν : *adj*, unique.

μοναθῶς : *adv*, uniquely.

μόνος -η -ον : *adj*, alone.

νοέω, **—**, *νόησα, νενόηκα, νενόημαι, ἐνοήθην* : *vb*, apprehend, conceive.

οἷος -α -ον : *pre*, such as, of what sort.

ὀκτάεδρος -ον : *adj*, eight-sided.

ὅλος -η -ον : *adj*, whole.

ὁμογενής -ές : *adj*, of the same kind.

ὅμοιος -α -ον : *adj*, similar.

ὁμοιοπληθής -ές : *adj*, similar in number.

ὁμοιοταγής -ές : *adj*, similarly arranged.

ὁμοιότης -ητος, ἡ : *no* similarity.

ὁμοίως : *adv*, similarly.

ὁμόλογος -ον : *adj*, corresponding, homologous.

ὁμοταγής -ές : *adj*, ranged in the same row or line.

ὁμώνυμος -ον : *adj*, having the same name.

ὄνομα -ατος, τό : *no*, name; *ἐκ δύο ὀνομάτων*, binomial.

ὀχυγώνιος -ον : *adj*, acute-angled; *τὸ ὀχυγώνιον*, *no*, acute angle.

ὀχύς -εῖα -ύ : *adj*, acute.

ὁποιοσοῦν = *ὁποῖος -α -ον* + *οὖν* : *adj*, of whatever kind, any kind whatsoever.

ὁπόσος -η -ον : *pro*, as many, as many as.

ὁποσοσδηποτοῦν = *ὁπόσος -η -ον* + *δή* + *ποτέ* + *οὖν* : *adj*, of whatever number, any number whatsoever.

ὁποσοσοῦν = *ὁπόσος -η -ον* + *οὖν* : *adj*, of whatever number, any number whatsoever.

ὁπότερος -α -ον : *pro*, either (of two), which (of two).

ὀρθογώνιον, τό : *no*, rectangle, right-angle.

ὀρθός -ή -όν : *adj*, straight, right-angled, perpendicular; *πρὸς ὀρθὰς γωνίας*, at right-angles.

ὅρος, ὁ : *no*, boundary, definition, term (of a ratio).

ὁσαδηποτοῦν = *ὅσα* + *δή* + *ποτέ* + *οὖν* : *ind*, any number whatsoever.

ὁσάκις : *ind*, as many times as, as often as.

ὁσαπλάσιος -ον : *pro*, as many times as.

ὅσος -η -ον : *pro*, as many as.

ὅσπερ, ἥπερ, ὅπερ : *pro*, the very man who, the very thing which.

ὅστιξ, ἥτιξ, ὅ τι : *pro*, anyone who, anything which.

ὅταν : *adv*, when, whenever.

ὁτιοῦν : *ind*, whatsoever.

οὐδείξ, οὐδεμία, οὐδέν : *pro*, not one, nothing.

οὐδέτεροξ -α -ον : *pro*, not either.

οὐθέτεροξ : see *οὐδέτεροξ*.

οὐθέν : *ind*, nothing.

οὖν : *adv*, therefore, in fact.

οὕτωξ : *adv*, thusly, in this case.

πάλιν : *adv*, back, again.

πάντωξ : *adv*, in all ways.

παρὰ : *prep* + *acc*, parallel to.

παραβάλλω : *vb*, apply (a figure); see *βάλλω*.

παραβολή, ἡ : *no*, application.

παράκειμαι : *vb*, lie beside, apply (a figure); see *κεῖμαι*.

παραλλάσσω, παραλλάχω, **—,** *παρήλλαθα*, **—, —** : *vb*, miss, fall awry.

παραλληλεπίπεδοξ, -ον : *adj*, with parallel surfaces; *τὸ παραλληλεπίπεδον*, *no*, parallelepiped.

παραλληλόγραμμοξ -ον : *adj*, bounded by parallel lines; *τὸ παραλληλόγραμμον*, *no*, parallelogram.

παράλληλοξ -ον : *adj*, parallel; *τὸ παράλληλον*, *no*, parallel, parallel-line.

παραπλήρωμα -ατοξ, τό : *no*, complement (of a parallelogram).

παρατέλευτοξ -ον : *adj*, penultimate.

παρέκ : *prep* + *gen*, except.

παρεμπίπτω : *vb*, insert; see *πίπτω*.

πάσθω, πείσομαι, ἔπαθον, πέπονθα, **—, —** : *vb*, suffer.

πεντάγωνοξ -ον : *adj*, pentagonal; *τὸ πεντάγωνον*, *no*, pentagon.

πενταπλάσιοξ -α -ον : *adj*, five-fold, five-times.

πεντεκαιδεκάγωνον, τό : *no*, fifteen-sided figure.

πεπερασμένοξ -η -ον : *adj*, finite, limited; see *περαίνω*.

περαίνω, περανῶ, ἐπέρανα, **—,** *πεπεράσμαι, ἐπερανάνθην* : *vb*, bring to end, finish, complete; *pass*, be finite.

πέραξ -ατοξ, τό : *no*, end, extremity.

περατόω, **—, —, —, —, —** : *vb*, bring to an end.

περιγράφω : *vb*, circumscribe; see *γράφω*.

περιέθω : *vb*, encompass, surround, contain, comprise; see *ἔθω*.

περιλαμβάνω : *vb*, enclose; see *λαμβάνω*.

περισσάκιξ : *adv*, an odd number of times.

περισσόξ -ή -όν : *adj*, odd.

περιφέρεια, ἡ : *no*, circumference.

περιφέρω : *vb*, carry round; see *φέρω*.

πηλικότηξ -ητοξ, ἡ : *no*, magnitude, size.

πίπτω, πεσοῦμαι, ἔπεσον, πέπτωκα, **—, —** : *vb*, fall.

πλάτοξ -εοξ, τό : *no*, breadth, width.

πλείων -ον : *adj*, more, several.

πλευρά, ἡ : *no*, side.

πλῆθοξ -εοξ, τό : *no*, great number, multitude, number.

πλήν : *adv & prep* + *gen*, more than.

ποιόξ -ά -όν : *adj*, of a certain nature, kind, quality, type.

πολλαπλασιάζω : *vb*, multiply.

πολλαπλασιασμόξ, ὁ : *no*, multiplication.

πολλαπλάσιον, τό : *no*, multiple.

πολύεδροξ -ον : *adj*, polyhedral; *τό πολύεδρον*, *no*, polyhedron.

πολύγωνοξ -ον : *adj*, polygonal; *τό πολύγωνον*, *no*, polygon.

πολύπλευροξ -ον : *adj*, multilateral.

πόρισμα -ατοξ, τό : *no*, corollary.

ποτέ : *ind*, at some time.

πρῖσμα -ατοξ, τό : *no*, prism.

προβαίνω : *vb*, step forward, advance.

προδείκνυμι : *vb*, show previously; see *δείκνυμι.*

προεκτίθημι : *vb*, set forth beforehand; see *τίθημι.*

προερέω : *vb*, say beforehand; *perf pass part*, *προειρημένοξ -η -ον*, *adj*, aforementioned; see *εἴρω*.

προσαναπληρόω : *vb*, fill up, complete.

προσαναγράφω : *vb*, complete (tracing of); see *γράφω*.

προσαρμόζω : *vb*, fit to, attach to.

προσεκβάλλω : *vb*, produce (a line); see *ἐκβάλλω*.

προσευρίσκω : *vb*, find besides, find; see *εὑρίσκω*.

προσλαμβάμω : *vb*, add.

πρόκειμαι : *vb*, set before, prescribe; see *κεῖμαι*.

πρόσκειμαι : *vb*, be laid on, have been added to; see *κεῖμαι*.

προσπίπτω : *vb*, fall on, fall toward, meet; see *πίπτω*.

προτασιξ -εωξ, ἡ : *no*, proposition.

προστάσσω : *vb*, prescribe, enjoin; *τὸ τροσταθθέν*, *no*, the thing prescribed; see *τάσσω*.

προστίθημι : *vb*, add; see *τίθημι*.

πρότεροξ -α -ον : *adj*, first (comparative), before, former.

προτίθημι : *vb*, assign; see *τίθημι*.

προθωρέω : *vb*, go/come forward, advance.

πρῶτοξ -α -ον : *adj*, first, prime.

πυραμίξ -ίδοξ, ἡ : *no*, pyramid.

ῥητόξ -ή -όν : *adj*, expressible, rational.

ῥομβοειδήξ -έξ : *adj*, rhomboidal; *τὸ ῥομβοειδέξ*, *no*, romboid.

ῥόμβοξ, ὁ *no*, rhombus.

σημεῖον, τό : *no*, point.

σκαληνόξ -ή -όν : *adj*, scalene.

στερεόξ -ά -όν : *adj*, solid; *τὸ στερεόν*, *no*, solid, solid body.

στοιθεῖον, τό : *no*, element.

στρέφω, -στρέψω, ἔστρεψα, **—,** *ἐσταμμαι, ἐστάφην* : *vb*, turn.

σύγκειμαι : *vb*, lie together, be the sum of, be composed; *συγκείμενοξ -η -ον*, *adj*, composed (ratio), compounded; see *κεῖμαι*.

σύγκρίνω : *vb*, compare; see *κρίνω*.

συμβαίνω : *vb*, come to pass, happen, follow; see *βαίνω*.

συμβάλλω : *vb*, throw together, meet; see *βάλλω*.

σύμμετροξ -ον : *adj*, commensurable.

σύμπαξ -αντοξ, ὁ : *no*, sum, whole.

συμπίπτω : *vb*, meet together (of lines); see *πίπτω.*

συμπληϱόω : *vb*, complete (a figure), fill in.

συνάγω : *vb*, conclude, infer; see *ἄγω.*

συναμφότεϱοι -αι -α : *adj*, both together; *ὁ συναμφότεϱοξ*, *no*, sum (of two things).

συναποδείκνυμι : *no*, demonstrate together; see *δείκνυμι.*

συναφή, ἡ : *no*, point of junction.

σύνδυο, οἱ, αἱ, τά : *no*, two together, in pairs.

συνεϑήξ -έξ : *adj*, continuous; *κατὰ τὸ συνεϑέξ*, continuously.

σύνϑεσιξ -εωξ, ἡ : *no*, putting together, composition.

σύνϑετοξ -ον : *adj*, composite.

συ[ν]ίστημι : *vb*, construct (a figure), set up together; *perf imperat pass 3rd sg*, *συνεστάτω*; see *ἵστημι.*

συντίϑημι : *vb*, put together, add together, compound (ratio); see *τίϑημι.*

σϑέσιξ -εωξ, ἡ : *no*, state, condition.

σϑῆμα -ατοξ, τό : *no*, figure.

σφαῖϱα -αξ, ἡ : *no*, sphere.

τάχιξ -εωξ, ἡ : *no*, arrangement, order.

ταϱάσσω, ταϱάχω, **—, —,** *τετάϱαγμαι, ἐταϱάϑϑην* : *vb*, stir, trouble, disturbe; *τεταϱαγμένοξ -η -ον*, *adj*, disturbed, perturbed.

τάσσω, τάχω, ἔταχα, τέταϑα, τέταγμαι, ἐτάϑϑην : *vb*, arrange, draw up.

τέλειοξ -α -ον : *adj*, perfect.

τέμνω, τεμνῶ, ἔτεμον, -τέτμηκα, τέτμημαι, ἐτμήϑην : *vb*, cut; *pres/fut indic act 3rd sg*, *τέμει.*

τεταϱτημοϱιον, τό : *no*, quadrant.

τετϱάγωνοξ -ον : *adj*, square; *τὸ τετϱάγωνον*, *no*, square.

τετϱάκιξ : *adv*, four times.

τετϱαπλάσιοξ -α -ον : *adj*, quadruple.

τετϱάπλευϱοξ -ον : *adj*, quadrilateral.

τετϱαπλόοξ -η -ον : *adj*, fourfold.

τίϑημι, ϑήσω, ἔϑηκα, τέϑηκα, κεῖμαι, ἐτέϑην : *vb*, place, put.

τμῆμα -ατοξ, τό : *no*, part cut off, piece, segment.

τοίνυν : *par*, accordingly.

τοιοῦτοξ -αύτη -οῦτο : *pro*, such as this.

τομεύξ -έωξ, ὁ : *no*, sector (of circle).

τομή, ἡ : *no*, cutting, stump, piece.

τόποξ, ὁ : *no*, place, space.

τοσαυτάκιξ : *adv*, so many times.

τοσαυταπλάσιοξ -α -ον : *pro*, so many times.

τοσοῦτοξ -αύτη -οῦτο : *pro*, so many.

τουτέστι **=** *τοῦτ᾽ ἔστι* : *par*, that is to say.

τϱαπέζιον, τό : *no*, trapezium.

τϱίγωνοξ -ον : *adj*, triangular; *τὸ τϱίγωνον*, *no*, triangle.

τϱιπλάσιοξ -α -ον : *adj*, triple, threefold.

τϱίπλευϱοξ -ον : *adj*, trilateral.

τϱιπλ -όοξ -η -ον : *adj*, triple.

τϱόποξ, ὁ : *no*, way.

τυγϑάνω, τεύχομαι, ἔτυϑον, τετύϑηκα, τέτευγμαι, ἐτεύϑϑην : *vb*, hit, happen to be at (a place).

ὑπάϱϑω : *vb*, begin, be, exist; see *ἄϱϑω.*

ὑπεχαίϱεσιξ -εωξ, ἡ : *no*, removal.

ὑπεϱβάλλω : *vb*, overshoot, exceed; see *βάλλω.*

ὑπεϱοϑή, ἡ : *no*, excess, difference.

ὑπεϱέϑω : *vb*, exceed; see *ἔϑω.*

ὑπόϑεσιξ -εωξ, ἡ : *no*, hypothesis.

ὑπόκειμαι : *vb*, underlie, be assumed (as hypothesis); see *κεῖμαι.*

ὑπολείπω : *vb*, leave remaining.

ὑποτείνω, ὑποτενῶ, ὑπέτεινα, ὑποτέτακα, ὑποτέταμαι, ὑπετάϑην : *vb*, subtend.

ὕψοξ -εοξ, τό : *no*, height.

φανεϱόξ -ά -όν : *adj*, visible, manifest.

φημί, φήσω, ἔφην, **—, —, —** : *vb*, say; *ἔφαμεν*, we said.

φέϱω, οἴσω, ἤνεγκον, ἐνήνοϑα, ἐνήνεγμαι, ἠνέϑϑην : *vb*, carry.

ϑώϱιον, τό : *no*, place, spot, area, figure.

ϑωϱίξ : *pre* + *gen*, apart from.

ψαύω : *vb*, touch.

ὡξ : *par*, as, like, for instance.

ὡξ ἔτυϑεν : *par*, at random.

ὡσαύτωξ : *adv*, in the same manner, just so.

ὥστε : *conj*, so that (causal), hence.

GREEK-ENGLISH LEXICON

ABBREVIATIONS: *act* - active; *adj* - adjective; *adv* - adverb; *conj* - conjunction; *fut* - future; *gen* - genitive; *imperat* - imperative; *impf* - imperfect; *ind* - indeclinable; *indic* - indicative; *intr* - intransitive; *mid* - middle; *neut* - neuter; *no* - noun; *par* - particle; *part* - participle; *pass* - passive; *perf* - perfect; *pre* - preposition; *pres* - present; *pro* - pronoun; *sg* - singular; *tr* - transitive; *vb* - verb.

ἄγω, *ἄχω, ἤγαγον, -ἦθα, ἦγμαι, ἤθθην* : *vb*, lead, draw (a line).

ἀδύνατοξ -ον : *adj*, impossible.

ἀεί : *adv*, always, for ever.

αἱρέω, αἱρήσω, ε[˜]λον, ἥρηκα, ἥρημαι, ἡρέθην : *vb*, grasp.

ἀιτέω, αἰτήσω, ἤτησα, ἤτηκα, ἤτημαι, ἠτήθη : *vb*, postulate.

αἴτημα -ατοξ, τό : *no*, postulate.

ἀκόλουθοξ -ον : *adj*, analogous, consequent on, in conformity with.

ἄκροξ -α -ον : *adj*, outermost, end, extreme.

ἀλλά : *conj*, but, otherwise.

ἄλογοξ -ον : *adj*, irrational.

ἅμα : *adv*, at once, at the same time, together.

ἀμβλυγώνιοξ -ον : *adj*, obtuse-angled; *τὸ ἀμβλυγώνιον*, *no*, obtuse angle.

ἀμβλύξ -εῖα -ύ : *adj*, obtuse.

ἀμφότεροξ -α -ον : *pro*, both.

ἀναγράφω : *vb*, describe (a figure); see *γράφω.*

ἀναλογία, ἥ : *no*, proportion, (geometric) progression.

ἀνάλογοξ -ον : *adj*, proportional.

ἀνάπαλιν : *adv*, inverse(ly).

αναπληρόω : *vb*, fill up.

ἀναστρέφω : *vb*, turn upside down, convert (ratio); see *στρέφω.*

ἀναστροφή, ἥ : *no*, turning upside down, conversion (of ratio).

ἀνθυφαιρέω : *vb*, take away in turn; see *αἱρέω.*

ἀνίστημι : *vb*, set up; see *ἴστημι.*

ἄνισοξ -ον : *adj*, unequal, uneven.

ἀντιπάσθω : *vb*, be reciprocally proportional; see *πάσθω.*

ἄχων -ονοξ, ὅ : *vb*, axis.

ἅπαχ : *adv*, once.

ἅπαξ, ἅπασα, ἅπαν : *adj*, quite all, the whole.

ἄπειροξ -ον : *adj*, infinite.

ἀπεναντίον : *ind*, opposite.

ἀπέθω : *vb*, be far from, be away from; see *ἔθω.*

ἀπλατήξ -έξ : *adj*, without breadth.

ἀπόδειχιξ -εωξ, ἥ : *no*, proof.

ἀποκαθίστημι : *vb*, re-establish, restore; see *ἴστημι.*

ἀπολαμβάνω : *vb*, take from, subtract from, cut off from; see *λαμβάνω.*

ἀποτέμνω : *vb*, cut off, subtend.

ἀπότμημα -ατος, τό : *no*, piece cut off, segment.

ἀποτομή, ἥ : *vb*, piece cut off, apotome.

ἅπτω, ἅψω, ἧψα, **—,** *ἧμμαι*, **—** : *vb*, touch, join, meet.

ἀπώτεροξ -α -ον : *adj*, further off.

ἄρα : *par*, thus, as it seems (inferential).

ἀριθμόξ, ὅ : *no*, number.

ἀρτιάκιξ : *adv*, an even number of times.

αρτιόπλευροξ -ον : *adj*, having a even number of sides.

ἄρθω, ἄρχω, ἦρχα, ἦρθα, ἦργμαι, ἦρθθην : *vb*, rule; *mid.*, begin.

ἀσύμμετροξ -ον : *adj*, incommensurable.

ἀσύμπτωτοξ -ον : *adj*, not touching, not meeting.

ἄρτιοξ -α -ον : *adj*, even, perfect.

ἄτμητοξ -ον : *adj*, uncut.

ἀτόποξ -ον : *adj*, absurd, paradoxical.

αὐτόθεν : *adv*, immediately, obviously.

ἀφαίρεω : *vb*, take from, subtract from, cut off from; see *αἱρέω.*

ἁφή, ἥ : *no*, point of contact.

βάθοξ -εοξ, τό : *no*, depth, height.

βαίνω, -βήσομαι, -έβην, βέβηκα, **—, —** : *vb*, walk; *perf*, stand (of angle).

βάλλω, βαλῶ, ἔβαλον, βέβληκα, βέβλημαι, ἐβλήθην : *vb*, throw.

βάσιξ -εωξ, ἥ : *no*, base (of a triangle).

γάρ : *conj*, for (explanatory).

γί[γ]νομαι, γενήσομαι, ἐγενόμην, γέγονα, γεγένημαι, **—** : *vb*, happen, become.

γνώμων -ονοξ, ἥ : *no*, gnomon.

γραμμή, ἥ : *no*, line.

γράφω, γράψω, ἔγρα[ψ/φ]α, γέγραφα, γέγραμμαι, ἐραψάμην : *vb*, draw (a figure).

γωνία, ἥ : *no*, angle.

δεῖ : *vb*, be necessary; *δεῖ*, it is necessary; *ἔδει*, it was necassary; *δέον*, being necessary.

δείκνυμι, δείχω, ἔδειχα, δέδειθα, δέδειγμαι, ἐδείθθην : *vb*, show, demonstrate.

δεικτέον : *ind*, one must show.

δεῖχιξ -εωξ, ἥ : *no*, proof.

δεκαγώνοξ -ον : *adj*, ten-sided; *τὸ δεκαγώνον*, *no*, decagon.

δέθομαι, δέχομαι, ἐδεχάμην, **—,** *δέδεγμαι, ἐδέθθην* : *vb*, receive, accept.

δή : *conj*, so (explanatory).

δηλαδή : *ind*, quite clear, manifest.

δῆλοξ -η -ον : *adj*, clear.

δηλονότι : *adv*, manifestly.

διάγω : *vb*, carry over, draw through, draw across; see *ἄγω.*

διαγώνιοξ -ον : *adj*, diagonal.

διαλείπω : *vb*, leave an interval between.

διάμετροξ -ον : *adj*, diametrical; *ἥ διάμετροξ*, *no*, diameter, diagonal.

διαίρεσιξ -εωξ, ἥ : *no*, division, separation.

διαιρέω : *vb*, divide (in two); *διαρεθέντοξ -η -ον*, *adj*, separated (ratio); see *αἱρέω.*

διάστημα -ατοξ, τό : *no*, radius.

διαφέρω : *vb*, differ; see *φέρω.*

δίδωμι, δώσω, ἔδωκα, δέδωκα, δέδομαι, ἐδόθην : *vb*, give.

διμοίροξ -ον : *adj*, two-thirds.

διπλασιάζω : *vb*, double.

διπλάσιοξ -α -ον : *adj*, double, twofold.

διπλασίων -ον : *adj*, double, twofold.

διπλοῦξ -ῆ -οῦν : *adj*, double.

δίξ : *adv*, twice.

δίϑα : *adv*, in two, in half.

διϑοϱομία, ἡ : *no*, point of bisection.

δυάξ -άδοξ, ἡ : *no*, the number two, dyad.

δύναμαι : *vb*, be able, be capable, generate, square, be when squared; *δυναμένη, ἡ*, *no*, square-root (of area)—*i.e.*, straight-line whose square is equal to a given area.

δύναμιξ -εωξ, ἡ : *no*, power (usually 2nd power when used in mathematical sence, hence), square.

δυνατόξ -ή -όν : *adj*, possible.

δωδεκάεδϱοξ -ον : *adj*, twelve-sided.

ἑαυτοῦ -ῆξ -οῦ : *adj*, of him/her/it/self, his/her/its/own.

ἐγγίων -ον : *adj*, nearer, nearest.

ἐγγϱάφω : *vb*, inscribe; see *γϱάφω.*

εἶδοξ -εοξ, τό : *no*, figure, form, shape.

εἰκοσάεδϱοξ -ον : *adj*, twenty-sided.

εἴϱω/λέγω, ἐϱῶ/εϱέω, εἶπον, εἴϱηκα, εἴϱημαι, ἐϱϱήϑην : *vb*, say, speak; *per pass part*, *ειϱημένοξ -η -ον*, *adj*, said, aforementioned.

εἴτε **. . .** *εἴτε* : *ind*, either . . . or.

ἕκαστοξ -η -ον : *pro*, each, every one.

ἑκατέϱοξ -α -ον : *pro*, each (of two).

ἐκβάλλω : *vb*, produce (a line); see *βάλλω.*

ἐκϑέω : *vb*, set out.

ἔκκειμαι : *vb*, be set out, be taken; see *κεῖμαι.*

ἐκτίϑημι : *vb*, set out; see *τίϑημι.*

ἐκτόξ : *pre + gen*, outside, external.

ἐλάσσων/ἐλάττων -ον : *adj*, less, lesser.

ἐλάϑιστοξ -η -ον : *adj*, least.

ἐλλείπω : *vb*, be less than, fall short of.

ἐμπίπτω : *vb*, meet (of lines), fall on; see *πίπτω.*

ἔμπϱοσϑεν : *adv*, in front.

ἐναλλάχ : *adv*, alternate(ly).

ἐναϱμόζω : *vb*, insert; *perf indic pass 3rd sg*, *ἐνήϱμοσται.*

ἐνδέϑομαι : *vb*, admit, allow.

ἕνεκεν : *ind*, on account of, for the sake of.

ἐνναπλάσιοξ -α -ον : *adj*, nine-fold, nine-times.

ἔννοια, ἡ : *no*, notion.

ενπεϱιέϑω : *vb*, encompass; see *ἔϑω.*

ἐνπίπτω : see *ἐμπίπτω.*

ἐντόξ : *pre + gen*, inside, interior, within, internal.

ἐχάγωνοξ -ον : *adj*, hexagonal; *τὸ ἐχάγωνον*, *no*, hexagon.

ἐχαπλάσιοξ -α -ον : *adj*, sixfold.

ἐχῆξ : *adv*, in order, successively, consecutively.

ἔχωϑεν : *adv*, outside, extrinsic.

ἐπάνω : *adv*, above.

ἐπαφή, ἡ : *no*, point of contact.

ἐπεί : *conj*, since (causal).

ἐπειδήπεϱ : *ind*, inasmuch as, seeing that.

ἐπιζεύγνῦμι, ἐπιζεύχω, ἐπέζευχα, **—,** *ἐπέζευγμαι, ἐπέζεύϑϑην* : *vb*, join (by a line).

ἐπιλογίζομαι : *vb*, conclude.

ἐπινοέω : *vb*, think of, contrive.

ἐπιπέδοξ -ον : *adj*, level, flat, plane; *τὸ ἐπιπέδον*, *no*, plane.

ἐπισκέπτομαι : *vb*, investigate.

ἐπίσκεψιξ -εωξ, ἡ : *no*, inspection, investigation.

ἐπιτάσσω : *vb*, put upon, enjoin; *τὸ ἐπιταϑϑέν*, *no*, the (thing) prescribed; see *τάσσω.*

ἐπίτϱιτοξ -ον : *adj*, one and a third times.

ἐπιφάνεια, ἡ : *no*, surface.

ἕπομαι : *vb*, follow.

ἔϱϑομαι, ἐλεύσομαι, ἦλϑον, ἐλήλυϑα, **—, —** : *vb*, come, go.

ἔσϑατοξ -η -ον : *adj*, outermost, uttermost, last.

ἑτεϱόμηκηξ -εξ : *adj*, oblong; *τὸ ἑτεϱόμηκεξ*, *no*, rectangle.

ἕτεϱοξ -α -ον : *adj*, other (of two).

ἔτι : *par*, yet, still, besides.

εὐϑύγϱαμμοξ -ον : *adj*, rectilinear; *τὸ εὐϑύγϱαμμον*, *no*, rectilinear figure.

εὐϑύξ -εῖα -ύ : *adj*, straight; *ἡ εὐϑεῖα*, *no*, straight-line; *ἐπ᾽ εὐϑεῖαξ*, in a straight-line, straight-on.

εὑϱίσκω, εὑϱήσκω, ηὗϱον, εὕϱεκα, εὕϱημαι, εὑϱέϑην : *vb*, find.

ἐφάπτω : *vb*, bind to; *mid*, touch; *ἡ ἐφαπτομένη*, *no*, tangent; see *ἅπτω.*

ἐφαϱμόζω, ἐφαϱμόσω, ἐφήϱμοσα, ἐφήμοκα, ἐφήμοσμαι, ἐφήμόσϑην : *vb*, coincide; *pass*, be applied.

ἐφεχῆξ : *adv*, in order, adjacent.

ἐφίστημι : *vb*, set, stand, place upon; see *ἵστημι.*

ἔϑω, ἕχω, ἔσϑον, ἔσϑηκα, -έσϑημαι, **—** : *vb*, have.

ἡγέομαι, ἡγήσομαι, ἡγησάμην, ἡγημαι, **—,** *ἡγήϑην* : *vb*, lead.

ἤδη : *ind*, already, now.

ἥκω, ἥχω, **—, —, —, —** : *vb*, have come, be present.

ἡμικύκλιον, τό : *no*, semi-circle.

ἡμιόλιοξ -α -ον : *adj*, containing one and a half, one and a half times.

ἥμισυξ -εια -υ : *adj*, half.

ἤπεϱ **=** *ἤ* **+** *πεϱ* : *conj*, than, than indeed.

ἤτοι **. . .** *ἤ* : *par*, surely, either . . . or; in fact, either . . . or.

ϑέσις -εωξ, ἡ : *no*, placing, setting, position.

ϑεωϱημα -ατοξ, τό : *no*, theorem.

ἴδιοξ -α -ον : *adj*, one's own.

ἰσάκιξ : *adv*, the same number of times; *ἰσάκιξ πολλαπλάσια*, the same multiples, equal multiples.

ἰσογώνιοξ -ον : *adj*, equiangular.

ἰσόπλευϱοξ -ον : *adj*, equilateral.

ἰσοπληϑήξ -έξ : *adj*, equal in number.

ἴσοξ -η -ον : *adj*, equal; *ἐχ ἴσου*, equally, evenly.

ἰσοσκελήξ -έξ : *adj*, isosceles.

ἵστημι, στήσω, ἔστησα, —, —, ἐστάϑην : *vb tr*, stand (something).

ἵστημι, στήσω, ἔστην, ἕστηκα, ἕσταμαι, ἐσταϑην : *vb intr*, stand up (oneself); Note: perfect *I have stood up* can be taken to mean present *I am standing*.

ἰσοϋψήξ -έξ : *adj*, of equal height.

καϑάπεϱ : *ind*, according as, just as.

κάϑετοξ -ον : *adj*, perpendicular.

καϑόλου : *adv*, on the whole, in general.

καλέω : *vb*, call.

κἀκεινοξ = καὶ ἐκεῖνοξ .

κ᾽άν = καὶ ᾽άν : *ind*, even if, and if.

καταγϱαφή, ἥ : *no*, diagram, figure.

καταγϱάφω : *vb*, describe/draw, inscribe (a figure); see γϱάφω.

κατακολουϑέω : *vb*, follow after.

καταλείπω : *vb*, leave behind; see λείπω; τὰ καταλειπόμενα, *no*, remainder.

κατάλληλοξ -ον : *adj*, in succession, in corresponding order.

καταμετϱέω : *vb*, measure (exactly).

καταντάω : *vb*, come to, arrive at.

κατασκευάζω : *vb*, furnish, construct.

κεῖμαι, κεῖσομαι, —, —, —, — : *vb*, have been placed, lie, be made; see τίϑημι.

κέντϱον, τό : *no*, center.

κλάω : *vb*, break off, inflect.

κλίνω, κλίνω, ἔκλινα, κέκλικα, κέκλιμαι, ἐκλίϑην : *vb*, lean, incline.

κλίσιξ -εωξ, ἥ : *no*, inclination, bending.

κοῖλοξ -η -ον : *adj*, hollow, concave.

κοϱυφή, ἥ : *no*, top, summit, apex; κατὰ κοϱυφήν, vertically opposite (of angles).

κϱίνω, κϱινῶ, ἔκϱῖνα, κέκϱικα, κέκϱιμαι, ἐκϱίϑην : *vb*, judge.

κύβοξ, ὅ : *no*, cube.

κύκλοξ, ὅ : *no*, circle.

κύλινδϱοξ, ὅ : *no*, cylinder.

κυϱτόξ -ή -όν : *adj*, convex.

κῶνοξ, ὅ : *no*, cone.

λαμβάνω, λήψομαι, ἔλαβον, εἴληφα εἴλημμαι, ἐλήφϑην : *vb*, take.

λέγω : *vb*, say; *pres pass part*, λεγόμενοξ -η -ον, *adj*, so-called; see ἔιϱω.

λείπω, λείψω, ἔλιπον, λέλοιπα, λέλειμμαι, ἐλείφϑην : *vb*, leave, leave behind.

λημμάτιον, τό : *no*, diminutive of λῆμμα.

λῆμμα -ατοξ, τό : *no*, lemma.

λῆψιξ -εωξ, ἥ : *no*, taking, catching.

λόγοξ, ὅ : *no*, ratio, proportion, argument.

λοιπόξ -ή -όν : *adj*, remaining.

μανϑάνω, μαϑήσομαι, ἔμαϑον, μεμάϑηκα, —, — : *vb*, learn.

μέγεϑοξ -εοξ, τό : *no*, magnitude, size.

μείζων -ον : *adj*, greater.

μένω, μενῶ, ἔμεινα, μεμένηκα, —, — : *vb*, stay, remain.

μέϱοξ -ουξ, τό : *no*, part, direction, side.

μέσοξ -η -ον : *adj*, middle, mean, medial; ἐκ δύο μέσων, bimedial.

μεταλαμβάνω : *vb*, take up.

μεταχύ : *adv*, between.

μετέωϱοξ -ον : *adj*, raised off the ground.

μετϱέω : *vb*, measure.

μέτϱον, τό : *no*, measure.

μηδείξ, μηδεμία, μηδέν : *adj*, not even one, (neut.) nothing.

μηδέποτε : *adv*, never.

μηδέτεϱοξ -α -ον : *pro*, neither (of two).

μῆκοξ -εοξ, τό : *no*, length.

μήν : *par*, truely, indeed.

μονάξ -άδοξ, ἥ : *no*, unit, unity.

μοναϑόξ -ή -όν : *adj*, unique.

μοναϑῶξ : *adv*, uniquely.

μόνοξ -η -ον : *adj*, alone.

νοέω, —, νόησα, νενόηκα, νενόημαι, ἐνοήϑην : *vb*, apprehend, conceive.

οἷοξ -α -ον : *pre*, such as, of what sort.

ὀκτάεδϱοξ -ον : *adj*, eight-sided.

ὅλοξ -η -ον : *adj*, whole.

ὁμογενήξ -έξ : *adj*, of the same kind.

ὅμοιοξ -α -ον : *adj*, similar.

ὁμοιοπληϑήξ -έξ : *adj*, similar in number.

ὁμοιοταγήξ -έξ : *adj*, similarly arranged.

ὁμοιότηξ -ητοξ, ἥ : *no* similarity.

ὁμοίωξ : *adv*, similarly.

ὁμόλογοξ -ον : *adj*, corresponding, homologous.

ὁμοταγήξ -έξ : *adj*, ranged in the same row or line.

ὁμώνυμοξ -ον : *adj*, having the same name.

ὄνομα -ατοξ, τό : *no*, name; ἐκ δύο ὀνομάτων, binomial.

ὀχυγώνιοξ -ον : *adj*, acute-angled; τὸ ὀχυγώνιον, *no*, acute angle.

ὀχύξ -εῖα -ύ : *adj*, acute.

ὁποιοσοῦν = ὁποῖοξ -α -ον + οὖν : *adj*, of whatever kind, any kind whatsoever.

ὁπόσοξ -η -ον : *pro*, as many, as many as.

ὁποσοσδηποτοῦν = ὁπόσοξ -η -ον + δή + ποτέ + οὖν : *adj*, of whatever number, any number whatsoever.

ὁποσοσοῦν = ὁπόσοξ -η -ον + οὖν : *adj*, of whatever number, any number whatsoever.

ὁπότεϱοξ -α -ον : *pro*, either (of two), which (of two).

ὀϱϑογώνιον, τό : *no*, rectangle, right-angle.

ὀϱϑόξ -ή -όν : *adj*, straight, right-angled, perpendicular; πϱὸξ ὀϱϑὰξ γωνίαξ, at right-angles.

ὅϱοξ, ὅ : *no*, boundary, definition, term (of a ratio).

ὁσαδηποτοῦν = ὅσα + δή + ποτέ + οὖν : *ind*, any number whatsoever.

ὁσάκιξ : *ind*, as many times as, as often as.

ὁσαπλάσιοξ -ον : *pro*, as many times as.

ὅσοξ -η -ον : *pro*, as many as.

ὅσπερ, ἥπερ, ὅπερ : *pro*, the very man who, the very thing which.

ὅστιξ, ἥτιξ, ὅ τι : *pro*, anyone who, anything which.

ὅταν : *adv*, when, whenever.

ὁτιοῦν : *ind*, whatsoever.

οὐδείξ, οὐδεμία, οὐδέν : *pro*, not one, nothing.

οὐδέτεροξ -α -ον : *pro*, not either.

οὐθέτεροξ : see *οὐδέτεροξ*.

οὐθέν : *ind*, nothing.

οὖν : *adv*, therefore, in fact.

οὕτωξ : *adv*, thusly, in this case.

πάλιν : *adv*, back, again.

πάντωξ : *adv*, in all ways.

παρά : *prep* + *acc*, parallel to.

παραβάλλω : *vb*, apply (a figure); see *βάλλω*.

παραβολή, ἡ : *no*, application.

παράκειμαι : *vb*, lie beside, apply (a figure); see *κεῖμαι*.

παραλλάσσω, παραλλάχω, **—**, *παρήλλαθα*, **—**, **—** : *vb*, miss, fall awry.

παραλληλεπίπεδοξ, -ον : *adj*, with parallel surfaces; *τὸ παραλληλεπίπεδον*, *no*, parallelepiped.

παραλληλόγραμμοξ -ον : *adj*, bounded by parallel lines; *τὸ παραλληλόγραμμον*, *no*, parallelogram.

παράλληλοξ -ον : *adj*, parallel; *τὸ παράλληλον*, *no*, parallel, parallel-line.

παραπλήρωμα -ατοξ, τό : *no*, complement (of a parallelogram).

παρατέλευτοξ -ον : *adj*, penultimate.

παρέκ : *prep* + *gen*, except.

παρεμπίπτω : *vb*, insert; see *πίπτω*.

πάσθω, πείσομαι, ἔπαθον, πέπονθα, **—**, **—** : *vb*, suffer.

πεντάγωνοξ -ον : *adj*, pentagonal; *τὸ πεντάγωνον*, *no*, pentagon.

πενταπλάσιοξ -α -ον : *adj*, five-fold, five-times.

πεντεκαιδεκάγωνον, τό : *no*, fifteen-sided figure.

πεπερασμένοξ -η -ον : *adj*, finite, limited; see *περαίνω*.

περαίνω, περανῶ, ἐπέρανα, **—**, *πεπεράσμαι, ἐπερανάνθην* : *vb*, bring to end, finish, complete; *pass*, be finite.

πέραξ -ατοξ, τό : *no*, end, extremity.

περατόω, **—**, **—**, **—**, **—**, **—** : *vb*, bring to an end.

περιγράφω : *vb*, circumscribe; see *γράφω*.

περιέθω : *vb*, encompass, surround, contain, comprise; see *ἔθω*.

περιλαμβάνω : *vb*, enclose; see *λαμβάνω*.

περισσάκιξ : *adv*, an odd number of times.

περισσόξ -ή -όν : *adj*, odd.

περιφέρεια, ἡ : *no*, circumference.

περιφέρω : *vb*, carry round; see *φέρω*.

πηλικότηξ -ητοξ, ἡ : *no*, magnitude, size.

πίπτω, πεσοῦμαι, ἔπεσον, πέπτωκα, **—**, **—** : *vb*, fall.

πλάτοξ -εοξ, τό : *no*, breadth, width.

πλείων -ον : *adj*, more, several.

πλευρά, ἡ : *no*, side.

πλῆθοξ -εοξ, τό : *no*, great number, multitude, number.

πλήν : *adv & prep* + *gen*, more than.

ποιόξ -ά -όν : *adj*, of a certain nature, kind, quality, type.

πολλαπλασιάζω : *vb*, multiply.

πολλαπλασιασμόξ, ὁ : *no*, multiplication.

πολλαπλάσιον, τό : *no*, multiple.

πολύεδροξ -ον : *adj*, polyhedral; *τό πολύεδρον*, *no*, polyhedron.

πολύγωνοξ -ον : *adj*, polygonal; *τό πολύγωνον*, *no*, polygon.

πολύπλευροξ -ον : *adj*, multilateral.

πόρισμα -ατοξ, τό : *no*, corollary.

ποτέ : *ind*, at some time.

πρῖσμα -ατοξ, τό : *no*, prism.

προβαίνω : *vb*, step forward, advance.

προδείκνυμι : *vb*, show previously; see *δείκνυμι*.

προεκτίθημι : *vb*, set forth beforehand; see *τίθημι*.

προερέω : *vb*, say beforehand; *perf pass part*, *προειρημένοξ -η -ον*, *adj*, aforementioned; see *εἴρω*.

προσαναπληρόω : *vb*, fill up, complete.

προσαναγράφω : *vb*, complete (tracing of); see *γράφω*.

προσαρμόζω : *vb*, fit to, attach to.

προσεκβάλλω : *vb*, produce (a line); see *ἐκβάλλω*.

προσευρίσκω : *vb*, find besides, find; see *εὑρίσκω*.

προσλαμβάμω : *vb*, add.

πρόκειμαι : *vb*, set before, prescribe; see *κεῖμαι*.

πρόσκειμαι : *vb*, be laid on, have been added to; see *κεῖμαι*.

προσπίπτω : *vb*, fall on, fall toward, meet; see *πίπτω*.

προτασιξ -εωξ, ἡ : *no*, proposition.

προστάσσω : *vb*, prescribe, enjoin; *τὸ τροσταθθέν*, *no*, the thing prescribed; see *τάσσω*.

προστίθημι : *vb*, add; see *τίθημι*.

πρότεροξ -α -ον : *adj*, first (comparative), before, former.

προτίθημι : *vb*, assign; see *τίθημι*.

προθωρέω : *vb*, go/come forward, advance.

πρῶτοξ -α -ον : *adj*, first, prime.

πυραμίξ -ίδοξ, ἡ : *no*, pyramid.

ῥητόξ -ή -όν : *adj*, expressible, rational.

ῥομβοειδήξ -έξ : *adj*, rhomboidal; *τὸ ῥομβοειδέξ*, *no*, romboid.

ῥόμβοξ, ὁ *no*, rhombus.

σημεῖον, τό : *no*, point.

σκαληνόξ -ή -όν : *adj*, scalene.

στερεόξ -ά -όν : *adj*, solid; *τὸ στερεόν*, *no*, solid, solid body.

στοιθεῖον, τό : *no*, element.

στρέφω, -στρέψω, ἔστρεψα, **—**, *ἐσταμμαι, ἐστάφην* : *vb*, turn.

σύγκειμαι : *vb*, lie together, be the sum of, be composed; *συγκείμενοξ -η -ον*, *adj*, composed (ratio), compounded; see *κεῖμαι*.

σύγκρίνω : *vb*, compare; see *κρίνω*.

συμβαίνω : *vb*, come to pass, happen, follow; see *βαίνω*.

συμβάλλω : *vb*, throw together, meet; see *βάλλω*.

σύμμετροξ -ον : *adj*, commensurable.

σύμπαξ -αντοξ, ὁ : *no*, sum, whole.

συμπίπτω : *vb*, meet together (of lines); see *πίπτω.*

συμπληϱόω : *vb*, complete (a figure), fill in.

συνάγω : *vb*, conclude, infer; see *ἄγω.*

συναμφότεϱοι -αι -α : *adj*, both together; *ὁ συναμφότεϱοξ*, *no*, sum (of two things).

συναποδείκνυμι : *no*, demonstrate together; see *δείκνυμι.*

συναφή, ἡ : *no*, point of junction.

σύνδυο, οἵ, αἵ, τά : *no*, two together, in pairs.

συνεϑήξ -έξ : *adj*, continuous; *κατὰ τὸ συνεϑέξ*, continuously.

σύνϑεσιξ -εωξ, ἡ : *no*, putting together, composition.

σύνϑετοξ -ον : *adj*, composite.

συ[ν]ίστημι : *vb*, construct (a figure), set up together; *perf imperat pass 3rd sg*, *συνεστάτω*; see *ἵστημι.*

συντίϑημι : *vb*, put together, add together, compound (ratio); see *τίϑημι.*

σϑέσιξ -εωξ, ἡ : *no*, state, condition.

σϑῆμα -ατοξ, τό : *no*, figure.

σφαῖϱα -αξ, ἡ : *no*, sphere.

τάχιξ -εωξ, ἡ : *no*, arrangement, order.

ταϱάσσω, ταϱάχω, **—, —,** *τετάϱαγμαι, ἐταϱάϑϑην* : *vb*, stir, trouble, disturbe; *τεταϱαγμένοξ -η -ον*, *adj*, disturbed, perturbed.

τάσσω, τάχω, ἔταχα, τέταϑα, τέταγμαι, ἐτάϑϑην : *vb*, arrange, draw up.

τέλειοξ -α -ον : *adj*, perfect.

τέμνω, τεμνῶ, ἔτεμον, -τέτμηκα, τέτμημαι, ἐτμήϑην : *vb*, cut; *pres/fut indic act 3rd sg*, *τέμει.*

τεταϱτημοϱιον, τό : *no*, quadrant.

τετϱάγωνοξ -ον : *adj*, square; *τὸ τετϱάγωνον*, *no*, square.

τετϱάκιξ : *adv*, four times.

τετϱαπλάσιοξ -α -ον : *adj*, quadruple.

τετϱάπλευϱοξ -ον : *adj*, quadrilateral.

τετϱαπλόοξ -η -ον : *adj*, fourfold.

τίϑημι, ϑήσω, ἔϑηκα, τέϑηκα, κεῖμαι, ἐτέϑην : *vb*, place, put.

τμῆμα -ατοξ, τό : *no*, part cut off, piece, segment.

τοίνυν : *par*, accordingly.

τοιοῦτοξ -αύτη -οῦτο : *pro*, such as this.

τομεύξ -έωξ, ὁ : *no*, sector (of circle).

τομή, ἡ : *no*, cutting, stump, piece.

τόποξ, ὁ : *no*, place, space.

τοσαυτάκιξ : *adv*, so many times.

τοσαυταπλάσιοξ -α -ον : *pro*, so many times.

τοσοῦτοξ -αύτη -οῦτο : *pro*, so many.

τουτέστι = *τοῦτʼ ἔστι* : *par*, that is to say.

τϱαπέζιον, τό : *no*, trapezium.

τϱίγωνοξ -ον : *adj*, triangular; *τὸ τϱίγωνον*, *no*, triangle.

τϱιπλάσιοξ -α -ον : *adj*, triple, threefold.

τϱίπλευϱοξ -ον : *adj*, trilateral.

τϱιπλ -όοξ -η -ον : *adj*, triple.

τϱόποξ, ὁ : *no*, way.

τυγϑάνω, τεύχομαι, ἔτυϑον, τετύϑηκα, τέτευγμαι, ἐτεύϑϑην : *vb*, hit, happen to be at (a place).

ὑπάϱϑω : *vb*, begin, be, exist; see *ἄϱϑω.*

ὑπεχαίϱεσιξ -εωξ, ἡ : *no*, removal.

ὑπεϱβάλλω : *vb*, overshoot, exceed; see *βάλλω.*

ὑπεϱοϑή, ἡ : *no*, excess, difference.

ὑπεϱέϑω : *vb*, exceed; see *ἔϑω.*

ὑπόϑεσιξ -εωξ, ἡ : *no*, hypothesis.

ὑπόκειμαι : *vb*, underlie, be assumed (as hypothesis); see *κεῖμαι.*

ὑπολείπω : *vb*, leave remaining.

ὑποτείνω, ὑποτενῶ, ὑπέτεινα, ὑποτέτακα, ὑποτέταμαι, ὑπετάϑην : *vb*, subtend.

ὕψοξ -εοξ, τό : *no*, height.

φανεϱόξ -ά -όν : *adj*, visible, manifest.

φημὶ, φήσω, ἔφην, **—, —, —** : *vb*, say; *ἔφαμεν*, we said.

φέϱω, οἴσω, ἤνεγκον, ἐνήνοϑα, ἐνήνεγμαι, ἠνέϑϑην : *vb*, carry.

ϑώϱιον, τό : *no*, place, spot, area, figure.

ϑωϱίξ : *pre* + *gen*, apart from.

ψαύω : *vb*, touch.

ὡξ : *par*, as, like, for instance.

ὡξ ἕτυϑεν : *par*, at random.

ὡσαύτωξ : *adv*, in the same manner, just so.

ὥστε : *conj*, so that (causal), hence.